Spring 2026: Algebraic Geometry II

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1 Schemes

For a scheme X, we will denote $|X|$ for its underlying topological space. We will assume all schemes are quasicompact and Noetherian.

1.1 Separatedness and Properness

A k-scheme $X \to \text{Spec } k$ is finite type if each affine piece is finitely generated k-algebra. More generally, a morphism $f : X \to Y$ is of finite type if $Y = \bigcup V_i, V_i = \text{Spec } S_i$, and cover $f^{-1}(V_i) = \bigcup U_{ij}, U_{ij} = \text{Spec } R_{ij}$, we require each ring map $S_i \to R_{ij}$ to be finite type.

Example. *The structure morphism $\mathbb{P}^n_R \to \text{Spec } R$ is of finite type.*

Theorem 1.1. *Let $f : X \to Y$ be of finite type. Then $f(|X|)$ is constructible subset of $|Y|$.*

Corollary 1.2. *Let $f : X \to \mathbb{A}^1_{\mathbb{C}}$ be of finite type. Then $f(|X|)$ is finite or cofinite set.*

Remark. *This contrasts with complex geometry. Consider:*

$$f : M = \{(x, y) \in \mathbb{C}^2 \mid (e^x - 1) \cdot y - 1\} \to \mathbb{C}, \quad (x, y) \mapsto x$$

Note that $\text{im } (f) = \mathbb{C}\backslash 2\pi\mathbb{Z}$ *is not finite nor cofinite.*

Let $X/\mathbb{C}$ be a prevariety. Recall we can put "Euclidean topology" on $X(\mathbb{C})$. The notion of separateness captures the "Hausdorff property" of $X(\mathbb{C})$.

Definition 1.3. Let $X \to S$ be a morphism. We call this morphism separated if:

$$\Delta_{X/S} : X \to X \times_S X$$

is a closed subscheme.

Example. *Any morphism between affine schemes* $\text{Spec } A \to \text{Spec } B$ *is separated. Indeed,*

$$A \otimes_B A \to A$$

is surjective.

Example. *Let's check $\mathbb{P}^1$ is separated, and affine line with double origin X is not. Consider:*

$$\Delta : X \to X \times_{\mathbb{C}} X$$

Let $U = \mathbb{A}^1 = \text{Spec } \mathbb{C}[u], V = \mathbb{A}^1 = \text{Spec } \mathbb{C}[v]$ be the two charts on X. Let's assess the map:

$$\text{Spec } \mathbb{C}[t, t^{-1}] = U \cap V \to U \times_{\mathbb{C}} V = \text{Spec } \mathbb{C}[u, v]$$

It is given by: $u \mapsto t, v \mapsto t$, which is not surjective. So that X is not separated. For similar reasoning, $\mathbb{P}^1$ is separated.

Proposition 1.4. *1. Separatedness is closed under base change.*

2. *Separatedness behaves well under composition.*

3. *Let $f : X \to \operatorname{Spec} R$ be separated. Let U, V be two affine opens in X, then $U \cap V$ is also affine open.*

4. *Let X, Y be S-schemes, with X reduced and Y separated. Suppose I have two regular S-morphism $f, g : X \to Y$ which agrees on some open dense subset $U \subset X$, then $f = g$.*

Proof. For the thrid statement, we look at the diagram

$$\begin{array}{ccc} U \cap V & \longrightarrow & U \times_S V \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times_S X \end{array}$$

We know the arrow up stairs is a closed embedding into an affine scheme. For the fourth statement, it suffices to show the lifting problem

$$\begin{array}{ccc} Z = X \times_{Y \times Y} Y & \longrightarrow & Y \\ \downarrow & \nearrow & \downarrow{\scriptstyle \Delta_{Y/S}} \\ X & \xrightarrow[(f,g)]{} & Y \times_S Y \end{array}$$

has a solution. This is equivalent to saying $Z = X$. But we know $|U| \subset |Z| \subset |X|$, but U is open dense, so that $|Z| = |X|$. Now that X is reduced, we conclude $Z = X$. □

Definition 1.5. A variety over a field k is a reduced, finite type, and separated scheme over k.

Let's now discuss properness. Recall in topology we say a map $g : M \to N$ is proper if the preimage of a compact set is compact.

Definition 1.6. Let $f : X \to Y$ be a morphism. We call f proper if it's separated, finite type, and universally closed.

Remark. *To check a morphism is closed, one can check this on an affine open cover of the target. As a consequence, to check universally closedness, one can check by only base change to affine.*

Remark. *The structure map $X \to \operatorname{Spec}(\mathbb{C})$ proper iff $X(\mathbb{C})$ is compact and Hausdorff.*

Example. *Closed embedding is proper.*

Example. $\mathbb{A}^1$ *is not universally closed. Consider the base change map*

$$\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$$

and look at the image of $xy = 1$.

Proposition 1.7. *A finite morphism if proper.*

Proof. It suffice to prove the affine version $f : \text{Spec } B \to \text{Spec } A$ for B finite over A. In fact, it suffices to show this map is closed, since any base change $A' \to B \otimes_A A'$ is still finite extension of rings. Let's consider irreducible Spec $B/Q \subset$ Spec B, we have $f(Q) = P = \varphi^{-1}(Q)$ for $\varphi : A \to B$ structure map. By going up, we see $f(\text{Spec } B/Q) = V(P) \subset \text{Spec } A$ is closed. □

Definition 1.8. A morphism $f : X \to Y$ is quasi-finite if all its fibres are finite sets. This is a strictly weaker notion than finiteness.

Proposition 1.9. *1. Composition of proper morphisms is proper.*

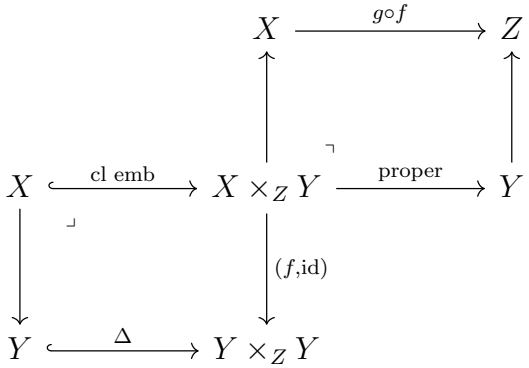
2. Base change of proper morphism is proper.

3. In the case

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

and assume $g \circ f$ *proper, and* g *separated, then* f *is proper.*

Proof. The third part is from the diagram:



□

Corollary 1.10. *Map from proper variety to any variety is automatically proper.*

Corollary 1.11. *If* X *is connected variety over* $k = \overline{k}$. *Then* $\mathcal{O}_X(X) = k$.

Proof. Given $f \in \mathcal{O}_X(X)$, we have a ring map $k[t] \to \mathcal{O}_X(X), t \mapsto f$. This gives a map:

$$X \to \text{Spec } k[t] \to \mathbb{P}^1$$

This is a proper morphism, so the image must be a single point. Hence the morphism:

$$k[t] \to \mathcal{O}_X(X)$$

factors as:

$$k[t] \to k \to \mathcal{O}_X(X)$$

□

Theorem 1.12. *The structure map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is proper. As a conseuqnece, each projective morphism is proper.*

Proof. (Vakil Chapter 8, Elimination theory). The only ingredient is to show $\mathbb{P}^n_R \to \operatorname{Spec} R$ is closed for each ring R. Consider any closed subset $Z = V(g_1, \cdots, v_r) \hookrightarrow \mathbb{P}^n_R$ for homogenous polynomial $g_i \in R[x_0, \cdots, x_n]$. So we need to construct $I \subset R$ such that $\pi(|Z|) = V(I)$. Consider:

$$\begin{array}{ccc} \mathbb{P}^n_\kappa \supset Z_t & \longrightarrow & Z \\ \downarrow & & \downarrow \\ t = \operatorname{Spec}(\kappa(P)) & \longrightarrow & \operatorname{Spec}(R) \end{array}$$

Explicitly

$$Z_t = V(\overline{g}_1, \cdots, \overline{g}_r), \quad g_i \in \kappa(P)[x_0, \cdots, x_n]$$

We ask when is $Z_t \neq \emptyset$. Indeed, this happens if

$$\sqrt{(\overline{g}_i)} \neq (x_0, \cdots, x_n) \iff \forall N, (x_0, \cdots, x_n)^N \nsubseteq (\overline{g}_i)$$

That happens iff for all N, there exists a monomial of degree N not in $(\overline{g}_i)$. Consider:

$$\kappa(P)[x_0, \cdots, x_n]_{N-d_i} \xrightarrow{\overline{g}_i} \kappa(P)[x_0, \cdots, x_n]_N$$

Collect all g_i:

$$\bigoplus_{i=1}^r \kappa(P)[x_0, \cdots, x_n]_{N-d_i} \xrightarrow{\oplus \overline{g}_i} \bigoplus_{i=1}^r \kappa(P)[x_0, \cdots, x_n]_N$$

This is given by some matrix $\overline{M}_N$. So that $t \in \operatorname{im}(Z) \iff \overline{M}_N$ is not surjective for all N. Hence the above condition is equivalent to: for all N, every determinant of $D_N \times D_N$ minor of $\overline{M}_N$ must vanish, where $\overline{M}_N$ is some reduction of some matrix M_N with coefficient in R. This is equivalent to:

$$\det(T_{N,i}) \in P, \text{for all minor } T_{N,i}$$

□

Let's conclude by stating valuation criterion of separateness and properness.

Theorem 1.13. *Let $f : X \to Y$ be a morphism of schemes, and assume X is noetherian. Then f is separated iff the following condition holds. For any valuaiton ring R with $K = \operatorname{Frac}(R)$, let $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$, then there is at most one solution for the lifting problem:*

$$\begin{array}{ccc} U & \longrightarrow & X \\ {\scriptstyle i}\downarrow & \nearrow & \downarrow \\ T & \longrightarrow & Y \end{array}$$

Assume further f is finite type. Then f is proper if the above lifting problem has a unique solution.

As an application, let's show show $\mathbb{P}^n_k$ is proper using valuation criterion. Suppose we have:

$$b : \text{Spec } K \to U_0 \subset \mathbb{P}^n$$

given by

$$b = [1 : b_1 : \cdots : b_n], \quad b_i \in K$$

Suppose that b_k has minimal valuation, then we can rewrite:

$$b = [\frac{1}{b_k} : \cdots \frac{b_n}{b_k}]$$

which extens to a moprhism Spec $R \to X$.

1.2 Quasi Coherent Sheaves on Projective Spaces

Recall we have line bundles $\mathcal{O}(a)$ on $\mathbb{P}^n = \text{Proj}(S_\bullet)$, for $S_\bullet = k[x_0, \cdots, x_n]$. Explicitly on affine open $\mathcal{O}(U_i) = (S_0[\frac{1}{x_i}])_{\deg 0}$, we put $M_i = \mathcal{O}(U_i)$. The gluing map $\varphi_{ij} : U_i \to U_j$ is multiplying by $(x_i/x_j)^a$. Then for example we have:

$$\{\frac{x_k}{x_i} \text{ on } M_i\} \subset \Gamma(\mathcal{O}(1))$$

Alternatively, one can use basis for open sets of $\mathbb{P}^n$: $D_+(f) = \{x \in \mathbb{P}^n \mid f(x) \neq 0\}$ for homogenous f of degree > 0. Now given a graded module $M_\bullet$ for $S_\bullet$, we obtain:

$$\widetilde{M_\bullet} \in \mathsf{QCoh}(\mathbb{P}^n)$$

$$\widetilde{M_\bullet}(D_+(f)) = (M_\bullet[\frac{1}{f}])_{\deg 0}$$

this is evidently an $\mathcal{O}(D_+(f)) = \text{Spec } (S_\bullet[\frac{1}{f}])_{\deg 0}$ module. Now give an point $[P] \in \mathbb{P}^n$ (so P is a homogenous prime ideal), and my stalk is:

$$\widetilde{M_\bullet}_{[P]} = ((M_\bullet)_P)_{\deg 0}$$

This construction depends on the grading of M. For example in general:

$$\widetilde{M_\bullet[a]} \neq \widetilde{M_\bullet}$$

Example. $\widetilde{S_\bullet[a]} = \mathcal{O}(a)$. *To see this, on U_i, we have:*

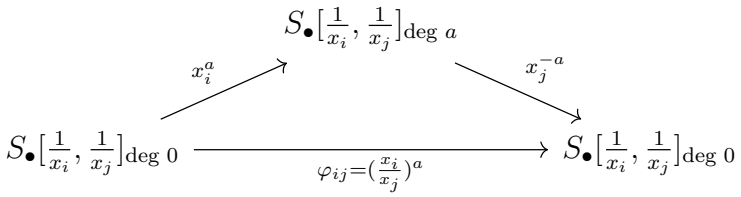
$$\widetilde{S_\bullet[a]}(U_i) = (S_\bullet[a][1/x_i])_{\deg 0} = S_\bullet[\frac{1}{x_i}]_{\deg a}$$

$$\mathcal{O}(a)(U_i) = S_\bullet[\frac{1}{x_i}]_{\deg 0}$$

Now I have an isomorphism:

$$S_\bullet[\frac{1}{x_i}]_{\deg 0} \xrightarrow{x_i^a} S_\bullet[\frac{1}{x_i}]_{\deg a}$$

On the overlap $U_i \cap U_j$:



is precisely the desired transition map.

In general, I have:

$$\widetilde{M_\bullet[a]} \simeq \widetilde{M_\bullet} \otimes \mathcal{O}(a)$$

Now let's consider the functor:

$$S_\bullet\text{-mod} \to \mathsf{QCoh}(\mathbb{P}^n)$$
$$M_\bullet \to \widetilde{M_\bullet}$$

This is not an equivalence of categories. For example if I have some $M_\bullet$ such that $M_N = 0$ for all sufficiently large N, then:

$$M_\bullet[\frac{1}{x_i}] = 0$$

In fact this is the only failure for this functor to be an equivalence. We have a functor in the other direction:

$$\Gamma_\bullet : \mathsf{QCoh}(\mathbb{P}^n) \to S_\bullet\text{-mod}$$
$$\mathcal{E} \mapsto \bigoplus_{a\in\mathbb{Z}} \Gamma(\mathbb{P}^n, \mathcal{E} \otimes \mathcal{O}(a))$$

To give RHS a $S_\bullet$-module structure, we need to produce a map:

$$S_b \otimes \Gamma(\mathcal{E} \otimes \mathcal{O}(a)) \to \Gamma(\mathcal{E} \otimes \mathcal{O}(a+b))$$

this is evident as $S_b = \Gamma(\mathcal{O}(b))$.

Lemma 1.14. $\Gamma_\bullet(\mathcal{O}_{\mathbb{P}^n}) = S_\bullet$.

Proof. Concretely, I need to compute $\Gamma(\mathbb{P}^n, \mathcal{O}(a))$ for each a. But:

$$\bigoplus_{a\in\mathbb{Z}} \Gamma(\mathbb{P}^n, \mathcal{O}(a)) = \ker(\bigoplus_{i,a} S_\bullet[\frac{1}{x_i}]_{\deg a} \rightrightarrows \bigoplus_{i,a} S_\bullet[\frac{1}{x_i}, \frac{1}{x_j}]_{\deg a}) = \bigcap_{i=0}^n S_\bullet[\frac{1}{x_i}]$$

View the last intersection inside $S_\bullet[\frac{1}{x_0}, \cdots, \frac{1}{x_n}]$. So it remains to see:

$$S_\bullet = \bigcap_{i=0}^n S_\bullet[\frac{1}{x_i}]$$

These rings have $\mathbb{Z}^{n+1}$-grading:

$$S_\bullet[\frac{1}{x_0}, \cdots, \frac{1}{x_n}] = \bigoplus_{(a_i)\in\mathbb{Z}^{n+1}} k x_0^{a_0} \cdots x_n^{a_n}$$

By looking at each grading piece the equality we desired is obvious. □

Theorem 1.15. $\widetilde{\Gamma_\bullet(\mathcal{E})} \simeq \mathcal{E}$

Definition 1.16. Let X be a scheme, $\mathcal{L} \in \mathrm{Pic}(X)$, and $s \in \Gamma(X, \mathcal{L})$. Given $x \in X$, picking a triviliazation gives me: $\varphi : \mathcal{L}_x \simeq \mathcal{O}_{X,x}$. Now that the map:

$$s \mapsto s_x \mapsto f_x = \varphi(s_x) \in \mathcal{O}_{X,x}$$

Note f_x is defined up to an element of $\mathcal{O}_{X,x}^\times$. We say s vanishes at x if $f_x \in \mathfrak{m}_x \subset \mathcal{O}_{X,x}$. This is independent of the trivialization we chose. We will denote $D(s) = \{x \in X \mid s \text{ does not vanish}\}$, this is an open subset of X.

Definition 1.17. Let $\mathcal{E} \in \mathsf{QCoh}(X), \mathcal{L} \in \mathrm{Pic}(X)$. Given two global sections $s \in \Gamma(\mathcal{L}), t \in \mathcal{E}$ I can form:

$$s^r \cdot t \in \Gamma(X, \mathcal{E} \otimes \mathcal{L}^r)$$

Lemma 1.18. *Given $\mathcal{E} \in \mathsf{QCoh}(X), t \in \Gamma(\mathcal{E})$ such that $t|_{D(s)} = 0$, then there exists $a > 0$ such that:*

$$s^a t = 0$$

as an element in $\Gamma(\mathcal{E} \otimes \mathcal{L}^a)$

Proof. Cover by charts and use the affine case. □

Lemma 1.19. *Let $t \in \Gamma(D(s), \mathcal{E})$, then there exists $b > 0$ such that*

$$s^b t \in \Gamma(D(s), \mathcal{E} \otimes \mathcal{L}^b)$$

extends to a global section.

Proof. (of theorem). We first need to construct a map:

$$\widetilde{\Gamma_\bullet(\mathcal{E})} \to \mathcal{E}$$

On U_i:

$$(\bigoplus \Gamma(\mathbb{P}^n, \mathcal{E}(a)) \otimes S_\bullet[\frac{1}{x_i}])_{\deg 0} \to \Gamma(U_i, \mathcal{E})$$

$$m \otimes \frac{1}{x_i^a} \mapsto \frac{1}{x_i^a} m|_{U_i}$$

This being an isomorphism follows from the two lemmas before. □

Corollary 1.20. *Let $Z \hookrightarrow \mathbb{P}_k^n$ closed. There exists a homogeneous ideal $I \subset S_\bullet$ cutting out Z.*

Definition 1.21. Let $\mathcal{L} \in \mathrm{Pic}(X)$. We say $\mathcal{L}$ is very ample if there exists a locally closed embedding $f : X \to \mathbb{P}^n$ such that $\mathcal{L} \simeq f^*\mathcal{O}(1)$.

Therefore, studying very ample divisors inside $\mathrm{Pic}(X)$ will help us to understand embeddings $X \hookrightarrow \mathbb{P}^n$.

Given $f : X \hookrightarrow \mathbb{P}^n$, and $s \in \Gamma(\mathbb{P}^n, \mathcal{O}(1))$, we can define:

$$f^*(s) = \Gamma(X, f^*\mathcal{O}(1))$$

View this as gluing together "$s \otimes 1$".

Definition 1.22. We call $\mathcal{L} \in \mathrm{Pic}(X)$ ample if $\mathcal{L}^{\otimes a}$ is very ample for some $a \geq 1$.

If X is smooth proper $/\mathbb{C}$, whether $\mathcal{L}$ is ample only depends on $c_1(\mathcal{L}) \in H^2(X(\mathbb{C}), \mathbb{Z})$. In some sense it has topological nature.

Definition 1.23. Let X be a scheme, $\mathcal{E} \in \mathrm{Coh}(X)$. We say $\mathcal{E}$ is generated by finitely many global sections $s_1, \cdots, s_r \in \Gamma(X, \mathcal{E})$ if for all $x \in X$, we have:

$$(s_i)_x \in \mathcal{E}_x$$

generate $\mathcal{E}_x$ as a $\mathcal{O}_{X,x}$-module. By Nakayama's lemma, it's equivalent to prove $s_i(x) \in \mathcal{E}_x/\mathfrak{m}_x\mathcal{E}_x$ generates. This is again equivalent to say:

$$\bigoplus_{i=1}^{r} \mathcal{O}_X \xrightarrow{(s_1,\cdots,s_r)} \mathcal{E}$$

is surjective.

Example. *On $\mathbb{P}^n$, we know $\mathcal{O}(a)$ is generated by finitely many global sections if for $a \geq 0$.*

For $\mathcal{L} \in \mathrm{Pic}(X)$, we see $s_1, \cdots, s_r \in \Gamma(X, \mathcal{L})$ globally generate iff $X = \bigcup D(s_i)$.

Theorem 1.24 (Serre)**.** *Let X/k be projective, $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X. Given $\mathcal{E} \in \mathrm{Coh}(X)$, there exists n_0 such that:*

$$\mathcal{E} \coloneqq \mathcal{E} \otimes \mathcal{O}_X(1)^{\otimes n}$$

is globally generated for $n \geq n_0$.

Proof. Say

$$i : X \hookrightarrow \mathbb{P}^n, \quad \mathcal{O}_X(1) = i^*\mathcal{O}_{\mathbb{P}^n}(1)$$

Note that $\{s_1, \cdots, s_r\}$ globally generate $\mathcal{E}$ iff $\{\overline{s_1}, \overline{s_2}, \cdots \overline{s_r}\}$ globally generate $i_*\mathcal{E}$. This is compatible with tensoring with $\mathcal{O}_{\mathbb{P}^n}(1)$ (resp. $\mathcal{O}_X(1)$) by projection formula. Therefore, it suffices to prove the statement for $\mathbb{P}^n$. We now cover $\mathbb{P}^n$ with U_i. We know:

$$\mathcal{E}|_{U_i} = \widetilde{M(i)}$$

for some finitely generated $\mathcal{O}(U_i)$. Let's pick $m_i^1, \cdots, m_i^r$ generating $M(i)$. Then

$$x_i m_i^j$$

extends to $\widetilde{m_i^{(j)}} \in \Gamma(\mathbb{P}^n, \mathcal{E}(a))$ for some large a. We can assume the same a for each U_i. Then it is straightforward to check the collection $\{\widetilde{m_i^{(j)}}\}$ globally generate $\mathcal{E} \otimes \mathcal{O}_X(a)$. □

Corollary 1.25. *Given $\mathcal{E} \in \mathrm{Coh}(X)$ for projective X, then there exists a surjection:*

$$\bigoplus_{i=1}^{m} \mathcal{O}_X(a_i) \to \mathcal{E} \to 0$$

Proof. Pick large a and consider:

$$\bigoplus \mathcal{O}_X \to \mathcal{E}(a) \to 0$$

We then tensor $\mathcal{O}_X(-a)$ back. □

Theorem 1.26. *Let X/k be projective, $\mathcal{E} \in \mathrm{Coh}(X)$. Then $\Gamma(X, \mathcal{E})$ is finite dimensional as a vector space.*

Remark. *Similar result holds for arbitrary ring base. In fact, this implies push forward a coherent sheaf along projective morphism remains coherent.*

Proof. (Sketched) Assume $X = \mathbb{P}^n$ and $\mathcal{E} = \widetilde{M_\bullet}$. We can assume $M_\bullet$ is finitely generated. We can then filter $M_\bullet$ by $S_\bullet/P[a]$. We can then study $A_\bullet = S_\bullet/P$: if $B_\bullet = \Gamma_\bullet(\widetilde{A_\bullet})$ is the integral closure of $A_\bullet$, so is finitely geneated (this is a theorem from commutative algebra). □

1.3 Divisors

We will introduce Weil divisor and Cartier divisors. The Weil divisor is about studying algebraic cycles, so in some sense it mimics the behavior of (Borel-Moore) homology. The theory of Cartier divisor is in most case identical to the study of $\mathrm{Pic}(X)$, this mimics the behavior of cohomology theory. Just like Poincaré duality in topology, these two theory coincides in many reasonable cases.

Consider a curve $C \hookrightarrow \mathbb{P}^2$. Let $\mathcal{L} = i^*\mathcal{O}_{\mathbb{P}^2}(1)$. This induces:

$$\Gamma(\mathbb{P}^2, \mathcal{O}(1)) \to \Gamma(C, \mathcal{L})$$

Now view s a hyperplane section, then its image is a finite collection of points (with multiplicities). Vary s, we get different collection of points (Weil divisors), but they should correspond to the same line bundle $\mathcal{L}$.

Let X be integral, separated and $\dim X = d$. Assume X is regular in codimension one ($\mathrm{Sing}(X) \subset X$ has codimension ≥ 2).

Definition 1.27. A **prime divisor** in X is a closed, integral codimension 1 subscheme of X. The Weil divisor group is the freee abelian group generated by these Y.

For each $f \in K(X)^*$, note we can define $\mathrm{div}(f) \in \mathrm{Div}(X)$ using the valuation of local rings. This gives me:

$$K(X)^* \to \mathrm{Div}(X) \to \mathrm{Cl}(X) \to 0$$

This is the definition of **class group**.

Example. *Let $X = \mathbb{P}^n_k$, then $\mathrm{Cl}(X) \simeq \mathbb{Z}$. To see this, consider:*

$$\deg : \mathrm{Div}(\mathbb{P}^n) \to \mathbb{Z}$$

$$\sum a_P[P] \to \sum a_P d_P$$

where if $[P] = V(g_P)$, then d_P is the degree for the homogeneous irreducible polynomial d_P. Then evidentally $K(X)^* = \ker(\deg)$, it follows that $\mathrm{Cl}(\mathbb{P}^n) = \mathbb{Z}$.

Proposition 1.28. $\mathrm{Cl}(X) \simeq \mathrm{Cl}(X \times \mathbb{A}^1)$

Proof. Note there is a natural map:

$$\mathrm{Div}(X) \to \mathrm{Div}(X \times \mathbb{A}^1)$$

$$[Y] \mapsto [\pi^{-1}(Y)]$$

This descends to a map:

$$\pi^* : \mathrm{Cl}(X) \to \mathrm{Cl}(X \times \mathbb{A}^1)$$

This map is obviously injective. To show surjectivity, let's look at the diagram:

$$\begin{array}{ccc} \mathbb{A}^1_{K(X)} & \longrightarrow & X \times \mathbb{A}^1 \\ \downarrow & & \downarrow \\ \mathrm{Spec}\ K(X) & \hookrightarrow & X \end{array}$$

Suppose I have any prime divisor $Z \subset X \times \mathbb{A}^1$. We may assume $\pi : Z \to X$ is dominant, otherwise Z necessarily has form $\pi^{-1}(Y)$ for some prime divisor Y in X. So the generic point of $\mathbb{Z}$ is mapped to the generic point of X, so that we can consider:

$$Z_\eta \subset \mathbb{A}^1_{K(X)}$$

Is another Weil divisor. But now any divisor in $\mathbb{A}^1_{K(X)}$ is principal, so that we can find $f \in K(X)(t)^* = K(X \times \mathbb{A}^1)^*$ such that $\mathrm{div}(f) = Z_\eta$. Let's then consider:

$$D = Z - (f)$$

Then the support of D is completely disjoint from the generic fibre, meaning that D is necessarily obtained from pullback of $\mathrm{Cl}(X)$. □

Proposition 1.29. *Let $Z \subset X$ be closed, with $U = X \setminus Z$. Then there exists a surjection map*

$$\mathrm{Cl}(X) \to \mathrm{Cl}(U) \to 0$$

Moreover:

1. *If Z has codimension ≥ 2, then $\mathrm{Cl}(X) \simeq \mathrm{Cl}(U)$.*
2. *If Z is a prime divisor, then:*

$$\mathbb{Z}[Z] \to \mathrm{Cl}(X) \to \mathrm{Cl}(U) \to 0$$

Corollary 1.30. *Let $C = V(f) \subset \mathbb{P}^2$ for some f of degree d. Then we have a short exact sequence:*

$$\mathbb{Z} \xrightarrow{d} \mathbb{Z} \to \mathrm{Cl}(U) \to 0$$

So that $\mathrm{Cl}(U) = \mathbb{Z}/d\mathbb{Z}$.

Definition 1.31. Let X be a scheme. A **Cartier divisor** is an element in $\Gamma(K^\times/\mathcal{O}_X^\times)$. This is given by the data:

$$(U_i, f_i)$$

for U_i open cover for X, $f_i \in K^\times$ and on $U_i \cap U_j$, we have:

$$f_i|_{U_i\cap U_j} = g_{ij} f_j|_{U_i\cap U_j}.$$

for $g_{ij} \in \mathcal{O}(U_i \cap U_j)^\times$. The **Cartier class group** is defined to be:

$$\Gamma(X, K^\times/\mathcal{O}^\times)/K^\times$$

By looking at the sequence:

$$0 \to \mathcal{O}_X^\times \to K_X^\times \to K_X^\times/\mathcal{O}_X^\times$$

we can see that:

$$\mathrm{CaCl}(X) \simeq H^1(X, \mathcal{O}_X^\times) \simeq \mathrm{Pic}(X)$$

Theorem 1.32. *If X is locally factorial, then:*

$$\mathrm{CaDiv}(X) \simeq \mathrm{Div}(X)$$

which descends to an isomorphism on the level of class groups.

Example. *Let $X = \mathrm{Spec}\, k[x,y,z]/(xy - z^2)$. Look at $D = \{y = 0, z = 0\}$. Then at $p = (0,0,0)$, D_p is not principal in the local ring. Now D_p corresponds to the prime ideal $\mathfrak{q} = (y, z)$ inside A_0, where*

$$A = (k[x,y,z]/(xy - z^2))_{(x,y,z)}$$

Now that:

$$\mathfrak{q}^2 = (y^2, yz, z^2) = (y^2, yz, xy) = (y) \in A_\mathfrak{q}$$

This shows $2D_p$ in principal in the local ring. In fact $2D$ will come from some Cartier divisor.

A very explicit dictionary between Cartier class group and Picard group is the following:

$$\mathrm{CaDiv}(X) \to \mathrm{Pic}(X)$$

$$D \mapsto \mathcal{O}(D)$$

where if $D = (U_i, f_i)$, then $\mathcal{O}(D)$ is a subsheaf of $\mathcal{K}$, with:

$$\mathcal{O}(D)|_{U_i} = \mathcal{O}_{U_i} \cdot f_i^{-1} \subset \mathcal{K}|_{U_i}$$

Let X be regular. Then

$$\mathrm{Cl}(X) \simeq \mathrm{CaCl}(X) \simeq \mathrm{Pic}(X)$$

$$D \mapsto \mathcal{O}_X(D)$$

On homeworks we see $\mathcal{O}(D)$ has a distinguished rational section:

$$s_D \in \Gamma(U, \mathcal{O}(D))$$

such that $\mathrm{div}(s_D) = D$.

Definition 1.33. We call D effective if $s_D \in \Gamma(X, \mathcal{O}(D))$ has no poles.

Let's now see some applications to curves. Let $X/k = \overline{k}$ be smooth, projective, connected. Consider:

$$\pi : X \to Y$$

be finite (this is true for any non constant morphism of curves). Recall that $\deg(\pi) = \deg K(X)/K(Y)$. We wish to define:

$$\pi^* : \mathrm{Div}(Y) \to \mathrm{Div}(X)$$

$$\pi^*[y] = \sum_{x \in \pi^{-1}(y)} \mathrm{val}_X(\pi^*(t))[x] \in \mathrm{Div}(X)$$

where $\mathfrak{m}_y = (t) \subset \mathcal{O}_{Y,y}$.

Definition 1.34. Let $\deg : \mathrm{Div}(X) \to \mathbb{Z}$ be:

$$\sum a_x[x] \mapsto \sum a_x$$

Theorem 1.35. $\deg(\pi^* D) = \deg(D) \cdot \deg(\pi)$.

Corollary 1.36. $\deg(\pi) = \deg(\pi^*[y])$ *for any closed point* $y \in Y$.

Proof. (of the theorem). We need to check it for $D = [y]$. Restrict to some open affine $V = \mathrm{Spec}\ R \subset Y$, so that:

$$\pi : U = \pi^{-1}(V) = \mathrm{Spec}\ S \to \mathrm{Spec}\ R$$

So that $R \subset S$. Let's look at the fibre carefully:

$$R_{\mathfrak{m}_y} \subset S \otimes_R R_{\mathfrak{m}_y}$$

This is a finite extension of DVR. But then $S \otimes_R R_{\mathfrak{m}_y}$ is torsion free, so that $S \otimes_R R_{\mathfrak{m}_y}$ is a free module over $R_{\mathfrak{m}_y}$. So we can write:

$$S \otimes_R R_{\mathfrak{m}_y} \simeq R_{\mathfrak{m}_y}^{\oplus \deg \pi}$$

The rank can be seen by tensoring by the fraction field. Now tensoring with $R_{\mathfrak{m}}/\mathfrak{m}$, I get:

$$S \otimes_R k(y) \simeq k(y)^{\oplus \deg \pi}$$

But using the structure of Artinian rings (Artinian rings is always direct sum of its (maximal) local rings), I have:

$$S \otimes_R k(y) = S/tS = \bigoplus_{x \mapsto y} S_{\mathfrak{m}_x}/t\mathfrak{m}_x = \bigoplus S_{\mathfrak{m}_x}/u^{\mathrm{val}} S_{\mathfrak{m}_x}$$

for $(t) = \mathfrak{m}_y$, $(u) = \mathfrak{m}_x$, and $(\pi^* t) = (u)^{\mathrm{val}}$. □

Corollary 1.37. *Degree of principal divisors on a curve is zero. Therefore* deg *actually descends to:*

$$\deg : \mathrm{Cl}(X) \to \mathbb{Z}$$

Proof. Given $f \in K(X)$ a nonconstant rational function. Since X is projective, it will define a map:

$$f : X \to \mathbb{P}^1$$

It follows that the principal divisor corresponding to f is nothing but:

$$\mathrm{div}(f) = f^*([0] - [\infty]) = 0$$

by the above theorem. □

Thus we get a exact sequence for any curve:

$$0 \to \mathrm{Pic}_0(X) \to \mathrm{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \to 0$$

Proposition 1.38. *If $X \neq \mathbb{P}^1$, then $\mathrm{Pic}_0(X) \neq 0$.*

Proof. If $[P] \simeq [Q]$, so that there exists a function $f \in K(X)$ such that $\mathrm{div}(f) = [P] - [Q]$. It defines a birational morphism to $\mathbb{P}^1$. But any curve birational to $\mathbb{P}^1$ actually is isomorphic to $\mathbb{P}^1$, we're done. □

Example. *Consider the elliptic curve $y^2z = x^3 - xz^2$. We claim there is a bijection:*

$$\alpha : X(k) \to \mathrm{Pic}_0(X)$$

In fact $\mathrm{Pic}_0(X)$ is always rational points on an abelian variety. The way to define the map α is to pick the $P_0 = [0 : 1 : 0] \in X(k)$. We just define:

$$\alpha(P) = [P] - [P_0]$$

for each rational point $P \in X(k)$. The injectivity is clear, since if $[P] \simeq [P_0]$, then this forces $X \simeq \mathbb{P}^1$, which is a contradiction.

1.4 Morphism to Projective Space

Theorem 1.39. *Given X/k. There exists a bijection between $f : X \to \mathbb{P}^n$ and*

$$\{L \in \mathrm{Pic}(X) \mid s_0, \cdots, s_n \in \Gamma(X, L) \text{ globally generated}\}$$

Proof. Given a morphism $f : X \to \mathbb{P}^n$. We can just set:

$$L = f^*\mathcal{O}(1)$$

$$s_i \coloneqq f^*(x_i) \in \Gamma(X, L)$$

So that:

$$D(s_i) = f^{-1}(D(x_1)) = f^{-1}(U_i)$$

and they will cover X so globally generated. Conversely, assume I'm equipped with a line bundle with generating global sections. I just take $V_i = D(s_i) \subset X$. We know $\cup V_i = X$ by assumptions. We use s_i to define:

$$f_i : V_i \to U_i = \text{Spec } k[\frac{x_0}{x_i}, \cdots \frac{x_n}{x_i}] \subset \mathbb{P}^n$$

and see they will glue. This is equivalent to give:

$$k[\frac{x_0}{x_i}, \cdots \frac{x_n}{x_i}] \to \mathcal{O}(V_i)$$

$$\frac{x_j}{x_i} \mapsto \frac{s_j}{s_i}$$

□

Definition 1.40. Given $W \subset \Gamma(X, L)$. We call W is globally generated if $X = \bigcup_{w \in W} D(w)$. By pick a basis of W, we get a map:

$$X \to \mathbb{P}^{\dim W - 1}$$

Now the question is that when is

$$f : X \to \mathbb{P}^n$$

a closed embedding. Here is a criterion when X is projective and $k = \overline{k}$.

Theorem 1.41. *Given L, $W = \text{Span}(s_0, \cdots, s_n) \in \Gamma(X, L)$ globally generates L. Then $f : X \to \mathbb{P}^n$ is a closed embedding iff:*

1. *For all $x \neq y \in X(k)$, there exists a section $s \in W$ such that $s(x) \neq 0, s(y) = 0$.*
2. *For $x \in X(k)$, the sections in $W_x = \{s \in W \mid s(x) = 0\}$ spans*

$$\mathfrak{m}_x L_x / \mathfrak{m}_x^2 L_x = L \otimes T^\vee_{X,x}$$

Remark. *The first condition reads that sections separate points. More geometrically, a section corresponds to a hyperplane in $\mathbb{P}^n$, and then:*

$$f(x) \neq f(y) \iff \exists H, f(x) \notin H, f(y) \in H \iff s(x) \neq 0, s(y) = 0.$$

This is checking that f is injective on closed points. The second condition geometrically reads:

$$T^\vee_{\mathbb{P}^n, f(x)} \to T^\vee_{X,x} \to 0$$

so equivalently this requires injectivitiy on the level of tangent spaces. The idea is that after trivialization, we can identify:

$$W_x \simeq T^\vee_{\mathbb{P}^n, f(x)}, L \otimes T^\vee_{X,x} \simeq T^\vee_{X,x}$$

The second condition is called separating tangent vectors.

Theorem 1.42. *Let X/k be a projective variety. Then L is ample iff for any $E \in \mathrm{Coh}(X)$ such that $E \otimes L^{\otimes a}$ is globally generated for $a \gg 0$.*

Proof. The direction $\implies$ is given by the fact was proved on $\mathbb{P}^n$. For the other direction:

Step 1: We replace L by some power of it, we can find $s_i \in \Gamma(X, L)$ such that $D(s_i)$ is affine and $X = \bigcup D(s_i)$. To see this, let's take $p \in U \subset X$ for some affine U. Let $Y = X \setminus U$, given by ideal sheaf $\mathcal{I}_Y$. By assumption, we see $\mathcal{I}_Y \otimes L^{\otimes n}$ is globally generated by large enough n. Therefore:

$$i : \mathcal{I}_Y \otimes L^{\otimes n} \hookrightarrow L^{\otimes n}$$

There exists $s \in \Gamma(\mathcal{I}_Y \otimes L^{\otimes n})$ which generates the stalk at p, since $\mathcal{I}_{Y,p} \simeq \mathcal{O}_{Y,p}$. Now set $\widetilde{s} = i(s) \in \Gamma(L^{\otimes n})$. Note that $D(\widetilde{s}) \subset U$ is affine and $p \in D(\widetilde{s})$, this is because $\widetilde{s}$ vanishes on Y by construction. Assume U chosen such that $L|_U$ is trivial, then $\widetilde{s}$ becomes a function, and then $D(\widetilde{s})$ is just a basic affine open in U.

Step 2: Step 1 implies L is ample. Now $D(s_i) = \mathrm{Spec}\ R_i$ is affine, finitely generated by $a_i^{(j)} \in R_i$. We can thus extend $s_i^{\otimes m} a_i^{(j)} \in \sigma_{ij}$ extends to $\Gamma(X, L^{\otimes m})$. The claim is that if we use:

$$s_i^m, \sigma_{ij}$$

to create:

$$X \to \mathbb{P}^N$$

Note that this is well-defined, since the $\{s_i^m\}$ already globally generates. In fact the image lies inside $\bigcup D(x_i)$. Now let's check things locally:

$$D(s_i) \to D(x_i)$$

$$a_i^{(j)} = \frac{\sigma_{ij}}{s_i^m} \leftarrow \frac{x_{ij}}{x_i}$$

from this we see this a closed embedding. □

1.5 Kähler Differentials

Gicen a morphism $X \to Y$, we wish to construct $\Omega_{X/Y} \in \mathrm{Coh}(X)$ the sheaf of one-forms on X. One of the motivation for introducing this notion to produce invariants of the variety.

Given a map of rings $i : A \to B$, we can define the module of differentials (M, d_M) is defined to be a B-module equipped with a map of an A-module morphism:

$$d_M : B \to M$$

satisifying the Leibniz rule.

Definition 1.43. The B-module $\Omega_{B/A}$ is the universal module of differentials. More concretely, I have:

$$d : B \to \Omega_{B/A}$$

such that for any other module of differential (M, d_M), d_M will factor through d.

We need to show its existence. Indeed, we should take:

$$\Omega_{B/A} = \bigoplus_{f \in B} B \cdot df / \langle d(f+f') = df + df', d(i(a)) = 0, d(fg) = fdg + gdf \rangle$$

From this presentation see that if B is generated by $f_1, \cdots, f_n/A$, then $\Omega_{B/A}$ is generated as B-modules by $df_1, \cdots, df_n$.

Example. *Let $A = k, B = k[x_1, \cdots, x_n]$. Then*

$$\Omega_{B/A} = \bigoplus_{i=1}^{n} B dx_i$$

is free of rank n.

Let's list some basic properties.

1. Localization. Let $T \subset B$ is multiplicative subset. Then:

$$\Omega_{T^{-1}B/A} = T^{-1}\Omega_{B/A}$$

More generally, if $S \subset A$ with $i(S) \subset T$, then:

$$\Omega_{T^{-1}B/S^{-1}A} = T^{-1}\Omega_{B/A}$$

2. Given $A \xrightarrow{i} B \xrightarrow{j} C$, then we have short exact sequence:

$$C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

3. (On homework) if $C = B/I$, then $\Omega_{C/B} = 0$. Then we have a short exact sequence:

$$I/I^2 \xrightarrow{\delta} C \otimes \Omega_{B/A} \to \Omega_{C/A} \to 0$$

The first map is given by:

$$f \mapsto 1 \otimes df$$

Example. *Let $B = A[x,y], C = A[x,y]/(f(x,y))$. Then use property 3 I have:*

$$C \cdot df \to Cdx \oplus Cdy \to \Omega_{C/A} \to 0$$

For example, let's take $A = k, f(x,y) = xy$. Then:

$$\Omega_{C/k} = C \oplus C/(ydx + xdy)$$

The fibre at $(x,y) \neq (0,0)$, I get one-dimensional; the fibre at $(0,0)$ is two dimensional.

In general, if $B = A[x_1, \cdots, x_n]/(f_1(x), \cdots, f_r(x))$, then:

$$\Omega_{B/A} = \text{coker } (J)$$

for $J : B^{\oplus r} \to B^{\oplus n}$ the Jacobian matrix.

Proposition 1.44. *Let L/K be finite separable extension of fields, then $\Omega_{L/K} = 0$.*

Example. *Let $k = \overline{\mathbb{F}_p}(t), L = k[x]/(x^p - t)$. Then $\Omega_{L/k} = L$.*

Recall Spec $B \to$ Spec A is smooth if the Jacobian matrix has full rank for at all $x \in$ Spec B.

Let $B_{\mathfrak{p}}$ be a local ring.

Lemma 1.45. *A square matrix T in $B_{\mathfrak{p}}$ is invertible $\iff$ $T \bmod \mathfrak{p}$ is invertible $\iff$ $\det(T) \neq 0 \in k(\mathfrak{p})$.*

Proof. Note that $\det T \in B_{\mathfrak{p}}^{\times} \iff T$ is invertible. □

Assume first $r \times r$ of Jacobian $J \otimes k(x)$ is invertible, then the first $r \times r$ minor is invertible for $J \otimes_B B_{\mathfrak{p}}$. Then:

$$B_{\mathfrak{p}}^{\oplus r} \to B_{\mathfrak{p}}^{\oplus n} \to \Omega_{B/A} \otimes_B B_{\mathfrak{p}} \to 0$$

It follows that $\Omega_{B/A} \otimes B_{\mathfrak{p}} = B_{\mathfrak{p}}^{\otimes n-r}$.

Corollary 1.46. *If* Spec $B \to$ Spec A *is smooth, then $\Omega_{B/A}$ is locally free of rank $n - r$.*

On homework, we will see if $k \hookrightarrow A$ and $\mathfrak{m} \subset A$ is a maximal ideal such that $A/\mathfrak{m} = k$, then:

$$\Omega_{A/k} \otimes_A k \simeq \mathfrak{m}/\mathfrak{m}^2 = T^{\vee}_{\text{Spec } A,\mathfrak{m}}$$

Hence $\Omega_{A/k}$ is a coherent sheaf on Spec A whose fibres at closed points recover cotangent space.

Now we can globalize the picture and define:

$$\Omega_{X/Y} \in \text{Coh}(X)$$

For $f : X \to Y$ a finite type morphism. This is well-defined because $\Omega_{B/A}$ localizes well.

Example. *Consider $\mathbb{P}^1/k$. We wish to describe $\Omega_{\mathbb{P}^1/k}$. We cover by two charts U_0, U_1. So:*

$$\Omega|_{U_0} = \mathcal{O}_{U_0} dt$$

$$\Omega|_{U_1} = \mathcal{O}_{U_1} d(\frac{1}{t})$$

On $U_0 \cap U_1$, we have:

$$d(\frac{1}{t}) = -\frac{1}{t^2} dt$$

Let's pick the rational section $\frac{1}{t}$ of Ω. It has no zeroes or poles on U_1, but it has a pole of order 2 on U_0. Therefore, we conclude:

$$\Omega_{\mathbb{P}^1/k} \simeq \mathcal{O}(-2)$$

Theorem 1.47. *We have a short exact sequence:*

$$0 \to \Omega_{\mathbb{P}^n/k} \to \mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus n+1} \xrightarrow{\widetilde{f}} \mathcal{O}_{\mathbb{P}^n} \to 0$$

this is the Euler sequence, where:

$$\widetilde{f} = (x_0, \cdots, x_n)$$

To visualize the sequence, we dualize the sequence:

$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus n+1} \to T_{\mathbb{P}^n} \to 0$$

We have a $\mathbb{C}^*$-fibration:

$$\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{CP}^n$$

We thus have the short exact sequence of vector bundles on $\mathbb{C}^{n+1} \setminus \{0\}$:

$$0 \to \mathbb{C} \cdot x_i \partial_i \to T_{\mathbb{C}^{n+1}\setminus\{0\}} \to T_{\mathbb{P}^n} \to 0$$

Proof. (of Euler sequence, sketched). Let $S_\bullet = k[x_0, \cdots, x_n]$. Let's look at:

$$\bigoplus_{i=0}^{n} S_\bullet e_i \xrightarrow{f} S_\bullet$$

$$e_i \mapsto x$$

where e_i has degree 1. When we turn into sheaves this becomes the surjective map:

$$\mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus n+1} \to \mathcal{O}_{\mathbb{P}^n}$$

Now Let $M_\bullet = \ker f$, we need to show that:

$$\widetilde{M_\bullet} = \Omega_{\mathbb{P}^n/k}$$

Let's identify these explicitly. Recall that:

$$\widetilde{M_\bullet}(U_i) = (M_\bullet[\frac{1}{x_i}])_{\deg=0}$$

But

$$M_\bullet[\frac{1}{x_i}] = \bigoplus S_\bullet(\frac{e_j}{x_i} - \frac{x_j}{x_i^2}e_i)$$

On the other hand:

$$\Omega_{\mathbb{P}^n/k}(U_i) = \bigoplus \mathcal{O}(U_i) d(\frac{x_j}{x_i})$$

This motivates us to send:

$$\varphi_i : d(\frac{x_j}{x_i}) \to \frac{e_j}{x_i} - \frac{x_j}{x_i^2}e_i$$

These φ_i glue together to get an isomorphism:

$$\widetilde{M_\bullet} \simeq \Omega_{\mathbb{P}^n/k}$$

□

Proposition 1.48. *Let $X \xrightarrow{\pi} Y \to Z$. We have short exact sequence:*

$$\pi^*\Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0$$

In the case $i : X \to Y$ is a closed embedding, and let $\mathcal{I} = \mathcal{I}_{X/Y}$, we have:

$$\mathcal{I}/\mathcal{I}^2 \to i^*\Omega_{Y/Z} \to \Omega_{X/Z} \to 0$$

Proof. If $X \to Y$ is smooth of relative dimension d, then $\Omega_{X/Y}$ is locally free of rank d. $\square$

Now let X/k be a smooth projective vareity of dimension d, then $\Omega_{X/k}$ is a locally free sheaf of rank d. Now for each $a \leq d$, we can form:

$$\Omega^a_{X/k} \coloneqq \bigwedge^a \Omega_{X/k}$$

locally free of rank $\binom{d}{a}$, this is the sheaf of a-forms. For $a = d$, we have:

$$K_{X/k} \coloneqq \Omega^d \in \mathrm{Pic}(X)$$

the canonical line bundle. It turns out that:

$$\Gamma(X, K_X^{\otimes b})$$

will be one of the fundamental invariants of classifying X.

Example. *$K_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$. By taking determinant of the Euler sequence we see*

$$K_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$$

Now let's $X \hookrightarrow \mathbb{P}^n$ be a smooth projective variety. We wish to calculate $K_X \in \mathrm{Pic}(X)$.

Theorem 1.49. *Let $i : X \hookrightarrow Y$ be closed embedding of smooth varieties. Suppose X is cut out by ideal sheaf $\mathcal{I}$. Recall we have a short exact sequence:*

$$\mathcal{I}/\mathcal{I}^2 \to i^*\Omega_Y \to \Omega_X \to 0$$

We have:

1. *$\mathcal{I}/\mathcal{I}^2 \in \mathrm{Coh}(X)$ is locally free of rank $\dim Y - \dim X$: this is the conormal bundle.*
2. *In fact we have:*
$$0 \to \mathcal{I}/\mathcal{I}^2 \to i^*\Omega_Y \to \Omega_X \to 0$$
If we dualize this sequence, we get the standard:
$$0 \to T_X \to T_Y \to N_{X/Y} \to 0$$

Let's denote $r = \dim Y - \dim X$.

Lemma 1.50. *The ideal sheaf $\mathcal{I}_p \subset \mathcal{O}_{Y,p}$ is generated by r elements.*

Lemma 1.51. *If $\mathcal{O}_{Y,p}$ is a local ring. Suppose I have a surjection:*

$$\mathcal{O}_{Y,p}^{\oplus a} \to \mathcal{O}_{X,p}^{\oplus b} \to 0$$

Then the kernel is also free, and the short exact sequence splits.

Proof. (of the theorem). Look at the stalk:

$$0 \to \ker \to (i^*\Omega_Y)_p \to \Omega_{X,p} \to 0$$

So by previous of the lemma we know the kernel is free. In addition I have:

$$(i^*\mathcal{I})_p \to \ker \to 0$$

However $(i^*\mathcal{I})_p$ is generated by r elements. We thus get:

$$\mathcal{O}_{X,p}^{\oplus r} \to (i^*\mathcal{I})_{X,p} \to \mathcal{O}_{X,p}^{\oplus r}$$

a surjection. By the second lemma, these maps must be an isomorphism. Then everything follows. □

Example. *Let $Y = \mathbb{P}^2$, and C be a smooth degree d curve. Note that:*

$$\mathcal{I}_C = \mathcal{O}(-C) = \mathcal{O}(-d) \subset \mathcal{O}_{\mathbb{P}^2}$$

Therefore the above conormal sequence reads:

$$0 \to i^*\mathcal{I}_C \to i^*\Omega_{\mathbb{P}^2} \to \Omega_C \to 0$$

$$\iff 0 \to i^*\mathcal{O}(-d) \to i^*\Omega_{\mathbb{P}^2} \to \Omega_C \to 0$$

Hence:

$$i^*K_{\mathbb{P}^2} = i^*\mathcal{O}(-d) \otimes K_C$$

Recall that we computeted:

$$\Omega^2(\mathbb{P}^2) = \mathcal{O}(-3)$$

Hence $K_C = i^*\mathcal{O}(d-3)$. This implies that $\deg(K_C) = d(d-3)$

Definition 1.52. For a smooth projective curve, the genus for the curve is defined by:

$$2g(C) - 2 = \deg K_C$$

Therefore for a curve of degree d,

$$g(C) = \frac{d(d-3)}{2} + 1 = \binom{d-1}{2}$$

More generally suppose D is a smooth divisor in a smooth variety Y. We still have the cotangent sequence:

$$0 \to i^*\mathcal{O}(-D) \to i^*\Omega_{Y/k} \to \Omega_{D/k} \to 0$$

Take determinant, I get:

$$K_D \otimes i^*\mathcal{O}(-D) = i^*K_Y$$

$$K_D = i^*(K_Y \otimes \mathcal{O}_Y(D))$$

This is the **adjunction formula** for canonical divisor.

1.6 Riemann-Hurwitz

Let X, Y be smooth projective curves, and $f : X \to Y$ finite map. The question is that how are g_X and g_Y related? Assume f is separable (the extension of fields is separable).

Example. *An example of nonseparable extension is:*

$$\overline{\mathbb{F}_p}(t^{\frac{1}{p}})/\overline{\mathbb{F}_p}(t)$$

Recall we have:

$$f^*\Omega_Y \to \Omega_X \to \Omega_{X/Y} \to 0$$

Theorem 1.53. *This is actually a short exact sequence in our setting, and the sheaf $\Omega_{X/Y}$ is a torsion sheaf (i.e. supported on finitely many points).*

Example. *Let $f : \mathbb{A}^1 \to \mathbb{A}^1$ be:*

$$k[t] \to k[u], \quad t \mapsto u^a$$

Now: $dt = d(u^a) = a \cdot u^{a-1}du$, which vanishes at $u = 0$. We have:

$$0 \to k[u]dt \to k[u]du \to k[u]/u^{a-1} \to 0$$

But if $char(k) = p$, then $dt = 0$ and so the above theorem fails.

Proof. (of the theorem) Recall $K(Y) \subset K(X) \implies \Omega_{K(X)/K(Y)} = 0$. Therefore:

$$\Omega_{X/Y}|_\eta = 0$$

This forces $\Omega_{X/Y}$ to be torsion. This will imply the fist statement. In fact, restrict to an affine open chart Spec A, we know both $f^*\Omega_Y$ and Ω_X are locally free of rank 1, and a A-module morphism $\varphi : A \to A$ with torsion cokernel must be injective. □

Definition 1.54. The ramification divisor is:

$$R = \sum \dim(\Omega_{X/Y})_x[x]$$

This is an effective Weil divisor.

We write:

$$0 \to f^*K_Y \to K_X \to \Omega_{X/Y} \to 0$$

Tensoring with K_X^{-1}:

$$0 \to f^*K_Y \otimes K_X^{-1} \to \mathcal{O}_X \to \mathcal{O}_R \to 0$$

Therefore we know:

$$\mathcal{O}(-R) = f^*K_Y \otimes K_X^{-1}$$

Hence:

$$K_X = f^*K_Y \otimes \mathcal{O}(R)$$

Take degree of this:

$$2g_X - 2 = \deg(f)(2g_Y - 2) + \deg R$$

This is the **Riemann Hurwitz formula**. The next question is how to compute R? Consider $y = f(x)$, then we have:

$$f^* : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$$

let t be a local coordinate at y, then the **ramification index** is:

$$e_x \coloneqq \mathrm{val}_{\mathcal{O}_{X,x}}(f^*t) \geq 1$$

So that:

$$f^*t = s^{e_x}u$$

for s a local coordinate near x, and u a unit in the local ring. Now we localize

$$0 \to f^*\Omega_Y \to \Omega_X \to \Omega_{X/Y} \to 0$$

at x:

$$0 \to \mathcal{O}_{X,x}dt \to \mathcal{O}_{X,x}ds \to (\Omega_{X/Y})_x \to 0$$

where the second map is:

$$dt \mapsto e_x s^{e_x - 1} u ds + s^{e_x} du$$

In the case of **tame ramificaiton** (i.e. $\mathrm{char}(k) \nmid e_x$), we have:

$$\dim(\Omega_{X/Y})_x = e_x - 1$$

In the case of **wild ramification**, we have:

$$\dim(\Omega_{X/Y})_x \geq e_x$$

Fix y, recall that:

$$\sum_{x \mapsto y} e_x = \deg f$$

Therefore if $\mathrm{char}(k) > \deg f$, the ramifications are all tame.

If all ramifications are tame, we know:

$$R = \sum_{x \in X} (e_x - 1)[x]$$

So that:

$$\deg R = \sum (e_x - 1) = \sum_{y \in Y} \deg(f) - |f^{-1}(y)|$$

Let's now work over $\mathbb{C}$.

Example. *Consider $\mathbb{P}^1 \to \mathbb{P}^1, z \mapsto z^n$. Then $\deg f = n$, it's clear that:*

$$\deg R = 2n - 2$$

In this case the Riemann-Hurwitz reads:

$$-2 = n(-2) + 2(n-1)$$

Example. *There is no connected finite étale cover of* $\mathbb{P}^1$.

Example. *If the base field is zero characteristic, then there is no connected finite étale cover of* $\mathbb{A}^1$. *Given* $f : C \to \mathbb{A}^1$, *this induces a map:*

$$\overline{f} : \overline{C} \to \mathbb{P}^1$$

with the only ramification happens at infinity. Assume we're in the tame world, then:

$$-2 = \deg f \cdot (2g_{\overline{C}} - 2) + \deg f - |f^{-1}(\infty)| = \deg f \cdot (2g_{\overline{C}} - 1) - |f^{-1}(\infty)|$$

It follows that $g_{\overline{C}} = 0, \deg f = 1$. *But in characteristic* p, *wild ramifications happen all the time:*

$$\text{Spec } k[x,t]/(x^p - x - t) \to \text{Spec } k[t]$$

Therefore:

$$\pi_1^{\text{ét}}(\mathbb{A}^1_k)$$

is huge in the case of wild ramification.

Let's now do an interesting application. Let $k = \mathbb{C}$, and $C \subset \mathbb{P}^2_{\mathbb{C}}$ a smooth plane curve of degree d. Recall we computed:

$$2g_C - 2 = d(d-3)$$

Let's assume $p = [0,0,1] \notin C$. I now ask: how many lines through p are tangent to C. Assume that there are no bitangents through p, and every tangent through p intersects C with multiplicity at most 2.

Consider the projection:

$$\pi : \mathbb{P}^2 \setminus p \to \mathbb{P}^1$$

$$[x : y : z] \mapsto [x : y]$$

This gives a map:

$$f : C \to \mathbb{P}^1$$

By Bezout's theorem, we know f has degree d. The map f ramifies at point q iff pq is the tangent line at q. By assumption, we know for each $r \in \mathbb{P}^1$, $f^{-1}(r)$ only ramifies at most one point, and ramification index is at most 2. In summary, $\deg R$ is exactly the number of tangent lines. Therefore:

$$2g_C - 2 = d(-2) + \deg R \implies \deg R = d^2 - d$$

There are exactly $d^2 - d$ tangent line through C.

2 Cohomology

Suppose I have $X/k = \overline{k}$, $\mathcal{L} \in \mathrm{Pic}(X)$. I want to ask is $\mathcal{L}$ globally generated. In other words, for each $x \in X$, I need to find $s \in \Gamma(X, \mathcal{L})$ such that $s(x) \neq 0$. I have:

$$0 \to \mathcal{I}_{X,x} \to \mathcal{O}_X \to i_* k_x \to 0$$

Tensor this sequence with $\mathcal{L}$:

$$0 \to \mathcal{L} \otimes \mathcal{I}_{X,x} \to \mathcal{L} \to i_* k_x \to 0$$

Take global sections:

$$0 \to \Gamma(\mathcal{L} \otimes \mathcal{I}_{X,x}) \to \Gamma(\mathcal{L}) \to \mathcal{L} \otimes k_x$$

We're exactly asking whether $\Gamma(\mathcal{L}) \to \mathcal{L} \otimes k_x$ is surjective. This is an instance for us to study sheaf cohomology: to measure the failure of the functor $\Gamma(X, -)$ to preserve exactness.

2.1 Definition and Properties

Let X/A be a scheme. For each $k \geq 0$, we wish to define:

$$H^k(X, -) : \mathsf{QCoh}(X) \to A\text{-mod}$$

with the properties:

1. $H^0(X, -) = \Gamma(X, -)$.
2. Given $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$, we have LES:

$$\cdots \to H^i(X, \mathcal{F}) \to H^i(X, \mathcal{G}) \to H^i(X, \mathcal{H}) \to H^{i+1}(X, \mathcal{F}) \to \cdots$$

3. Given $X \xrightarrow{\pi} Y$, and $\mathcal{F} \in \mathsf{QCoh}(X)$, there exists:

$$H^k(Y, \pi_* \mathcal{F}) \to H^k(X, \mathcal{F})$$

   extending the identity map:

$$H^0(Y, \pi_* \mathcal{F}) \to H^0(X, \mathcal{F})$$

   Given $\mathcal{G} \in \mathsf{QCoh}(Y)$, I have the adjunction morphism:

$$\mathcal{G} \to \pi_* \pi^* \mathcal{G}$$

   Combining with property 3:

$$H^k(Y, \mathcal{G}) \to H^k(Y, \pi_* \pi^* \mathcal{G}) \to H^k(X, \pi^* \mathcal{G})$$

   This is pullback of sheaf.

4. If π is affine, then:

$$H^k(Y, \pi_* \mathcal{F}) \simeq H^k(X, \mathcal{F})$$

5. Suppose X is covered by n affine open sets, then $H^k(X, -) = 0$ for $k \geq n$.

6. Commutes with arbitrary direct sum.

Key computation is the following:

Theorem 2.1. *We have:*

$$H^0(\mathbb{P}^n_A, \mathcal{O}(m)) = \begin{cases} A[x_0, \cdots, x_n]_m, & \text{if } m \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

$$H^n(\mathbb{P}^n_A, \mathcal{O}(m)) = H^0(\mathbb{P}^n, \mathcal{O}(-m-n-1))^\vee$$

and vanishes otherwise.

Theorem 2.2. *Let X be a projective scheme over* Spec A*, and $\mathcal{F} \in \mathrm{Coh}(X)$. Then $H^*(X, \mathcal{F})$ are finitely generated.*

Proof. This reduce to the case $X = \mathbb{P}^n$. We will prove $H^k(X, \mathcal{F})$ is finitely by descending induction on k. For $k \geq n+1$ this is clear. Given $\mathcal{F} \in \mathrm{Coh}(\mathbb{P}^n)$, I write:

$$0 \to \mathcal{G} \to \mathcal{O}(m)^{\oplus a} \to \mathcal{F} \to 0$$

LES reads:

$$\cdots \to H^{n-1}(\mathcal{G}) \to H^{n-1}(\mathcal{O}(m)^{\oplus a}) \to H^{n-1}(\mathcal{F}) \to H^n(\mathcal{G}) \to H^n(\mathcal{O}(m)^{\oplus a}) \to H^n(\mathcal{F}) \to 0$$

This proves the case for $k = n$. More careful analysis of the LES tells me this holds for each k. □

Let X be a separated scheme over A. Pick an affine open cover of X: $\mathcal{U} = \{U_1, \cdots, U_n\}$. For $I = \{i_1, \cdots, i_k\} \subset \{1, 2, \cdots, n\}$, I can define:

$$U_I = U_{i_1} \cap \cdots \cap U_{i_k}$$

I can define the **Čech complex**, for each $k \geq 0$:

$$C^k_{\mathcal{U}}(\mathcal{F}) = \prod_{I \subset [n], |I| = k+1} \Gamma(U_I, \mathcal{F})$$

So we have:

$$0 \to C^0_{\mathcal{U}}(\mathcal{F}) \xrightarrow{d_0} C^1_{\mathcal{U}}(\mathcal{F}) \xrightarrow{d_1} C^2_{\mathcal{U}}(\mathcal{F}) \to \cdots \to C^{n-1}_{\mathcal{U}}(\mathcal{F}) \to 0$$

Let me specify:

$$d_k : \Gamma(U_I, \mathcal{F}) \to \Gamma(U_J, \mathcal{F})$$

for $|I| = k+1, |J| = k+2$, it reads:

$$d_k(s) = \begin{cases} 0, & \text{if } I \nsubseteq J; \\ (-1)^{a-1} s, & \text{if } J = I \cup \{j\}, j \text{ the a-th element of } J. \end{cases}$$

One can check that $d_{k+1} \circ d_k = 0$, so we indeed get a chain complex.

A few basic things to observ: $H^k_{\mathcal{U}}$ is covariant functor; $H^0_{\mathcal{U}} = \ker(d_0) = \Gamma(X, \mathcal{F})$; if X can be covered n-affine charts, then $H^k_{\mathcal{U}}(X, -) = 0$ for $k \geq n$.

Let's see how we get the long exact sequence. Suppose I have:

$$0 \to \mathcal{E} \to \mathcal{F} \to \mathcal{G} \to 0$$

I will obtain:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C^{k-1}(\mathcal{E}) & \longrightarrow & C^{k-1}(\mathcal{F}) & \longrightarrow & C^{k-1}(\mathcal{G}) & \longrightarrow & 0 \\
 & & \downarrow d & & \downarrow d & & \downarrow d & & \\
0 & \longrightarrow & C^{k}(\mathcal{E}) & \longrightarrow & C^{k}(\mathcal{F}) & \longrightarrow & C^{k}(\mathcal{G}) & \longrightarrow & 0 \\
 & & \downarrow d & & \downarrow d & & \downarrow d & & \\
0 & \longrightarrow & C^{k+1}(\mathcal{E}) & \longrightarrow & C^{k+1}(\mathcal{F}) & \longrightarrow & C^{k+1}(\mathcal{G}) & \longrightarrow & 0
\end{array}$$

Then the result follows from Snake lemma.

Lemma 2.3. *If $\pi : X \to Y$ is an affine morphism, $\mathcal{F} \in \mathsf{QCoh}(X)$, then:*

$$H^k(Y, \pi_* \mathcal{F}) = H^k(X, \mathcal{F})$$

Theorem 2.4. *The groups $H^k_{\mathcal{U}}(\mathcal{F})$ is independent of the cover $\mathcal{U}$. More precisely, if $\{U_i\} \subset \{V_j\}$, I have a natural projection:*

$$\pi : C^k_{\mathcal{V}} \to C^k_{\mathcal{U}}$$

and it is a quasi-isomorphism of complexes.

Proof. Let $\mathcal{U} = \{U_1, \cdots, U_n\}, \mathcal{U}' = \{U_0, \cdots, U_n\}$. To prove the theorem, let's break into several steps:

1. Step 1: Reduce to the case when X is affine. Look at:

   $$0 \to \ker^\bullet \to C^\bullet_{\mathcal{U}'} \to C^\bullet_{\mathcal{U}} \to 0$$

   This looks like:

   $$0 \to \prod_{I \subset \{0,\cdots,n\}, 0 \in I} \Gamma(U_I, \mathcal{F}) \to \prod_{I \subset \{0,\cdots,n\}} \Gamma(U_I, \mathcal{F}) \to \prod_{I \subset \{0,\cdots,n\}} \Gamma(U_I, \mathcal{F}) \to 0$$

   Hence:

   $$\ker^k = \prod_{I' \subset \{1,\cdots,n\}, |I'| = k} \Gamma(U_{I'} \cap U_0, \mathcal{F})$$

   Now if let's consider $\mathcal{V} = \{U_i \cap U_0\}$ cover of U_0. Then $\ker^\bullet$ is the augemented Cech complex:

   $$0 \to \Gamma(U_0, \mathcal{F}) \to C^0_{\mathcal{V}}(\mathcal{F}) \to C^1_{\mathcal{V}}(\mathcal{F}) \to \cdots$$

Now the long exact sequence for:

$$0 \to \ker^\bullet \to C^\bullet_{\mathcal{U}'} \to C^\bullet_{\mathcal{U}} \to 0$$

$$\cdots \to H^k(\ker^\bullet) \to H^k(C_{\mathcal{U}'}) \to H^k(C_{\mathcal{U}}) \to H^{k+1}(\ker^\bullet) \to \cdots$$

So we will get what we desire by showing that $\ker^\bullet$ has vanishing cohomology at every degree.

2. Now X is affine, I have the augemented chain complex:

$$C^{\text{aug}}_{\mathcal{U}}(X, \mathcal{F}) : 0 \to \Gamma(X, \mathcal{F}) \to \prod \Gamma(U_i, \mathcal{F}) \to \cdots$$

I need to see this has no cohomology.

(a) If $U_1 = X$. Compare $C^{\text{aug}}_{\mathcal{U}}$ with Cech complex for $\mathcal{U}' = \mathcal{U} \backslash \{U_1\}$. We have a short exact sequence of complexes:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & 0 & \longrightarrow & \Gamma(U_1, \mathcal{F}) & \longrightarrow & \prod_{|I|=2,1\in I} \Gamma(U_I, F) & \longrightarrow & \prod_{|I|=3,1\in I} \Gamma(U_I, F) \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma(X, F) & \longrightarrow & C^0_{\mathcal{U}} & \longrightarrow & C^1_{\mathcal{U}} & \longrightarrow & C^2_{\mathcal{U}} \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma(X, F) & \longrightarrow & C^0_{\mathcal{U}'} & \longrightarrow & C^1_{\mathcal{U}'} & \longrightarrow & C^2_{\mathcal{U}'}
\end{array}$$

Look at the long exact sequence, we see that all cohomology for $H^k(C^{\text{aug}}_{\mathcal{U}})$ vanishes and the middle row is actually a direct sum of the top and bottom rows.

(b) In the general case. Let $X = \text{Spec } R \to \text{Spec } A$. Then $C^{\text{aug}}_{\mathcal{U}}(X, \mathcal{F})$ is a complex of R-modules. We need the following fact:

Lemma. *Let $M^\bullet$ is a complex of R-mdoules. Suppose* Spec $R = \bigcup_{i=1}^n D(f_i)$*, then if*

$$M^\bullet \otimes_R R[\frac{1}{f_i}]$$

is exact for all f_i, then $M^\bullet$ is exact.

Now pick $X = \bigcup D(f_i)$, such that at each $x \in X$ there exists f_i such that $x \in D(f_i) \subset U_j$ for some j. Now for each f_i, $D(f_i) \cap \mathcal{U}$, one of these sets is all of $D(f_i)$. Now let's look at: $C^{\text{aug}}_{\mathcal{U}\cap D(f_i)}(D(f_i), \mathcal{F}) = C^{\text{aug}}_{\mathcal{U}}(X, \mathcal{F}) \otimes_R R[\frac{1}{f_i}]$. But by case (a), we know $C^{\text{aug}}_{\mathcal{U}\cap D(f_i)}(D(f_i), \mathcal{F})$ has no cohomology. Now we're done by the commutative algebra lemma.

□

2.2 Cohomology of Projective Schemes

.

Our next goal is to compute:

$$H^k(\mathbb{P}^n_A, \mathcal{O}(m)) = \begin{cases} A[x_0, \cdots, x_n]_m, & \text{if } k = 0; \\ (x_0 \cdots x_n)^{-1} A[x_0^{-1}, \cdots, x_n^{-1}]_{m+n+1}, & \text{if } k = n; \\ 0, & \text{otherwise.} \end{cases}$$

Let's calculate using standard cover of $\mathbb{P}^n$. Note:

$$\Gamma(U_I, \mathcal{O}(m)) = A[x_0, \cdots, x_n, \frac{1}{x_i} \mid i \in I]_{\deg = m}$$

Let's work with $\bigoplus_{m \in \mathbb{Z}} \mathcal{O}(m)$. Let's demonstrate in the case in $\mathbb{P}^2$. The augmented complex reads:

$$0 \to A[x_0, x_1, x_2] \to \prod_{i=0}^{2} A[x_0, x_1, x_2, x_i^{-1}] \to \prod_{i \neq j} A[x_0, x_1, x_2, x_i^{-1}, x_j^{-1}] \to A[x_0^{\pm 1}, x_1^{\pm 1}, x_3^{\pm 1}] \to 0$$

Some combinatorics show that the only cohomology appears in the last place, consisting of monomials with purely negative degrees.

Now let X/k be a projective scheme, and $\mathcal{F} \in \mathrm{Coh}(X)$. We know each cohomology group $H^j(X, \mathcal{F})$ is finite dimensional, so we can define the Euler characteristic:

$$\chi(X, \mathcal{F}) = \sum_{j \geq 0} (-1)^j \dim_k H^j(X, \mathcal{F}) \in \mathbb{Z}$$

Remark. *Note that:*

$$H^j(X_{k'}, F \otimes_k k') = H^j(X, \mathcal{F}) \otimes_k k'$$

Hence the Euler characteristic is only sensitive to the geometric information (i.e. we can assume $k = \overline{k}$)

It is clear that χ descends to a map on the K-groups:

$$\chi(X, -) : K(\mathrm{Coh}(X)) \to \mathbb{Z}$$

Example. *We can compute:*

$$\chi(\mathbb{P}^n, \mathcal{O}(m)) = \begin{cases} \binom{m+n}{n} = \dfrac{(m+1)\cdots(m+n)}{n!}, & \text{if } m \geq 0; \\ (-1)^n \binom{-m-1}{-m-n-1} = \dfrac{(m+1)\cdots(m+n)}{n!}, & \text{if } m \leq -n-1; \\ 0, & \text{otherwise.} \end{cases}$$

In fact they're the same expression.

Let $\mathcal{F} \in \mathrm{Coh}(X)$, we can define the **Hilbert function**:

$$h_{\mathcal{F}}(m) = \dim H^0(X, \mathcal{F} \otimes \mathcal{O}(m))$$

For example let $Z \subset X$ be a closed subscheme, we will have long exact sequence:

$$0 \to H^0(\mathcal{I}_Z \otimes \mathcal{O}(m)) \to H^0(\mathcal{O}(m)) \to H^0(\mathcal{O}(m) \otimes \mathcal{O}_Z) \to \cdots$$

In the case $X = \mathbb{P}^n$, then $h_{\mathcal{I}_Z}(m)$ measures the space of degree m homogenous polynomials vanishing along Z. This is very sensitive to Z.

Example. *Let $X = \mathbb{P}^2$, Z_1 be three generic points on X, Z_2 be three collinear points on X. It follows that:*

$$0 = h_{\mathcal{I}_{Z_1}}(1) < h_{\mathcal{I}_{Z_2}}(1) = 1$$

Instead, one can consider the **Hilbert polynomial**:

$$p_{\mathcal{F}}(m) = \chi(X, \mathcal{F} \otimes \mathcal{O}(m))$$

Note this depends on $(X, \mathcal{F}, \mathcal{O}(1))$. For $m \gg 0$, we know $p_{\mathcal{F}}(m) = h_{\mathcal{F}}(m)$ due to higher cohomology vanishing.

Example. *In the above example, for $Z = Z_1$ or Z_2, we have short exact sequence:*

$$0 \to \mathcal{I}_Z \otimes \mathcal{O}(m) \to \mathcal{O}_X(m) \to \mathcal{O}_Z \to 0$$

So that:

$$p_{\mathcal{I}_Z}(m) = \chi(X, \mathcal{O}_X(m)) - 3 = \binom{m+2}{2} - 3$$

This is insensitive to Z.

Theorem 2.5. *Hilbert polynomial is a polynomial (in $\mathbb{Q}[t]$).*

Proof. Let's induct on $\dim(\mathrm{supp}\ \mathcal{F})$. We may assume assume $X = \mathbb{P}^n$. Let's take $s \in \Gamma(\mathbb{P}^n, \mathcal{O}(1))$, and $V(s) = H$. Let's look at:

$$0 \to \mathcal{O}(-1) \xrightarrow{s} \mathcal{O} \to \mathcal{O}_H \to 0$$

Tensoring with $\mathcal{F} \otimes \mathcal{O}(m)$, I get:

$$\mathcal{F} \otimes \mathcal{O}(m-1) \to \mathcal{F} \otimes \mathcal{O}(m) \to \mathcal{F} \otimes \mathcal{O}_H(m) \to 0$$

It turns out we can pick s such that this is also exact on the left, and $\dim(\mathrm{supp}\ \mathcal{F} \otimes \mathcal{O}_H) = \dim(\mathrm{supp}\ \mathcal{F}) - 1$. Assuming this, we know:

$$\chi(\mathcal{F} \otimes \mathcal{O}(m)) - \chi(\mathcal{F} \otimes \mathcal{O}(m-1)) = \chi(\mathcal{F} \otimes \mathcal{O}_H(m)) = \chi(\mathcal{F}|_H(m))$$

We are therefore done by induction: we know $\chi(\mathcal{F}|_H(m))$ is polynomial of degree $= \dim \mathrm{supp}\ \mathcal{F} - 1$, and the result follows.

Let's now discuss why can we pick such s.

1. Let's find finite number of (possibly non-closed) points of $\mathbb{P}^n$ $y_1, \cdots, y_r$ such that if $y_i \notin H$ for all i, then the claim holds. The second condition is clear if we include generic points of supp $\mathcal{F}$. For the exactness, let's work on an affine chart Spec A. We're given $M \in A$-mod and are looking for:

$$\mathfrak{p}_1, \cdots, \mathfrak{p}_r \in \text{Spec } A$$

such that $f \notin \bigcup \mathfrak{p}_i$, then: $M \xrightarrow{f} M$ is injective. These are precisely the associated primes of M (or $\mathcal{F}$).

Fact: Let's find

$$0 \subset M_1 \subset M_2 \subset \cdots \subset M_a = M$$

such that $M_i/M_{i-1} = A/\mathfrak{p}_i$ for some i. These $\mathfrak{p}_i$ are what we want. If $f \notin \bigcup \mathfrak{p}_i$, one can show $M \xrightarrow{f} M$ is injective by showing it inductively on each M_i.

2. Given $y_1, \cdots, y_r$, we can find $y_i \notin H$ (see homwork 8).

□

Remark. *The degree of $p_{\mathcal{F}}(t)$ equals the dimension of the support of $\mathcal{F}$. The constant term $p_{\mathcal{F}}(0) = \chi(\mathcal{F})$ is again independent of the embedding into $\mathbb{P}^n$.*

Definition 2.6. Let $(X, \mathcal{F}, \mathcal{O}(1)) = (X, \mathcal{O}_X, \mathcal{O}(1))$, we call this the Hilbert polynomial $X \hookrightarrow \mathbb{P}^n$.

Example. *We have:*

$$p_{\mathbb{P}^n, \mathcal{O}(1)}(m) = \chi(\mathcal{O}(m)) = \binom{m+n}{n}$$

Example. *Consider degree d hypersurface: $X_d \hookrightarrow \mathbb{P}^n$. We have:*

$$0 \to \mathcal{O}(-d) \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{X_d} \to 0$$

So:

$$p_{X_d}(m) = p_{\mathbb{P}^n}(m) - p_{\mathbb{P}^n}(m-d) = \binom{m+n}{n} - \binom{m+n-d}{n}$$

Note this has degree $n-1$.

Let $X \hookrightarrow \mathbb{P}^n$. We claim that:

$$p_X(t) = \frac{a}{(\dim X)!} t^{\dim X} + O(t^{\dim X - 1})$$

we will call $\deg(X) \coloneqq a$ the **degree** of the projective variety $X \hookrightarrow \mathbb{P}^n$. As a sanity check $\deg(X_d) = d$.

Now let's define:

$$p_a(X) := 1 - \chi(X, \mathcal{O}_X) = 1 - p_X(0)$$

this is the arithmetic genus.

2.3 More Properties on Sheaf Cohomology

Theorem 2.7 (Künneth)**.** *Let $X, Y/k$, given $\mathcal{F} \in \mathsf{QCoh}(X), \mathcal{G} = \mathsf{QCoh}(Y)$, then we have:*

$$H^m(X \times_k Y, \mathcal{F} \boxtimes \mathcal{G}) \simeq \bigoplus_{p+q=m} H^p(X, \mathcal{F}) \otimes_k H^q(Y, \mathcal{G})$$

Let's discuss cup product. Let $\mathcal{F}, \mathcal{G} \in \mathsf{QCoh}(X)$, we can form:

$$\Gamma(X, \mathcal{F}) \otimes \Gamma(X, \mathcal{G}) \to \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$$

One can generalize this to:

$$H^p(X, \mathcal{F}) \times H^q(X, \mathcal{G}) \to H^{p+q}(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$$

Theorem 2.8 (Serre duality)**.** *Let X be smooth projective over k, with dimension n. Let $\omega_X \in \mathrm{Pic}(X)$ be the canonical bundle. Then for any locally free sheaf $\mathcal{F}$, I have a perfect pairing:*

$$H^i(X, \mathcal{F}) \times H^{n-i}(X, F^\vee \otimes \omega_X) \to k$$

As a special case, if $X = \mathbb{P}^n, \mathcal{F} = \mathcal{O}(a)$ then:

$$\dim H^0(\mathbb{P}^n, \mathcal{O}(a)) \simeq \dim H^n(\mathbb{P}^n, \mathcal{O}(-a-n-1))$$

3 Curves and Surfaces

We will focus on smooth projective curves over algebraically closed fields. We have a basic classification:

Theorem 3.1. *The category of curves finite over C is equivalent to the category of finite field extensions of $k(C)$.*

Remark. *This is obviously false for higher dimensions. Consider $\mathbb{P}^2$ and blowup $\mathbb{P}^2$ at a point p.*

We will study the cohomology of line bundles on curves. First note that:

$$H^1(C, \mathcal{L}) = H^0(C, \mathcal{L}^\vee \otimes K_C)^*$$

by Serre duality.

3.1 Riemann-Roch

Theorem 3.2 (Riemann-Roch)**.** *Let C be a curve and $\mathcal{L} \in \mathrm{Pic}(C)$, then:*

$$\chi(C, \mathcal{L}) = \deg(\mathcal{L}) + \chi(C, \mathcal{O}) = \deg \mathcal{L} + 1 - p_a(C)$$

Proof. Note every line bundle has form $\mathcal{O}(D)$ for some divisor D. Then what we only need to show is that if $p \in C$, then: $\chi(\mathcal{L} \otimes \mathcal{O}(-p)) = \chi(\mathcal{L}) - 1$. We're therefore done by looking at the sequence:

$$0 \to \mathcal{L}(-p) \to \mathcal{L} \to \mathcal{O}_p \to 0$$

□

Note in our setting:

$$p_a(C) = 1 - \chi(\mathcal{O}_C) = 1 - \dim H^0 + \dim H^1 = \dim H^1(C, \mathcal{O}_C)$$

Let $h^* = \dim H^*$. Then by Serre duality:

$$h^0(C, \mathcal{L}) - h^0(C, \mathcal{L}^\vee \otimes K_C) = \deg \mathcal{L} + 1 - p_a(C)$$

$$h^0(K_C) = h^1(K_C^\vee \otimes K_C) = h^1(\mathcal{O}_0) = p_a(C)$$

By Riemann-Roch:

$$\deg K_C + 1 - p_a(C) = h^0(K_C) - h^1(K_C) = p_a(C) - h^0(\mathcal{O}_C) = p_a(C) - 1$$

Therefore:

$$\deg K_C = 2(p_a(C) - 1)$$

Therefore in fact: $g(C) = p_a(C)$. In summary:

$$g(C) = \frac{1}{2} \deg K_C + 1 = h^0(K_C) = h^1(\mathcal{O}_C)$$

Let's discuss the interaction between Riemann-Roch and Hilbert polynomials. Consider:

$$C \hookrightarrow \mathbb{P}^n$$

given by very ample $\mathcal{A} = i^*\mathcal{O}(1)$. Then:

$$p_{C,\mathcal{L},\mathcal{A}}(m) = \chi(\mathcal{L} \otimes \mathcal{A}^{\otimes m}) = \deg(\mathcal{L} \otimes \mathcal{A}^{\otimes m}) + 1 - g = m \cdot \deg(C) + \deg \mathcal{L} + 1 - g$$

Let's derive some easy consequences of Riemann-Roch. Let C be a curve, $\mathcal{L} \in \mathrm{Pic}(C)$.

1. If $\deg \mathcal{L} < 0$, $h^0(C, \mathcal{L}) = 0$.
2. If $\deg \mathcal{L} = 0$, then $h^0(\mathcal{L}) = \begin{cases} 1, & \text{if } \mathcal{L} = \mathcal{O}; \\ 0, & \text{otherwise.} \end{cases}$
3. If $\deg \mathcal{L} > 2g - 2$, then $h^0(\mathcal{L}) = \deg \mathcal{L} + 1 - g$.
4. If $\deg \mathcal{L} = 2g - 2$, then $h^0(\mathcal{L}) = \begin{cases} g, & \text{if } \mathcal{L} = K_C; \\ g - 1, & \text{otherwise.} \end{cases}$

Now let's discuss globally generated line bundles on a curve. Fore very closed point p, let's consider:

$$0 \to \mathcal{O}_C(-p) \to \mathcal{O}_C \to \mathcal{O}_p \to 0$$

Tensoring with $\mathcal{L}$, we get:

$$0 \to H^0(\mathcal{L}(-p)) \to H^0(\mathcal{L}) \xrightarrow{s \mapsto s(p)} H^0(\mathcal{O}_p) \to H^1(\mathcal{L}(-p)) \to H^1(\mathcal{L}) \to 0$$

From this, we see $\mathcal{L}$ is gloablly generated iff $h^0(\mathcal{L}(-p)) = h^0(\mathcal{L}) - 1$ for all p.

Corollary 3.3. *If* $\deg \mathcal{L} > 2g - 1$*, then* $\mathcal{L}$ *is globally generated.*

The natural follow up question: is $\mathcal{L}$ very ample? Now given $\mathcal{L}$ globally generated, we get a map $f : C \to \mathbb{P}^n$, for $n + 1 = h^0(C, \mathcal{L})$. We need to check whether this is a closed embedding. Recall we need to check the sections separate points and tangent spaces.

Let $V = \Gamma(C, \mathcal{L})$, and $V_p \subset V$ be the global sections vanishing at p. We know in this our case V_p has codimension 1. I have a map:

$$\varphi_p : V_p \to \mathcal{L} \otimes \mathfrak{m}/\mathfrak{m}^2$$

Now $T_{C,p} \hookrightarrow T_{\mathbb{P}^n, f(p)}$ iff φ_p is surjective. Hence if $\mathcal{L}$ separates tangent vectors at p, if there exists $s \in \Gamma(\mathcal{L} \otimes \mathcal{O}(-p))$ but $s \notin \Gamma(\mathcal{L} \otimes \mathcal{O}(-2p))$.

1. Note $h^0(\mathcal{L}) - h^0(\mathcal{L} \otimes \mathcal{O}(-p - q)) = 0, 1, 2$. Then $\mathcal{L}$ separates point is requiring the above equals 2 for all $p \neq q$.

2. $\mathcal{L}$ separates tangent vectors if:

   $$h^0(\mathcal{L}) - h^0(\mathcal{L}(-2p)) = 2$$

   for all p.

Corollary 3.4. *The line bundle* $\mathcal{L}$ *is very ample, if for all* $p, q \in C$, $h^0(\mathcal{L}) = h^0(\mathcal{L} \otimes \mathcal{O}(-p - q)) + 2$

Corollary 3.5. *If* $\deg \mathcal{L} > 2g$*, then* $\mathcal{L}$ *is very ample.*

Corollary 3.6. $\deg \mathcal{L} > 0 \iff \mathcal{L}$ *is ample.*

In higher dimensions over $\mathbb{C}$, whether $\mathcal{L} \in \mathrm{Pic}(X)$ is ample only depends on $c_1(\mathcal{L}) \in H^2(X, \mathbb{Q})$. A natural question to ask is that, in general, is there a bound on how big a power to take for ample bundle to be very ample? This is Fujita's conjecture.

3.2 Lower Genus Curves

Let's assume $k = \overline{k}$.

Proposition 3.7. *The only curve of genus* 0 *is* $\mathbb{P}^1$.

Proof. Let C be a genus 0 curve. Pick $p \in C$. Then $\mathcal{L} = \mathcal{O}(p)$ is very ample, and $h^0(\mathcal{L}) = 2$. So we have a closed embedding $C \hookrightarrow \mathbb{P}^1$. We're done. □

Assume now $g(C) \geq 1$.

Lemma 3.8. *Let* $\mathcal{L} \in \mathrm{Pic}(C)$ *with* $\deg \mathcal{L} = 1$, *then* $h^0(C, \mathcal{L}) \leq 1$.

Proof. If not, then pick $s_1, s_2 \in H^0(C, \mathcal{L})$ not proportional. Now let $\mathrm{div}(s_1) = [p], \mathrm{div}(s_2) = [q]$, we must have $p \neq q$. So s_1, s_2 globally generate $\mathcal{L}$, and thus we have a degree one map $C \to \mathbb{P}^1$. Contradiction. □

Corollary 3.9. *If* $p \neq q$, *then* $\mathcal{O}(p) \not\simeq \mathcal{O}(q)$. *Hence* $\mathcal{O}(p - q)$ *is degree* 0 *but not trivial.*

Let's start from genus 1 curves. In this case:

$$\deg K_C = 0, h^0(K_C) = 1 \implies K_C = \mathcal{O}_C$$

Take $\mathcal{L} \in \mathrm{Pic}(C)$ degree one, and note that $1 > 2g - 2$, we get:

$$h^0(\mathcal{L}) = \chi(\mathcal{L}) = 1 + 1 - g = 1$$

Therefore $\mathcal{L} \simeq \mathcal{O}(p)$ for a unique $p \in C(k)$. Hence if we fix a bas point $p_0 \in C(k)$, we have a bijection:

$$C(k) \xrightarrow{\sim} \mathrm{Pic}_0(C)$$
$$p \mapsto \mathcal{O}(p - p_0)$$

In this case we call (C, p_0) an elliptic curve.

If $\deg \mathcal{L} = 2 > 2g - 1$. Then $\mathcal{L}$ is globally generated, and $h^0(\mathcal{L}) = 2$. Therefore it gives me a degree two map $C \to \mathbb{P}^1$. For example, if I take $\mathcal{L} = \mathcal{O}(2p_0)$, then the corresponding $C \to \mathbb{P}^1$ ramified at p_0.

If $\deg \mathcal{L} = 3$, say $\mathcal{L} = \mathcal{O}(3p_0)$. Then $\mathcal{L}$ is very ample, and $h^0(\mathcal{L}) = 3$. Therefore I have a degree 3 closed embedding:

$$C \hookrightarrow \mathbb{P}^2$$

I want to claim that there is a cubic polynomial $f \in k[x, y, z]_3$ such that $C = V(f)$. Start from:

$$0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$$

Tensoring with $\mathcal{O}(3)$:

$$0 \to \mathcal{I}_C(3) \to \mathcal{O}_{\mathbb{P}^2}(3) \to i_*(\mathcal{L}^{\otimes 3}) \to 0$$

Hence:

$$0 \to \Gamma(\mathcal{I}_3(3)) \to \Gamma(\mathcal{O}_{\mathbb{P}^2}(3)) \to \Gamma(C, \mathcal{L}^{\otimes 3})$$

But the second term has dimension 10, and the third term has dimension 9. So we can find $f \in \Gamma(\mathcal{I}_3(3))$ and it follows that $C \subset V(f)$. But now:

$$p_C(m) = 3m = p_{V(f)}(m)$$

By a homweork question we see that $C = V(f)$.

Let's now discuss for $g \geq 2$.

Proposition 3.10. *The canonical bundle K_C is globally generated.*

Proof. Note that $h^0(K_C) = g$. We need $h^0(K_C \otimes \mathcal{O}(-p)) = g - 1$ for all $p \in C$. So we need $h^1(\mathcal{O}(p)) = g - 1$. But this is equivalent to say $h^0(\mathcal{O}(p)) = 1$, which is true. □

Therefore we have a map $f : C \to \mathbb{P}^{g-1}$ with the property $f^*\mathcal{O}(1) = K_C$. For $g = 2$, we get a degree 2 map: $f : C \to \mathbb{P}^1$.

Definition 3.11. A curve (of genus ≥ 2) is hyperelliptic if it admits a degree 2 map to $\mathbb{P}^1$.

Let's assume $\text{char}(k) \neq 2$.

Theorem 3.12.
1. *Given $p_1, \cdots, p_r \in \mathbb{P}^1$ with r even, there exists a unique $C \xrightarrow{\pi} \mathbb{P}^1$ hyperelliptic, branch points at $p_1, \cdots, p_r$. If r is odd, then no such cover exists.*
2. *Riemann-Hurwitz: $r = 2g(C) + 2$.*
3. *If C is hyperellitpic, then the degree 2 map $C \to \mathbb{P}^1$ is uniquely determined by C, up to PGL_2*

Proof. Let $p_1, \cdots, p_r \in \mathbb{P}^1$. Assume $p_1, \cdots, p_r \neq 0, \infty$. Let's look at:

$$f(x) = \prod (x - p_i)$$

and $K(C) = k(x)[y]/(y^2 - f(x))$. We can in fact explicitly write down the curve: $C_0 = \text{Spec } k[x, y]/(y^2 - f(x)) \subset C$. Finally, we need to see the uniqueness of the hyperelliptic map. Let's take:

$$C \xrightarrow{\pi} \mathbb{P}^1$$

We claim that we can recover π using "canonical map". Now $\pi^*\mathcal{O}(1) = \mathcal{L}$, $\deg \mathcal{L} = 2$, $h^0(\mathcal{L}) = 2$. Then I look at:

$$f : C \to \mathbb{P}^1 \xrightarrow{v} \mathbb{P}^{g-1}$$

for v the degree $g - 1$ Veronese embedding. Then

$$f^*\mathcal{O}_{\mathbb{P}^{g-1}}(1) = \mathcal{L}^{\otimes g-1} \implies \deg(f^*\mathcal{O}(1)) = 2g - 2$$

One can show that:

$$H^0(\mathbb{P}^{g-1}, \mathcal{O}(1)) \simeq H^0(C, f^*\mathcal{O}(1))$$

Therefore $h^0(C, f^*\mathcal{O}(1)) = g$ and $\deg f^*\mathcal{O}(1) = 2g - 2$. This forces $f^*\mathcal{O}(1) = K_C$, and f is the canonical map. □

Corollary 3.13. *There exists curve of any genus $g \geq 0$.*

Corollary 3.14. *Every curve of genus 2 is obtained by picking 6 points on $\mathbb{P}^1$, up to PGL_2.*

I can use PGL_2 action to set $p_1 = 0, p_2 = 1, p_3 = \infty$. We get a 3-dimensional family of genus 2 curves, by picking p_4, p_5, p_6. Therefore, let $\mathcal{C}$ be the moduli space of all genus 2 curves, we have a natural map:

$$\mathcal{C} \to U \subset \mathbb{A}^3$$

Every genus curve occurs finitely many times in this family.

Now let's consider $g \geq 3$.

Theorem 3.15. *Given a curve $g(C) \geq 3$, either C is elliptic or K_C is very ample. In the second case, we have canonical embedding:*

$$C \hookrightarrow \mathbb{P}^{g-1}$$

defining using $H^0(C, K_C)$. This is unique after choosing a basis / up to choosing $\mathrm{PGL}(g)$.

Question: How to relate abstract properties of C to the equations of C inside $\mathbb{P}^{g-1}$? There is a conjecture called Green's conjecture, which is known for generic curves.

Proof. Recall that K_C is very ample iff $h^0(K_C \otimes \mathcal{O}(-p-q)) = h^0(K_C) - 2$ for all $p, q \in C$. Therefore K_C is not very ample $\iff$ there exists p, q such that:

$$h^0(K_C \otimes \mathcal{O}(-p-q)) = h^0(K_C) - 1$$

$$\iff h^1(\mathcal{O}_C(p+q)) = g - 1 \iff h^0(\mathcal{O}(p+q)) = 2$$

$\iff \mathcal{O}(p+q)$ is globally generated, and therefore $C \to \mathbb{P}^1$ is a degree 2 map. Thus C is hyperelliptic. □

Therefore, for $g = 3$ curves, we know it is:

1. either hyperelliptic, determined by 8 points on $\mathbb{P}^1$. This gives me a 5-dimensional family of these.
2. or I have the canonical embedded: $C \hookrightarrow \mathbb{P}^2$. This is a degree 4 curve.

Moreover, every smooth degree 4 plane curve comes from a canonical map. In other words, given a degree 4 curve $i : C \hookrightarrow \mathbb{P}^2$ we want to show $i^*\mathcal{O}(1) = K_C$.

Indeed:

$$H^0(\mathbb{P}^2, \mathcal{O}(1)) \hookrightarrow H^0(C, i^*\mathcal{O}(1))$$

$$\implies h^0(i^*\mathcal{O}(1)) \geq 3$$

Now $i^*\mathcal{O}(1)$ has degree $2g - 2 = 4$, and $h^0(i^*\mathcal{O}(1)) \geq g$ so that $i^*\mathcal{O}(1) = K_C$. Let's do a näive dimension count. The space of plane quartics has dimension:

$$h^0(\mathbb{P}^2, \mathcal{O}(4)) - 1 - \dim \mathrm{PGL}_3 = 6$$

In fact we should view hyperelliptic $g = 3$ curves are in closure of the other ones.

Let's now deal with genus 4 curves.

1. C is hyperelliptic. It branched at 10 points, so gives a 7-dimensional of these.

2. Or $C \hookrightarrow \mathbb{P}^3$ a degree 6 embedding.

   Claim: C is a complete intersection by two hypersurfaces $X_2 \cap X_3$, where X_2 is uniquely determined degree 2 irreducible hypersurfaces, and X_3 is a degree 3 irreducible hypersurface. Moreover, every such complete interesection comes from a canonical embedding.

   Proof. Look at:
   $$0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_C \to 0$$
   Tensoring with $\mathcal{O}(2)$:
   $$0 \to \mathcal{I}_C(2) \to \mathcal{O}(2) \to K_C^{\otimes 2} \to 0$$
   Therefore I get:
   $$0 \to \Gamma(\mathcal{I}_C(2)) \to \Gamma(\mathcal{O}_{\mathbb{P}^3}(2)) \to H^0(C, K_C^{\otimes 2}) \to \cdots$$
   But $\deg K_C^{\otimes 2} = 12 \implies h^0(K_C^{\otimes 2}) = 12 + 1 - g = 9$, and $h^0(\mathcal{O}_{\mathbb{P}^3}(2)) = 10$. This shows
   $$h^0(\mathcal{I}_C(2)) \geq 1$$
   This gives a hypersurface $C \hookrightarrow X_2$. Note that since:
   $$H^0(\mathbb{P}^3, \mathcal{O}(1)) \simeq H^0(C, i^*\mathcal{O}(1))$$
   so that C is not contained in any hyperplane, and therefore X_2 is irreducible. To see uniqueness of X_2, suppose there exists other X_2' such that:
   $$C \hookrightarrow X_2 \cap X_2'$$
   so that C is contained in a (not necessarily smooth) curve of degree 4, which is a contradiction.

   Now let's seek for cubic surfaces containing C. We still have:
   $$0 \to \Gamma(\mathcal{I}_C(3)) \to \Gamma(\mathcal{O}_{\mathbb{P}^3}(3)) \to \Gamma(C, K_C^{\otimes 3}) \to \cdots$$
   Now:
   $$h^0(\mathbb{P}^3, \mathcal{O}(3)) = 20, h^0(C, K_C^{\otimes 3}) = 15$$
   So $h^0(\mathbb{P}^3, \mathcal{I}_C(3)) \geq 5$. But I already have:
   $$F_2 \cdot x_0, \cdots, F_2 \cdot x_3$$
   for F_2 the defining equation for X_3. Now I choose F_3 not in the 4-dimensional subspace above. So $C \subset V(F_3) = X_3$. Now I'm in the setting:
   $$C \hookrightarrow C' = X_2 \cap X_3$$
   But then $p_a(C) = p_a(C') = 4$, $\deg(C') = \deg(C) = 6$. So $C \hookrightarrow C'$ has the same Hilbert polynomial, and therefore $C = C'$. □

In this case, we get a 9-dimensional family.

Remark. *In this case, we proved that for $g = 4$, we can find $U \subset \mathbb{A}^N$ and a family $\mathcal{C} \to U$ such that "most" $g = 4$ curves (except hyperelliptic ones) occurs in this family. This is true for $g \leq 15$, and this definitely fails for $g \geq 22$.*

Remark. *The dimension of the moduli space of genus g curves equal $3g - 3$.*

Remark. *More interesting materials related to these discussions can be found in the book by Arbarello, Cornalba, Griffiths, Harris.*

3.3 Serre Duality for Curves

The goal of this section is to prove Serre duality for curves. We will only prove hte version for line bundles:

$$H^1(C, \mathcal{O}(D))^\vee = H^0(C, \Omega_C \otimes \mathcal{O}(-D))$$

We wish to to describe $H^1(\mathcal{O}(D))$ explicitly and pairing also. Let's define:

$$R = \{(a_p)_{p \in C} \mid a_p \in k(C) = \mathrm{Frac}(\mathcal{O}_{C,p}) \text{ all but finitely many } a_p \in \mathcal{O}_{C,p}\}$$

Similarly, let's define:

$$R(D) = \{(a_p) \in R \mid \mathrm{val}(a_p) \geq -c_p, \forall p\}$$

for $D = \sum c_p[p]$ a Weil divisor.

Proposition 3.16. *We have:*

$$H^1(\mathcal{O}(D)) \simeq R/(R(D) + k(C))$$

Proof. Let's consider:

$$0 \to \mathcal{O}(D) \to k(C) \to S \to 0$$

Take cohomology:

$$0 \to H^0(\mathcal{O}(D)) \to k(C) \to H^0(S) \to H^1(\mathcal{O}(D)) \to 0$$

We want $H^0(S) = R/R(D)$. Now let's look at the stalk of S_p:

$$k(C)/\mathcal{O}(D)_p = \frac{k(C)}{\{f \mid \mathrm{val}_p(f) \geq -c_p\}} = (R/R(D))_p$$

We claim that

$$\Gamma(S) = \bigoplus_{p \in C} S_p = R/R(D)$$

But one can directly check:

$$s \mapsto (s_p)$$

gives the isomorphism. □

Now let $I(D) = R/(R(D) + k(C))$, then we can consider:

$$J(D) = I(D)^* = \mathrm{Hom}(I(D), k)$$

I can view $\theta \in J(D)$ as a map $\theta : R \to k$ such that $\theta|_{k(C)} = \theta|_{R(D)} = 0$. If $D' < D$, then $J(D) \subset J(D')$. Let's take $J = \bigcup J(D)$. Then Serre duality will follows from:

Proposition 3.17. *We have:*

$$J(D) \simeq \Gamma(\Omega_C(-D)), J \simeq \Omega \otimes k(C)$$

Proof. We use the residue pairing. Let's start by taking completion:

$$\widehat{\mathcal{O}_{C,p}} \simeq k[[t]]$$

for some uniformizer t. Moreover:

$$k(C) \otimes \widehat{\mathcal{O}_{C,p}} \simeq k((t))$$

Now we can expand a 1-form into Luarent series:

$$\Omega_{C,p} \otimes k(C) \to \Omega_{C,p} \otimes k(C) \otimes \widehat{\mathcal{O}_{C,p}} = k((t))dt$$

$$\omega \mapsto \sum a_n t^n dt$$

Now we have the **residue pairing**:

$$\mathrm{res}_{p,t}(\omega) = a_{-1}$$

Claim: The residue $\mathrm{Res}_{p,t}(\omega)$ is independent of the coordinate t.

This claim is motivated by the complex analytic fact:

$$\mathrm{res}(\omega) = \frac{1}{2\pi i} \oint_\gamma \omega$$

is independent of the choice of coordinate.

Claim: If $\omega \in \Omega \otimes k(C)$, then:

$$\sum_p \mathrm{res}_p(\omega) = 0$$

This is motivated by Stoke's theorem:

$$\sum_i \oint_{\gamma_i} \omega = \iint_{C \setminus U} d\omega = 0$$

for since ω is closed as a real 2-form (from Cauchy-Riemann equation). From these two claims we can now conclude. Let $\omega \in \Omega \otimes k(C)$, let's consider $\theta_\omega : R \to k$ given by:

$$\theta_\omega((a_p)) = \sum_{p \in C} \mathrm{res}_p(a_p \cdot \omega)$$

so $\theta_\omega|_{k(C)}= 0$ from the second claim. On the other hand, it is straightforward to check that $\text{res}_p(a_p\omega) = 0 \implies \theta_\omega|_{R(D)}= 0$. Therefore, we have a well-defined map:

$$\theta : \Gamma(\Omega_C(-D)) \to J(D)$$

Run through all D we get:

$$\theta : \Omega_C \otimes k(C) \to J$$

We claim that θ is an isomorphism (for both statements).

We first show that if $\omega \in \Omega \otimes k(C)$ satisfies $\theta_\omega \in J(D)$, then $\omega \in \Gamma(\Omega_C(-D))$. Indeed, if $\omega \notin \Gamma(\Omega_C(-D))$, then there exists p such that $\omega_p = t^{c_p-k}dt + \cdots$ for some $k \geq 1$. Now pick $a_p = t^{-c_p+k-1}$ and a_q everywhere else, so that $(a_p) \in R(D)$. It follows that:

$$\theta_\omega((a_p)) = \text{res}_p(a_p\omega) \neq 0 \implies \theta_\omega \notin J(D)$$

This is a contradiction. Now suppose $\omega \in \Omega \otimes k(C)$ satisfies $\theta_\omega = 0$, then $\theta \in J(D)$ for all D for all D, and therefore $\omega \in \Gamma(\Omega(-D))$ for all D. This forces $\omega = 0$. This is saying $\theta : \Omega \otimes k(C) \hookrightarrow J$.

On the other hand, we should view J is a $k(C)$ vector space of dimension 1. The action is given by: for $f \in k(C)$ and $\sigma \in J$, we have

$$(f \cdot \sigma)((a_p)) = \sigma((f \cdot a_p))$$

Let's check $\dim_{k(C)J} \leq 1$. Given $\sigma_1, \sigma_2 \in J(D)$, we wish to show they're linearly independent. Let's pick $f_1, f_2 \in \Gamma(\mathcal{O}(\Delta))$ for $\Delta \gg 0$. Suppose they're independent, we have:

$$\Gamma(\mathcal{O}(\Delta)) \oplus \Gamma(\mathcal{O}(\Delta)) \hookrightarrow J(D - \Delta)$$

But LHS has dimension of growth rate $2 \deg \Delta$, while RHS has dimension of growth rate $\deg \Delta$. Contradiction. □

3.4 Baby Intersection Theory

The Motivation is the following. Bezout's theorem states that for $C, D \subset \mathbb{P}^2$ irreducible curves, we have:

$$\#(C \cap D) = \deg C \cdot \deg D$$

One can interpret this as the fact that the intersection number only depends on the divisor class, not the divisor itself. Intersection theory aims to generalize this to higher dimensions / other surfaces.

Let $X/k = \overline{k}$ be smooth projective. Let's introduce $N^1(X)$ as a quotient of $\text{Pic}(X)$. First, suppose C is integral projective (not necessarily smooth) curve, with $\mathcal{L} \in \text{Pic}(C)$. We can define:

$$\deg_C \mathcal{L} = \chi(\mathcal{L}) - \chi(\mathcal{O}_C)$$

This agrees with our usual definition by Riemann-Roch when C is smooth. Another concrete way to think about it is using normalization:

$$f : C' \to C$$

and then $\deg_C \mathcal{L} \coloneqq \deg_{C'} f^*\mathcal{L}$.

Definition 3.18. We call $\mathcal{L} \in \mathrm{Pic}(X)$ **numerically trivial** if for all $i : C \hookrightarrow X$, we have $\deg i^*\mathcal{L} = 0$. Let's denote $\mathrm{Pic}^\tau(X)$ to be the subgroup of numerically trivial line bundles. Then we define:

$$N^1(X) = \mathrm{Pic}(X)/\mathrm{Pic}^\tau(X)$$

Theorem 3.19. *$N^1(X)$ is a finitely generated, torsion free abelian group.*

Example. *If C is a smooth projective curve, then $\mathrm{Pic}(C)$ can be very large. But $\mathrm{Pic}^\tau(C)$ is just the collection of all degree 0 line bundles. Therefore:*

$$N^1(C) \xrightarrow{\deg} \mathbb{Z}$$

is an isomorphism.

Remark. *Assume $X/\mathbb{C}$, then I have:*

$$c_1^{\mathrm{top}} : \mathrm{Pic}(X) \to H^2(X, \mathbb{Z})$$

Then:

$$\deg_C i^*\mathcal{L} = (c_1(\mathcal{L}) \cap [C]) \in \mathbb{Z}$$

where $[C] \in H_2(X)$ is the fundamental class. In particular:

$$\mathrm{Pic}^\tau \supset \ker c_1$$

Therefore we have:

$$\mathrm{Pic}(X)/\ker c_1 \hookrightarrow H^2(X, \mathbb{Z})$$

Now N^1 is a quotient of $\mathrm{Pic}/\ker c_1$ is is finitely generated.

Remark. *$\rho(X) = \mathrm{rank}(N^1)$ is a subtle invariant of X. For example, take $X_d \subset \mathbb{P}^3$ hypersurface of degree of $d \geq 4$, then $\rho(X_d)$ depends on which X_d. Most of these have $\rho = 1$. However, there is a countable union, everywhere dense family on which ρ jumps to 2.*

Remark. *There is a notion of algebraically trivial line bundles, fitting into*

$$\text{algebraically trivial line bundles} \subset \ker c_1 \subset \mathrm{Pic}^\tau$$

Now we hope to use N^1 to define intersection numbers. Let X be projective, $\mathcal{F} \in \mathrm{Coh}(X)$ with $\dim(\mathrm{supp}\ (\mathcal{F})) \leq n$. Let $\mathcal{L}_1, \cdots, \mathcal{L}_n \in \mathrm{Pic}(X)$.

Definition 3.20. We define $(\mathcal{L}_1 \circ \mathcal{L}_2 \cdots \circ \mathcal{L}_n \mid \mathcal{F})$ to be:

$$\sum_{\{i_1, \cdots, i_m\} \subset [n]} (-1)^m \chi(\mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}) \in \mathbb{Z}$$

If $\mathcal{F} = \mathcal{O}_X$, we will use notation:

$$(\mathcal{L}_1 \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid X)$$

Example. *For $n = 1$, and $\mathcal{F} = \mathcal{O}_C$. Then:*

$$(\mathcal{L} \mid C) = \chi(\mathcal{O}_C) - \chi(\mathcal{L}^\vee) = -\deg \mathcal{L}^\vee = \deg \mathcal{L}$$

Example. *Let $n = 2, X = \mathbb{P}^2$. Let's say $\mathcal{L}_1 = \mathcal{O}(a), \mathcal{L}_2 = \mathcal{O}(b)$. Then:*

$$(\mathcal{L}_1 \circ \mathcal{L}_2 \mid \mathbb{P}^2) = \chi(\mathcal{O}) - \chi(\mathcal{O}(-a)) - \chi(\mathcal{O}(-b)) + \chi(\mathcal{O}(-a-b)) = ab$$

Lemma 3.21. *Suppose $D \hookrightarrow X$ be a hypersurface. So I have:*

$$0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$$

Suppose it stays exact when tensoring with $\mathcal{F}$ (support of D needs to avoid associated primes of $\mathcal{F}$, which is generically true), then:

$$(\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_{n-1} \circ \mathcal{O}(D) \mid \mathcal{F}) = (\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_{n-1} \mid i^*\mathcal{F})$$

Proof. Using:

$$0 \to \mathcal{F} \otimes \mathcal{O}(-D) \to \mathcal{F} \to i_* i^* \mathcal{F} \to 0$$

So that:

$$\chi(i^*\mathcal{F}) = \chi(\mathcal{F}) - \chi(\mathcal{F} \otimes \mathcal{O}(-D))$$

We can immediately deduce the statement from this. □

Given $D_1, \cdots, D_n \subset X$ such that the condition holds inductively, i.e. D_k and $\mathcal{O}_{D_{k+1} \cap \cdots \cap D_n}$ is transversal for each k. We can then apply the lemma repeatedly and get:

$$(\mathcal{O}(D_1) \circ \cdots \circ \mathcal{O}(D_n) \mid X) = (- \mid \mathcal{O}_{D_1 \cap \cdots \cap D_n}) = \text{length}(\mathcal{O}_{D_1 \cap \cdots \cap D_n}) = \#(D_1 \cap \cdots \cap D_n)$$

As a result, we call:

$$(\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_n \mid X)$$

the **intersection product**. In particular, this tells us that intersections of n hypersurfaces D_i only depends on $\mathcal{O}(D_i)$.

Proposition 3.22. *Fix $\mathcal{F}$, then:*

$$(\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F})$$

is symmetric, multilinear, and only depends on $[\mathcal{L}_i] \in N^1(X)$. Therefore, for any $\mathcal{F}$, we have a multilinear map:

$$N^1(X) \times \cdots \times N^1(X) \to \mathbb{Z}$$

determines all intersections of hypersurfaces.

Proof. Symmetry is by definition. Let's show multilinearity:

$$(\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) + (\mathcal{L}_1' \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) = (\mathcal{L}_1 \otimes \mathcal{L}_1' \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F})$$

From definition and regrouping:

$$(\mathcal{L}_1 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) + (\mathcal{L}_1' \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) - (\mathcal{L}_1 \otimes \mathcal{L}_1' \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) = (\mathcal{L}_1 \circ \mathcal{L}_1' \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F})$$

We need to show RHS vanishes. One can show (these two equivalent statements) by induction on n. For $n = 0$, the statement is vacuous. By induction we know if one of the line bundle equals $\mathcal{O}(D)$ for some hypersurface D satisfying the lemma, then we have the vanishing (again by the above lemma). In other words, if there is a very ample line bundle then the result is true. Now I need:

Lemma. *(From next homework) Given any $\mathcal{L} \in \mathrm{Pic}(X)$, there exists $\mathcal{A}_1, \mathcal{A}_2$ very ample such that $\mathcal{L} = \mathcal{A}_1 \otimes \mathcal{A}_2^\vee$.*

Let's take $\mathcal{L}_1 = \mathcal{A} \otimes \mathcal{B}^\vee$ and $\mathcal{L}_1' = \mathcal{A}' \otimes (\mathcal{B}')^\vee$ for $\mathcal{A}, \mathcal{A}', \mathcal{B}, \mathcal{B}'$ very ample bundles. Then:

$$\begin{aligned}(\mathcal{L}_1 \otimes \mathcal{L}_1' \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) &= ((\mathcal{A} \otimes \mathcal{A}' \otimes \mathcal{B}^\vee \otimes \mathcal{B}'^\vee) \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) \\ &= ((\mathcal{A} \otimes \mathcal{A}') \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) - ((\mathcal{B} \otimes \mathcal{B}') \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) \\ &= (\mathcal{A} \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) + (\mathcal{A}' \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) \\ &\quad - (\mathcal{B} \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) - (\mathcal{B}' \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) \\ &= (\mathcal{L}_1 \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) + (\mathcal{L}_1' \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F})\end{aligned}$$

as desired.

Finally, we need to show that if $\mathcal{L}_1 \sim 0$, then:

$$(\mathcal{L}_1 \circ \mathcal{L}_2 \circ \cdots \circ \mathcal{L}_n \mid \mathcal{F}) = 0$$

If $n = 1$, then:

$$(\mathcal{L} \mid \mathcal{F}) = \chi(\mathcal{F}) - \chi(\mathcal{F} \otimes \mathcal{L}^\vee) = 0$$

by Riemann-Roch (for nonreduced curves). Now use induction and linearity, we may assume $\mathcal{L}_n$ is very ample and $\mathcal{L}_n = \mathcal{O}(D)$ for D satsifying the lemma. Now I apply the lemma to set myself in one dimension lower. □

Finally let's specialize to the case of surfaces. In this case:

$$\mathrm{Div}(X) \times \mathrm{Div}(X) \to N^1 \times N^1 \to \mathbb{Z}$$

So that we can define:

$$C \cdot D = (\mathcal{O}(C) \cdot \mathcal{O}(D) \mid X)$$

If C, D are integral, I can apply the lemma:

$$C \cdot D = \deg_C(\mathcal{O}(D)|_C) = \deg_D(\mathcal{O}(C)|_D)$$

If $C \cap D$ is zero-dimensional, then:

$$C \cdot D = \mathrm{length}(\mathcal{O}_{C \cap D})$$

Example. *Let $X = \mathbb{P}^2$, then:* $\mathrm{Pic} = N^1 = \mathbb{Z} \cdot [\ell]$. *So that:*

$$N^1 \times N^1 \to \mathbb{Z}$$

is given by:

$$[a\ell] \cdot [b\ell] = ab$$

Example. *Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, then:*

$$\mathrm{Pic}(X) = N^1(X) = \mathbb{Z} \cdot [C] \oplus \mathbb{Z} \cdot [D]$$

for $C = 0 \times \mathbb{P}^1, D = \mathbb{P}^1 \times 0$. To describe the intersection pairing:

$$C \cdot D = D \cdot C = 1$$

$$C \cdot C = C \cdot C_t = 0$$

$$D \cdot D = D \cdot D_t = 0$$

where $C_t = t \times \mathbb{P}^1$, $D_t = \mathbb{P}^1 \times t$. Therefore I get a bilinear form

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Therefore for $E \subset \mathbb{P}^1 \times \mathbb{P}^1$ a hypersurface given by

$$f(x, y, z, w)$$

which is homogeneous of degree a in x, y and homogeneous of degree b in z, w. Then:

$$\mathcal{O}(E) = \mathcal{O}(aC + bD) \implies [E] = a[C] + b[D] \in \mathrm{Pic}(\mathbb{P}^1 \times \mathbb{P}^1)$$

Now if $E \subset \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree (a, b), and $E' \subset \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree (a', b') then:

$$\#(E \cap E') = ba' + ab'$$

This is the version of Bezout's theorem for $\mathbb{P}^1 \times \mathbb{P}^1$.

Remark. *If we diagonalize the above blinear pairing on $N^1(X) \otimes_{\mathbb{Z}} \mathbb{R}$, we're getting:*

$$\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

We thus get a positive part and a negative part. In general there will be only one positive part (given by a choice of very ample line bundle), and the rest is negative definite. This is **Hodge Index Theorem**.

Remark. *There's another way to think about $C \cdot C$ for some smooth curve $i : C \hookrightarrow X$ in a surface. Note that in this case: $\mathcal{I}_{C/X} = \mathcal{O}(-C)$. Therefore we have:*

$$i^*\mathcal{O}(-C) = \mathcal{I}/\mathcal{I}^2 = \mathcal{N}_C^\vee \implies i^*\mathcal{O}(C) = \mathcal{N}_C$$

Therefore $C \cdot C = \deg_C \mathcal{N}_C$ is the degree of the normal bundle of C. Note that the normal bundle fits into short exact sequence of sheaves on C:

$$0 \to \mathcal{T}_C \to i^*\mathcal{T}_X \to \mathcal{N}_C \to 0$$

3.5 Asymptotic Riemann-Roch

Let X be smooth projective, with $\mathcal{F} \in \mathrm{Coh}(X)$, and $\mathcal{L}$ be very ample. We showed that $\chi(\mathcal{F} \otimes \mathcal{L}^{\otimes m})$ is a polynomial function of m. In fact for any $\mathcal{L} \in \mathrm{Pic}(X)$, $\chi(\mathcal{F} \otimes \mathcal{L}^{\otimes m})$ is again a polynomial function of m.

Theorem 3.23 (Asymptotic Riemann-Roch)**.** *Let $p(m) \coloneqq \chi(\mathcal{F} \otimes \mathcal{L}^{\otimes m})$, then $p(m)$ is a polynomial function of m of degree at most $n = \dim \operatorname{supp} \mathcal{F}$. Moreover, the coefficient of the term of degree n is:*

$$\frac{(\mathcal{L} \circ \cdots \circ \mathcal{L} \mid \mathcal{F})}{n!}$$

Remark. *We can compute $p(m)$ precisely in terms of Chern classes of $\mathcal{L}$ and $\mathcal{F}$.*

Proof. Given a function $g : \mathbb{Z} \to \mathbb{Z}$, consider the successive difference:

$$\Delta g(m) \coloneqq g(m) - g(m-1)$$

Then g is a polynomial of degree $\leq n$ iff $\Delta^{n+1} g = 0$. Moreover, in this case the coefficient of m^n is $\frac{\Delta^n g}{n!}$. Recall that we proved last time:

$$0 = (\mathcal{L} \circ \cdots \circ \mathcal{L} \mid \mathcal{F} \otimes \mathcal{L}^{\otimes m})$$

where $\mathcal{F}$ is a sheaf of support dimension $\leq n$. Let's now unravel the definition of this statement:

$$\chi(\mathcal{F} \otimes \mathcal{L}^{\otimes m}) - (n+1)\chi(\mathcal{F} \otimes \mathcal{L}^{\otimes m-1}) + \binom{n+1}{2}\chi(\mathcal{F} \otimes \mathcal{F}^{\otimes m-2}) - \cdots = 0$$

In other words:

$$p(m) - \binom{n+1}{1}p(m-1) + \binom{n+1}{2}p(m-2) - \cdots = 0$$

But RHS is just Δ^{n+1}, therefore we know $p(m)$ is a polynomial function of degree $\leq n$, with leading coefficient equals:

$$\frac{1}{n!}\Delta^n p(m) = \frac{1}{n!}(\mathcal{L} \circ \cdots \circ \mathcal{L} \mid \mathcal{F})$$

□

3.6 Hodge Index Theorem

Let X be a smooth projective surface over $k = \overline{k}$. We consider the intersection pairing on $N^1(X) \otimes \mathbb{R}$.

Theorem 3.24 (Hodge Index Theorem)**.** *Suppose $N^1(X) = \mathbb{Z}^{\rho(X)}$, then the signature for the inersection pairing is: $(1, \rho - 1)$.*

Remark. *Let $X/\mathbb{C}$ be a complex manifold, and we view it has a 4-dimensional manifold. Note that $H^2(X,\mathbb{R})$ has signature. Note that we have:*

$$N^1(X)\otimes\mathbb{R}\subset H^2(X,\mathbb{R})$$

Then the signature on $H^2(X,\mathbb{R})$ is more subtle. It may take form:

$$(1+a,\rho-1+b)$$

But the point is that on its algebraic part the signature is well-understood.

If $\mathcal{A}=\mathcal{O}(D)$ is very ample on X, and therefore

$$\mathcal{A}\cdot\mathcal{A}=\deg_D i^*\mathcal{A}>0$$

What we need to show is that if $\mathcal{A}$ is ample, with $\mathcal{L}\in\mathrm{Pic}(X)$ with $\mathcal{L}\cdot\mathcal{A}=0$. Then:

$$\mathcal{L}\cdot\mathcal{L}\leq 0$$

Moreover if we have $\mathcal{L}\cdot\mathcal{L}=0$ then $[\mathcal{L}]=0$ in $N^1(X)$.

Proof. We will break the proof into 4 steps. Let $\mathcal{A}$ be very ample.

1. Given $\mathcal{M}\in\mathrm{Pic}(X)$ such that $\mathcal{M}\cdot\mathcal{A}>K_X\cdot\mathcal{A}$. We claim that $h^2(\mathcal{M})=0$. Let's now prove this claim. If $H^2(\mathcal{M})\neq 0$, then Serre duality implies that:

   $$H^0(\mathcal{M}^\vee\otimes K)\neq 0$$

   Therefore I can find an effective divisor D such that $M^\vee\otimes K=\mathcal{O}(D)$. Therefore:

   $$\mathcal{A}\cdot\mathcal{O}(D)=\deg_D\mathcal{A}\geq 0\implies\mathcal{A}\cdot(M^\vee\otimes K_X)\geq 0\implies\mathcal{A}\cdot K_X\geq\mathcal{A}\cdot\mathcal{M}$$

   Contradictions.

2. Given $\mathcal{L}\in\mathrm{Pic}(X)$ such that $\mathcal{L}\cdot\mathcal{L}>0$ and $\mathcal{L}\cdot\mathcal{A}>0$. Then for $n\gg 0$, $h^0(\mathcal{L}^{\otimes n})>1$. If $n\gg 0$, then:

   $$\mathcal{L}^{\otimes n}\cdot\mathcal{A}=n(\mathcal{L}\cdot\mathcal{A})>K_X\cdot\mathcal{A}$$

   We're now in the setting in the first step. So thaat $h^2(\mathcal{L}^{\otimes n})=0$. Now by asymptotic polynomial:

   $$\chi(\mathcal{L}^{\otimes n})=\frac{\mathcal{L}\cdot\mathcal{L}}{2}n^2+\cdots\to\infty$$

   Combining with the vanshing of $h^2(\mathcal{L}^{\otimes n})$ we're done.

3. Suppose $\mathcal{L}\cdot\mathcal{A}=0$. We need to show $\mathcal{L}\cdot\mathcal{L}\leq 0$. Let me replace $\mathcal{A}$ by:

   $$\mathcal{A}'=\mathcal{L}\otimes\mathcal{A}^{\otimes N}$$

   to be very ample for large N. If $\mathcal{L}\cdot\mathcal{L}>0$, then $\mathcal{L}\cdot\mathcal{A}'=N(\mathcal{L}\cdot\mathcal{A})+\mathcal{L}\cdot\mathcal{L}>0$. So the second part applies, and therefore:

   $$h^0(\mathcal{L}^{\otimes n})>1$$

   for some large n. So $\mathcal{L}^{\otimes n}=\mathcal{O}(D)$ for some effective D. Therefore $\mathcal{A}\cdot\mathcal{O}(D)>0\implies\mathcal{A}\cdot\mathcal{L}>0$. This is a contradiction. Therefore $\mathcal{L}\cdot\mathcal{L}\leq 0$.

4. Finally let's show if $\mathcal{L} \cdot \mathcal{L} = 0$ and $\mathcal{L} \cdot \mathcal{A} = 0$, then:

$$[\mathcal{L}] = 0 \in N^1(X)$$

If not, then there is a line bundle $\mathcal{M} \in \text{Pic}(X)$ such that $\mathcal{L} \cdot \mathcal{M} > 0$. Let's replace $\mathcal{M}$ with $\mathcal{M}' \in N^1 \otimes \mathbb{Q}$ such that:

$$\mathcal{L} \cdot \mathcal{M}' > 0, \mathcal{A} \cdot \mathcal{M}' = 0$$

If I take $[\mathcal{L}'] = n[\mathcal{L}] + [\mathcal{M}']$, then:

$$\mathcal{L}' \cdot \mathcal{A} = \mathcal{M}' \cdot \mathcal{A} = 0$$

$$\mathcal{L}' \cdot \mathcal{L}' = 2n(\mathcal{L} \cdot \mathcal{M}') + \mathcal{M}' \cdot \mathcal{M}' > 0$$

for $n \gg 0$. Now I just take some power of $\mathcal{L}'$ and I get an actual line bundle, and we see a contradiction from the third part.

□

Now let's do an application. Let $C/k = \overline{k}$ be a smooth projective curve. Consider:

$$\varphi : C \to C$$

be an endomorphism. Suppose I want to count:

$$C^\varphi = \{p \in C(k) \mid \varphi(p) = p\}$$

I want to bound this in terms of $g(C) = g$ and $\deg(f) = d$. Let $X = C \times C$, then we have:

$$\Delta_C \subset X, \Gamma_\varphi \subset X$$

be the diagonal embedding and the graph of φ. Then:

$$C^\varphi = \Gamma_\varphi \cap \Delta_C$$

Actually we want to compute:

$$\Gamma_\varphi \cdot \Delta$$

Now let

$$e = C \times \{p\} \in N^1(X), f = \{p\} \times C \in N^1(X)$$

then we know:

$$e \cdot e = f \cdot f = 0, e \cdot f = 1$$

Note that $e + f \in N^1(X)$ is ample, so if we write:

$$N^1 \otimes \mathbb{Q} = \langle e, f \rangle \oplus \langle e, f \rangle^\perp$$

and the intersection pairing on the second bit is negative definite by Hodge index theorem. Now let's write:

$$[\Gamma] = c_1 e + c_2 f + \Gamma'$$

$$[\Delta] = c_3 e + c_4 f + \Delta'$$

Note that:

$$c_1 = \Gamma \cdot f = \#\{(c, \varphi(c)) \mid c = p\} = 1$$

Similarly:

$$c_2 = \#\{\varphi^{-1}(p)\} = \deg \varphi = d$$

Therefore:

$$\Gamma = e + df + \Gamma'$$

$$\Delta = e + f + \Delta'$$

It follows that:

$$\Gamma \cdot \Delta = (d+1) + \Gamma' \cdot \Delta'$$

Now by Cauchy Schwarz, I have:

$$|\Gamma' \cdot \Delta'| \leq \sqrt{-\Gamma' \cdot \Gamma'} \cdot \sqrt{-\Delta' \cdot \Delta'}$$

It remains to study $\Gamma' \cdot \Gamma'$. First note that the self intersection:

$$\Gamma_\varphi \cdot \Gamma_\varphi = \deg_{\Gamma_\varphi} \mathcal{N}_{\Gamma_\varphi}$$

is the degree of the normal bundle. Let $i = 1 \times \varphi : C \hookrightarrow C \times C$, then we have the tangent sequence:

$$0 \to T_C \to i^* T_{C \times C} \to \mathcal{N}_{\Gamma_\varphi} \to 0$$

But the middle term is simply:

$$i^* T_{C \times C} = T_C \oplus \varphi^* T_C$$

and therefore $\mathcal{N}_{\Gamma_\varphi} = \varphi^* T_C$. It follows that:

$$\Gamma_\varphi \cdot \Gamma_\varphi = \deg(\varphi^* T_C) = d \cdot (2 - 2g)$$

On the othar hand:

$$\Gamma_\varphi \cdot \Gamma_\varphi = 2d + \Gamma' \cdot \Gamma' \implies \Gamma' \cdot \Gamma' = -2gd$$

Hence:

$$|\Gamma' \cdot \Delta'| \leq \sqrt{-\Gamma' \cdot \Gamma'} \cdot \sqrt{-\Delta' \cdot \Delta'} = 2g\sqrt{d}$$

Therefore we proved:

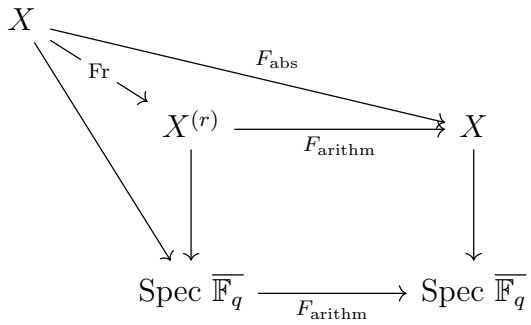
Theorem 3.25. *Let C be a smooth projective curve over an algebraically closed field k. Let $g = g(C)$ and $\varphi : C \to C$ be a degree d endomorphism of C. Suppose the set of fixed points is zero dimensional. Then the number r of fixed points of φ, counted with multiplicity, satisfies:*

$$|r - (d+1)| \leq 2g\sqrt{d}$$

In particular, an automorphism fixes at most $2g + 2$ points.

3.7 Weil Conjecture For Curves

To start with, we need to introduce Frobenius map in positive characteristic. Let p be a prime number and $q = p^r$. Let $X_0/\mathbb{F}_q$ be a smooth projective geometrically connected variety, and $X = (X_0)_{\overline{\mathbb{F}_q}}$. We define the **geometric Frobenius** map to be the Fr below:



where both the bottom arithmetic Frobenius and the absolute Frobenius are given by raising to q-th power. Since X is defined over finite field $\mathbb{F}_q$ (by X_0), therefore in fact $X^{(r)} \simeq X$ as Spec $\overline{\mathbb{F}_q}$-schemes. So that we can view Fr as an endomorphism on X.

Example. *Let $X = \mathbb{A}^n_{\overline{\mathbb{F}_q}}$, then $X \simeq X^{(r)}$, and:*

$$\mathrm{Fr} : \mathbb{A}^n \to \mathbb{A}^n$$

is given by:

$$\overline{\mathbb{F}_q}[x_1, \cdots, x_n] \to \overline{\mathbb{F}_q}[x_1, \cdots, x_n]$$
$$x_i \mapsto x_i^q$$

but fixes $\overline{\mathbb{F}_q}$. On the level of $X(\overline{\mathbb{F}_q})$, it is sending:

$$(\alpha_1, \cdots, \alpha_n) \mapsto (\alpha_1^q, \cdots, \alpha_n^q)$$

Proposition 3.26. *Let $k = \mathbb{F}_q$, and X_0/k be a smooth projective geometrically connected variety, and $X = (X_0)_{\overline{k}}$. Then:*

$$\mathrm{Fix}(\mathrm{Fr}^d)(\overline{k}) \simeq X_0(\mathbb{F}_{q^d})$$

More precisley, this is saying that:

$$\Delta_X \cap \Gamma_{\mathrm{Fr}^d} \simeq \prod_{X_0(\mathbb{F}_{q^d})} \mathrm{Spec}\ (\overline{\mathbb{F}_q})$$

as schemes, where the left hand side is scheme theoretic intersection.

Proof. Set theoretically they're clearly the same. To see LHS has no reduced point, note that:

$$T_p(\mathrm{Fix}(\mathrm{Fr}^d)) \subset T_p(\Delta_X) \cap T_p(\Gamma_{\mathrm{Fr}^d}) \subset T_p(X \times X) = T_pX \oplus T_pX$$

The former term in the intersection is the set (v, v) and the latter is the set $(v, d\mathrm{Fr}^d(v))$ for some $v \in T_pX$. But in fact $d\mathrm{Fr} = 0$, because for any derivation $v \in T_pX$ we have:

$$(d\mathrm{Fr}, v) = v(\mathrm{Fr}) = 0$$

this shows our intersection of tanent space is indeed transverse. □

It can also be shown that $\mathrm{Fr} : X \to X$ is a degree $q^{\dim X}$ morphism between varieties.

Now let us set $X_0 = C_0$ be a smooth projective and geometrically connected curve. Then our previous discussion immediately implies:

Corollary 3.27. *We have:*

$$|\#C_0(\mathbb{F}_{q^d}) - (q^d + 1)| \leq q^{\frac{d}{2}} \cdot 2g$$

Remark. *This implies that any genus* 0 *curve over* $\mathbb{F}_q$ *is again* $\mathbb{P}^1_{\mathbb{F}_q}$.

Let $C_0/\mathbb{F}_q$ be a curve. We defined the zeta function to be:

$$Z_{C_0}(t) = \exp \sum_{d \geq 1} \frac{N_d(C_0)}{d} t^d$$

where $N_d(C_0) = \#C_0(\mathbb{F}_{q^d})$. It is clear that:

$$N_d(C_0) = \sum_{x \in |C_0|, \deg(x) | d} \deg(x)$$

Therefore, we can rewrite:

$$\begin{aligned} Z_{C_0}(t) = \exp(\sum_{x \in |C_0|} \sum_{a \geq 1} \frac{t^{\deg a} \cdot a}{a}) &= \prod_{x \in |C_0|} \frac{1}{1 - t^{\deg x}} \\ = \prod_{x \in |C_0|} \frac{1}{1 - (q^{\deg(x)})^{-s}} &= \prod_{x \in |C_0|} \frac{1}{1 - (\#k(x))^{-s}} \end{aligned}$$

Remark. *Note this definition is parallel to the classical Riemann zeta function:*

$$\zeta(s) = \prod_{p \in \mathrm{Spec}\ \mathbb{Z}} \frac{1}{1 - p^{-s}}$$

Theorem 3.28 (Weil)**.** *Let* $C_0/\mathbb{F}_q$ *be a smooth projective curve. Then:*

1. $Z_{C_0}(t)$ *is a rational funciton. In fact it looks like:*

$$\frac{f(t)}{(1-t)(1-qt)}$$

   for f *a polynomial of degree* $2g$.

2. *Fnctional equation:*

$$Z_{C_0}(q^{-1}t^{-1}) = q^{1-g}t^{2-2g} Z_{C_0}(t)$$

3. *Riemann Hypothesis. If we write:*

$$f(t) = \prod_{i=1}^{2g} (1 - \alpha_i)$$

   then $|\alpha_i| = q^{\frac{1}{2}}$.

Remark. *The first statement is more or less Riemann-Roch; the second statement is Serre duality; the third statement is Hasse-Weil bound.*

Deligne generalizes this to higher dimension. Let $X_0/\mathbb{F}_q$ be a smooth projective variety. Then we can still define the zeta function as before, then the zeta function will take form:

$$Z_{X_0}(t) = \frac{P_1(t)P_3(t)\cdots P_{2d-1}(t)}{P_0(t)P_2(t)\cdots P_{2d}(t)}$$

then $P_i(t)$ has degree the i-th Betti number of X_0, and its root having magnitude $q^{\frac{i}{2}}$. There's also a version of functional equation.

4 Homeworks

4.1 Homework 1

Exercise 4.1 (Hartshorne Ex II.4.2)**.** *Given examples when there are two morphisms $f, g : X \to Y$ between two S-schemes which agree on an open dense subset of X, but they don't agree globally in the following cases:*

1. *Y is separated.*
2. *X is reduced.*

Proof. We always set $S = \text{Spec } k$. For the first example, we need to construct a nonreduced X. Consider:

$$X = \text{Spec } A = \text{Spec } k[\varepsilon, x]/(\varepsilon^2, \varepsilon x), \quad Y = \text{Spec } B = \text{Spec } k[t]$$

Note that X has an open dense subset $U = \text{Spec } A[\frac{1}{x}] \simeq \text{Spec } k[x, \frac{1}{x}]$. Now let's consider f, g to be defined by ring maps $B \to A$ given by: $t \mapsto 0, t \mapsto \varepsilon$ respectively. So that $f \neq g$. But f, g both restrict to the zero map:

$$\text{Spec } k[x, \frac{1}{x}] \to \text{Spec } k[t]$$

This satisfies the requirement. For the second example, let's take our classical example of nonseparated schemes:

$$Y = (\mathbb{A}^1 \coprod \mathbb{A}^1)/\sim$$

be the affine line with two origins. We pick $X = \mathbb{A}^1$. Then consider f, g be the morphism into the first copy and second copy of $\mathbb{A}^1$ in Y respectively. So that $f \neq g$. But if $U = \mathbb{A}^1 \backslash \{0\}$, then $f|_U = g|_U$, we're done.

□

Exercise 4.2. *We work over an algebraically closed field k. Let C be the curve defined by $y^2 = x^2(x+1)$ in $\mathbb{A}^2$ Consider the map:*

$$f : U = \mathbb{A}^1 \backslash \{1\} \to C \subset \mathbb{A}^2$$

given by

$$a \mapsto (a^2 - 1, a(a^2 - 1))$$

Wrtie down the corresponding ring maps and show that f is bijection on closed points. Moreover, show f is not universally closed.

Proof. The corresponding ring map is:

$$\varphi : k[x, y]/(y^2 - x^2(x+1)) \to k[t, \frac{1}{t-1}]$$

$$x \mapsto t^2 - 1, y \mapsto t(t^2 - 1)$$

To check the map is surjective on the level of closed points, we pick any $a \in k\backslash\{1\}$, and compute:

$$\varphi^{-1}((t-a)) = (x-(a^2-1), y-(a(a^2-1))) \in \text{Specmax } k[x,y]/(y^2-x^2(x+1))$$

Hence surjective. To check injectivity, Hilbert Nullstellensatz means we only need to show if: for $a, b \in k\backslash\{1\}$

$$a^2-1 = b^2-1, \quad a(a^2-1) = b(b^2-1) \implies a = b$$

but this is clear since $a, b \neq 1$. It remains to check f is not universally closed. Consider the following base change:

$$\begin{array}{ccc} U \times_C \mathbb{A}^1 & \xrightarrow{\tilde{f}} & \mathbb{A}^1 = \text{Spec } k[s] \\ \downarrow & & \downarrow{\scriptstyle s \mapsto (s^2-1, s(s^2-1))} \\ U & \xrightarrow{f} & C \end{array}$$

Note that $U \times_C \mathbb{A}^1$ is affine scheme given by ring

$$k[t, \frac{1}{t-1}] \otimes_{\mathcal{O}(C)} k[s] \simeq k[t, s, \frac{1}{t-1}]/(t^2-1 = s^2-1, t(t^2-1) = s(s^2-1))$$

This is the subscheme of $\mathbb{A}^2$ defined by:

$$\{(u,v) \in \mathbb{A}^2 \mid u = v \neq 1 \text{ or } u = -v = -1\} = \{(u,v) \in \mathbb{A}^2 \mid u = v \neq 1\} \cup \{(-1,1)\}$$

Therefore the image of the first component (which is closed in $U \times_C \mathbb{A}^1$) in $\mathbb{A}^1$ is $\mathbb{A}^1\backslash\{1\}$, which is not closed. □

Exercise 4.3 (Hartshorne Exercise II.3.16)**.** *Let X be a noetherian topological space, and let $\mathcal{P}$ be a property of closed subsets of X. Assume for any closed subset Y of X, if $\mathcal{P}$ holds for every proper closed subset of Y, then $\mathcal{P}$ holds for Y. Then $\mathcal{P}$ holds for X.*

Proof. Assume for contradiction $\mathcal{P}$ fails for $V_0 = X$. Then by assumption we can find a proper closed subset $V_1 \subsetneq X$ such that $\mathcal{P}$ fails for V_1. Similarly, we can find a proper subset $V_2 \subseteq V_1$ such that $\mathcal{P}$ fails. Continue in this way we get a infinity descending chain of closed subsets:

$$X = V_0 \supset V_1 \supset V_2 \supset \cdots$$

Taking complement, we get an infinite ascending chain of open subsets, contradicting the noetherian hypothesis. □

Exercise 4.4 (Harthsorne Exercise II.3.18(a))**.** *We call a topological space Zariski space if it is noetherian and every nonempty closed irreducible subset has a unique generic point. Let X be a Zariski space. A constructible subset of X is a subset belongs to the smallest family $\mathfrak{F}$ such that:*

1. *Every open subset is constructible.*

2. *It is closed under finite intersection and complement.*

Prove every constructible set can be written as finite disjoint union of locally closed subsets.

Proof. It is clear that finite disjoint union of locally closed subsets should be constructible. Hence it suffices to show that the collection of all such that satisfies the two conditions above. The first condition is trivial. We only need to check it's closed under finite intersection and complement.

First, it's clear that finite intersection of locally closed sets remain locally closed. So the same property holds for disjoint unions of such. Now let's deal with complement. Let $Y = \coprod_{i\in I}(U_i \cap Z_i)$ be such a subset, with each U_i open and Z_i closed. Note that:

$$Y^c = \bigcap (U_i \cap Z_i)^c = \bigcap ((U_i^c \cap Z_i^c) \sqcup (U_i^c \cap Z_i) \sqcup (U_i \cap Z_i^c))$$

Denote the three sets inside the rightmost bracket A_i^1, A_i^2, A_i^3, and they're all locally closed by definition. But then:

$$Y^c = \coprod_{I_1,I_2,I_3 \text{ partition } I} (\cap_{i_1\in I_1} A_{i_1}^1) \cap (\cap_{i_1\in I_2} A_{i_2}^2) \cap (\cap_{i_3\in I_3} A_{i_3}^3)$$

is disjoint union of locally closed sets. □

Exercise 4.5 (Hartshorne Exercise II.3.19)**.** *Prove Chevalley's theorem: Let $f : X \to Y$ be a morphism of finite type of noetherian schemes, then the image of a constructible set is constructible.*

Proof. Follow the hint in the book.

(a) Let's start with some reductions. Let $Z \subset X$ be a constructible subset, note that we can always endow Z a scheme structure, by presenting Z as disjoint union of open subschemes in (say reduced) closed subschemes of X. The restriction: $f|_Z \colon Z \to Y$ is again a morphism of noetherian schemes of finite type. So we may assume $Z = X$ and show $f(X)$ is constructible.

Note that a subset of X (resp. Y) is constructible iff its intersection with each irreducible component of X (resp. Y) is constructible, hence we may assume X, Y are both irreducible. Moreover, since the setting is topological we may also assume they're reduced. So we may assume X, Y are integral.

Now choose an affine open cover U_i of Y, and let $f^{-1}(U_i) = \cup V_{ij}$ be affine open cover of U_i, then:

$$f(X) = \bigcup f(V_{ij})$$

So it will be constructible if we can prove $f(V_{ij})$ is constructible in U_i (and hence constructible in Y). So in fact we only care about the map between affine schemes: $V_{ij} \to U_i$. We're thus free to assume $f : X = \operatorname{Spec} B \to Y = \operatorname{Spec} A$ is a morphism

between affine schemes. Now say the corresponding ring map is $\varphi : A \to B$, this map factors as $A \to A/\ker(\varphi) \hookrightarrow B$, on the level of schemes we get a map:

$$X = \text{Spec } (B) \to Z = \text{Spec } A/\ker \varphi \hookrightarrow Y = \text{Spec } A$$

for a closed subscheme Z of Y. So that it suffices to show the image of X in Z is constructible. So we may replace Y by Z and assume $X \to Y = Z$ is dominant.

(b) Follow the hint, we need to prove the following commutative algebra result.

Lemma. *Let $A \hookrightarrow B$ be inclusion of noetherian integral domains, with B finite type over A. Then for each $0 \neq b \in B$, we can find $0 \neq a \in A$ such that: for any algebraically closed field K and $\varphi : A \to K$ with $\varphi(a) \neq 0$, we can find an extension $\varphi' : B \to K$ with $\varphi'(b) \neq 0$.*

First observe that we can drop the requirement $\varphi'(b) = 0$ since this is equivalent to ask for an extension of φ to a map $B[\frac{1}{b}] \to K$, and $B[\frac{1}{b}]$ is still finitely type over A. We prove the lemma by induction on the number of generators of B/A. Suppose B/A is generated by one element, so that $B \simeq A[x]/I$ for some ideal I. Now localize at $S = A\backslash 0$, we get a ring extension:

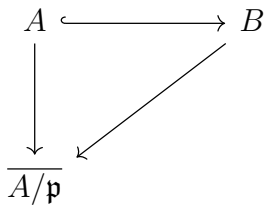
$$k \hookrightarrow k[x]/(k[x] \cdot I), \quad k = \text{Frac}(A)$$

There are two possibilities. If $k[x] \cdot I = 0$, then $I = 0$ so that $B = A[x]$. In this case we can just pick $a = 1$ and the extension problem is trivial (we can send x to any element). So we may assume $k[x] \cdot I \neq 0 \implies k[x] \cdot I = (f)$ for some $f \in k[x]$. Now let a be the product of all denominators and numerators appearing in the coefficients of some representative of f, so that $f \in A[\frac{1}{a}][x]$ and therefore:

$$B[\frac{1}{a}] \simeq A[\frac{1}{a}][x]/(f)$$

Now given any morphism $\varphi : A \to K$ with $\varphi(a) \neq 0$, this is the same data as a morphism $\varphi : A[\frac{1}{a}] \to K$ (we abuse the notation here). Now $\varphi(f) \in K[x]$ is nonzero by construction. We can thus sending $x \in B[\frac{1}{a}] \simeq A[\frac{1}{a}][x]/(f)$ to any root of $\varphi(f)$ in K, and this defines a map $B[\frac{1}{a}] \to K$. This proves the case for one generator. In general suppose $A \hookrightarrow B$ can be generated by n generators. We can pick C such that $A \hookrightarrow C \hookrightarrow B$ with C/A generated by $n-1$ generators and 1 generators respectively. So the lemma follows by induction hypothesis applies to C/A and B/C.

With the lemma, we can show for $f : X = \text{Spec } B \to Y = \text{Spec } A$, $f(X)$ contains a nonempty open subset of Y. In fact, in the setting of the above lemma, we pick $b = 1$ and pick corresponding a. We will see $f(X) \supset D(a)$, namely for each $\mathfrak{p} \in D(a) \subset \text{Spec } A$, we can find $\mathfrak{P} \in \text{Spec } B$ such that $\mathfrak{P} \cap A = \mathfrak{p}$. Consider the map $A \to K = \overline{A/\mathfrak{p}}$, we're thus in the situation:



Since $\overline{A/\mathfrak{p}}$ is a field, so that the map $B \to \overline{A/\mathfrak{p}}$ must factor through a domain $B/\mathfrak{P}$ for some prime ideal $\mathfrak{P}$. Moreover the commutative of the diagram ensures $\mathfrak{P}$ lies above $\mathfrak{p}$. We're done.

(c) Let's use noetherian induction on Y with respect to the property: for closed $Z \subset Y$, $f(X) \cap Z$ is constructible. Let's check the condition for noetherian induction holds. Let $Z = \text{Spec } A/I$ be a closed subscheme, then f restricts to

$$f|_{f^{-1}(Z)}\colon \text{Spec } B/I \cdot B \to \text{Spec } A/I$$

This still a finite type morphism between noetherian schemes, and its image equals $f(X) \cap Z$. Restricts to some irreducible components, we can apply the previous part and see that $f(X) \cap Z$ contains an open subset U of Spec A/I. Now that $Z \backslash U$ is proper closed subset, by noetherian induciton hypothesis we may assume $f(X) \cap (Z \backslash U)$ is constructible. It follows that

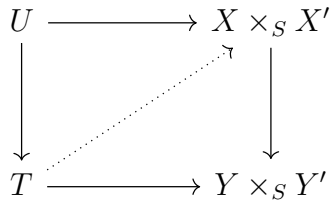
$$f(X) \cap Z = U \cup (f(X) \cap (Z \backslash U))$$

is constructible. We therefore check the noetherian induction hypothesis and therefore the property holds for Y, namely $f(X)$ is constructible in Y.

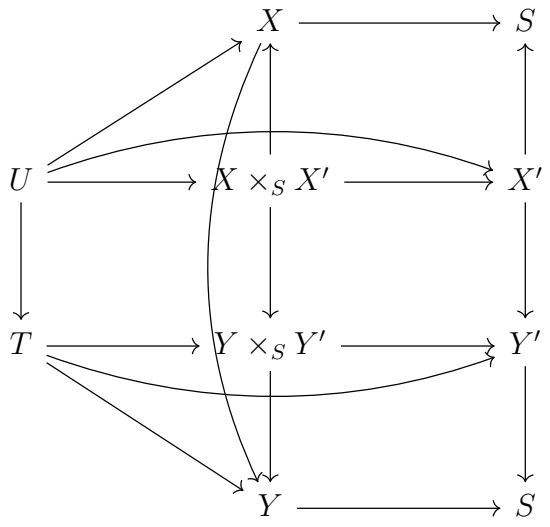
□

Exercise 4.6. *Use Valuation criterion to show product of proper morphisms is proper.*

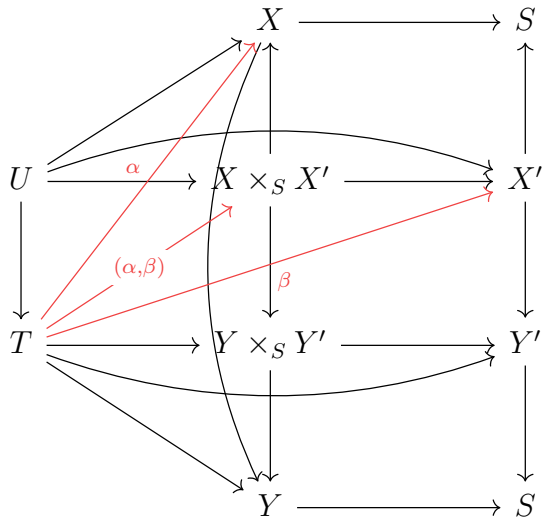
Proof. Let $f : X \to Y, g : X' \to Y'$ be proper morphisms over S. We wish to show $f \times g : X \times_S X' \to Y \times_S Y'$ is proper. Note it is clearly finite type, so it suffices to check the valuation criterion. The remaining is just a diagram chase. We are asking the lifting problem:



This extends to a diagram:



The quadrilateral $UXYT$ equips me with a unique lift $\alpha : T \to X$, the quadrilateral $UX'Y'T$ equips me with a unique lift $\beta : T \to X'$. And (α, β) defines the unique map $T \to X \times_S X'$.



We're done. □

4.2 Homework 2

Exercise 4.7. *Let $S_\bullet = k[x_0, \cdots, x_n]$ as in class, and take $X = \mathbb{P}^n_k$ by gluing together affine schemes U_i. Given a homogenous polynomial $f \in S_d$ with $d > 0$, we can take the open subscheme $D_+(f) \subset X$, which is also defined by gluing together open subschemes of U_i. Show that $D_+(f)$ is affine and given by* $\operatorname{Spec} S[\frac{1}{f}]_{\deg=0}$.

Proof. Recall that U_i is defined by:

$$U_i = \operatorname{Spec}\ k[\frac{x_0}{x_i}, \frac{x_1}{x_i}, \cdots, \frac{x_n}{x_i}]$$

and the open subscheme $U_i \cap D_+(f)$ is given by:

$$U_i \cap D_+(f) = \operatorname{Spec}\ k[\frac{x_0}{x_i}, \frac{x_1}{x_i}, \cdots, \frac{x_n}{x_i}][\frac{x_i^d}{f}]$$

Now let $V = \operatorname{Spec}\ S[\frac{1}{f}]_{\deg=0}$. Now that: V has open subschemes:

$$V_i = \operatorname{Spec}\ (S[\frac{1}{f}]_{\deg=0})[\frac{f}{x_i^d}] \simeq \operatorname{Spec}\ (S[\frac{1}{f}, \frac{1}{x_i}])_{\deg=0} \simeq U_i \cap D_+(f)$$

Now that (V_i) covers V, since if for some $\mathfrak{p} \in \operatorname{Spec}\ (S[\frac{1}{f}]_{\deg=0})$ containing each $\frac{x_i^d}{f}$, it follows that for each monomial $x_0^{r_0} \cdots x_n^{r_n}$ of degree d, we have:

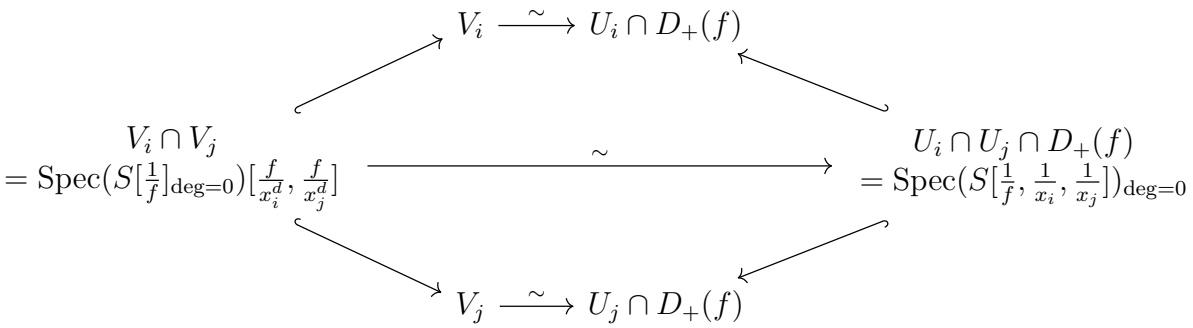
$$(\frac{x_0^{r_0} \cdots x_n^{r_n}}{f})^d = (\frac{x_0^d}{f})^{r_0} \cdots (\frac{x_n^d}{f})^{r_n} \in \mathfrak{p}$$

It follows that $\frac{x_0^{r_0} \cdots x_n^{r_n}}{f} \in \mathfrak{p}$ since $\mathfrak{p}$ is prime. It then forces $1 \in \mathfrak{p}$, contradiction.

Now that we have a bunch of maps:

$$V_i \simeq U_i \cap D_+(f) \hookrightarrow D_+(f)$$

It remains to check these maps glue (so that it gives an isomorphsim $V \to D_+(f)$). The gluing condition can be summarized by the commutativity of the following diagram:



□

Exercise 4.8. *Given an example of $f : X \to Y$ and a coherent sheaf E on X such that $f_*(E)$ is not coherent on Y. If f is finite morphism, show that $f_*(E)$ is always coherent.*

Proof. First consider open immersion:

$$f : X = \text{Spec } k[x, \frac{1}{x}] \hookrightarrow Y = \text{Spec } k[x]$$

and pick $E = \mathcal{O}_X$, it follows that $f_*(E)$ corresponds to the $k[x]$-module $k[x, \frac{1}{x}]$, which is obviously not finitely generated. So that $f_*(E)$ is not coherent as desired.

Now assume $f : X \to Y$ is finite, in particular it is affine. Now the question is local with respect to the target, and the affineness of the morphism enables us to reduce the problem to the case of $f : \text{Spec } B \to \text{Spec } A$ for B a finite A-algebra. Now a coherent $\mathcal{O}_{\text{Spec } B}$-module E corresponds a finite B-module M, and then $f_*(E)$ is just viewing M as an A-module. But since M/B is finite, B/A is finite, we know M/A is finite. So that $f_*(E)$ is cohrent. □

Exercise 4.9. *Consider a scheme X and a coherent sheaf E on X. Given a point $x \in X$, consider:*

$$\varphi(x) = \dim E_x \otimes \mathcal{O}_{X,x}\kappa(x)$$

the stalk dimension.

1. *Use Nakayama's lemma to show that $Z_n = \{x \mid \varphi(x) \geq n\}$ is Zariski-closed.*
2. *If X is reduced, and $\varphi(x)$ is constant on X. Show that E is locally free.*
3. *Given an example of X nonreduced where $\varphi(x)$ is constant but E is not locally free.*

Proof. 1. The question is local. So we may assume $X = \text{Spec } A$ and $E = \tilde{M}$ for some finite A-module M. We are asked to show:

$$Z_n = \{\mathfrak{p} \mid \dim_{\kappa(\mathfrak{p})} M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} \geq n\}$$

is closed. By Nakayama, we know:

$$\mathfrak{p} \in Z_n \iff \nexists\, m_1, \cdots, m_{n-1} \text{ generate } M_\mathfrak{p} \text{ as a } A_\mathfrak{p}\text{-module}$$

Therefore if $\mathfrak{p} \notin Z_n$, so that we can find $m_1, \cdots, m_{n-1}$ generates. Pick a collection of generator $s_1, \cdots, s_m$ for M, we know there exists $a_{ij} \in A_\mathfrak{p}$ such that:

$$\sum_{i=1}^{n-1} a_{ij} m_i = s_j$$

It follows that for each $\mathfrak{q} \in D(\prod a_{ij})$, we have $m_1, \cdots, m_{n-1}$ generate $M_\mathfrak{q}$. It follows that $D(\prod a_{ij}) \cap Z_n = \emptyset$. This shows Z_n is closed.

2. Again we may assume $X = \text{Spec } A$ is affine, and $E = \tilde{M}$. Let d be the commmon value of $\varphi(x)$. Suppose first there exists a surjection

$$\pi : A^d \twoheadrightarrow M$$

By hypothesis we know for all $\mathfrak{p} \in \text{Spec } A$:

$$\pi \otimes \kappa(\mathfrak{p}) : \kappa(\mathfrak{p})^d \simeq M \otimes \kappa(\mathfrak{p})$$

So that for all $\mathfrak{p}$:

$$(A/\mathfrak{p})^d \hookrightarrow M/\mathfrak{p}M$$

Therefore $\ker \pi \subset \bigcap \mathfrak{p}^d = 0$ since A is reduced. So in this case $\pi : A^d \simeq M$ is free, we're done. In general, we need to show for each maximal ideal $\mathfrak{m}$, we can find f with $\mathfrak{m} \in D(f)$, so that M_f is a free A_f-module. By hypothesis we can pick $m_1, \cdots, m_d \in M$ such that:

$$M \otimes_A \kappa(\mathfrak{m}) = \bigoplus_{i=1}^{d} m_i \otimes 1$$

Now put $M' = \sum Am_i \subset M$. So that we know $M' + \mathfrak{m}M = M$. By Nakayama, there exists $f \in 1 + \mathfrak{m}$ such that $f(M/M') = 0$. Therefore these m_i induces a surjection:

$$A_f^d \to M_f$$

This is forced to be an isomrphism from the first part.

3. Consider $X = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. Note topologically X is singleton, so that φ is tautological constant. But consider $k[\varepsilon]/(\varepsilon^2)$-module $k = k[\varepsilon]/(\varepsilon)$, this is not free and in fact not locally free (again since X is singleton).

□

Exercise 4.10 (Hartshorne Exercise II.5.9, simplified but essentially the same)**.** *Let $S_\bullet = k[x_0, \cdots, x_n]$ be graded ring. Let $M_\bullet$ be a graded $S_\bullet$-module.*

1. *Show there is a natural map:*

$$\alpha : M_\bullet \mapsto \Gamma_\bullet((\widetilde{M_\bullet})_{\deg 0})$$

2. *Assume $M_\bullet$ is finitely generated over $S_\bullet$. Show the map α is an isomorphism for all large degrees.*

3. *With the same hypothesis, we define an equivalence relation $\approx$ on graded $S_\bullet$-modules by saying $M_\bullet \approx M'_\bullet \iff M_{\geq d} \simeq M'_{\geq d}$ for some d. We say M is quasi-finitely generated if it is equivalent to a finitely generated module. Show that the functors $\widetilde{(-)}_{\deg 0}$ and $\Gamma_\bullet$ induces an equivalence of categories:*

$$\{\text{quasi-finitely generated graded } S\text{-mod}\}/\approx \simeq \operatorname{Coh}(\mathbb{P}^n)$$

Proof. 1. Given $m \in M_a$, we need to specify an elment:

$$\alpha(m) \in \Gamma(\mathbb{P}^n, (\widetilde{M_\bullet})_{\deg 0} \otimes \mathcal{O}(a)) = \Gamma(\mathbb{P}^n, (\widetilde{M_\bullet}[a])_{\deg 0})$$

On each U_i we have:

$$\Gamma(U_i, (\widetilde{M_\bullet}[a])_{\deg 0}) = M_a[\frac{1}{x_i}]$$

therefore the only reasonable choice is:

$$\alpha(m)|_{U_i} = m \in M_a[\frac{1}{x_i}]$$

These $\alpha(m)|_{U_i}$obviously glue and get an elment $\alpha(m) \in \Gamma(\mathbb{P}^n, (\widetilde{M_\bullet}[a])_{\deg 0})$.

2. Assume $\alpha(m) = 0$, so that $\alpha(m)|_{U_i} = 0 \implies m = 0 \in M_a[\frac{1}{x_i}] \implies x_i^{N_i} m = 0 \in M_{a+N_i}$. This is saying that m is a nilpotent element, i.e. $S_\bullet \cdot m$ has only finitely many nonzero graded pieces. Now let $N_\bullet \subset M_\bullet$ be the graded-submodule consisting of nilpotent elements. Under noetherian hypothesis we know $N_\bullet$ is also finitely generated, say by $n_1, \cdots, n_k$. But then this implies $N_\bullet$ only has finitely many nonzero graded pieces by looking at each individual n_i and take maximum. Now assume $N_t = 0$ for all $t > T$. Then we know for each $a > T$, we have α_a is injective.

   Now let's assess surjectivity. We know:

$$\Gamma(\mathbb{P}^n, (\widetilde{M_\bullet}[a])_{\deg 0}) = \bigcap_{i=0}^{n} M_a[\frac{1}{x_i}]$$

   It suffices to prove that the natural map:

$$M_\bullet \to \bigcap_{i=0}^{n} M_\bullet[\frac{1}{x_i}]$$

   is an surjective. Indeed, by picking a set of generators we have the following diagram:

$$\begin{array}{ccc} S_\bullet^{\oplus N} & \xrightarrow{f} & \bigcap S_\bullet^{\oplus N}[1/x_i] = (\bigcap S_\bullet[1/x_i])^{\oplus N} \\ \downarrow & & \downarrow \\ M_\bullet & \longrightarrow & \bigcap M_\bullet[1/x_i] \end{array}$$

   It suffices to prove the top map f is surjective. But we've shown this during the lecture (for the case $N = 1$).

3. The statement is now obvious since α is an honest natural isomorphism in the category of quasi-finite generated graded S-modules modulo the equivalence relation. The functor of the other direction being an natural isomorphism was checked during class. □

Exercise 4.11 (Hartshorne Exercise II.5.17 (c),(d))**.**

1. *Let Y be a scheme and $\mathcal{A}$ is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebra. Show that there is a unique scheme X together with an affine morphism $f : X \to Y$ such that for each affine open $V \hookrightarrow Y$, we have $f^{-1}(V) \simeq \operatorname{Spec} \mathcal{A}(V)$. Moreover, these isomorphisms are coherent in the sense that for $U \hookrightarrow V$ affine opens, the induced map:*

$$f^{-1}(U) \hookrightarrow f^{-1}(V)$$

   are given by restriction maps:

$$\mathcal{A}(V) \to \mathcal{A}(U)$$

   The scheme X is called **Spec** $\mathcal{A}$

2. *If $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_Y$-algebra, then:*

$$f : X = \mathbf{Spec}\ \mathcal{A} \to Y$$

is affine morphism, and $\mathcal{A} = f_*\mathcal{O}_X$. Conversely, if $f : X \to Y$ is affine, then $f_*\mathcal{O}_X$ is quasi-coherent sheaf of $\mathcal{O}_Y$ algebra, and $X \simeq \mathbf{Spec}\ \mathcal{A}$.

Proof. 1. First we claim that for every affine opens: $U \hookrightarrow V$, the map Spec $\mathcal{A}(U) \to$ Spec $\mathcal{A}(V)$ induced by restriction map: $\mathcal{A}(V) \to \mathcal{A}(U)$ is open. Indeed we have the following commutativediagram:

$$\begin{array}{ccc} \mathcal{A}(V) & \longrightarrow & \mathcal{A}(U) \\ \uparrow & & \uparrow \\ \mathcal{O}(V) & \longrightarrow & \mathcal{O}(U) \end{array}$$

The condition that $\mathcal{A}$ is quasi-coherent implies that:

$$\mathcal{A}(U) \simeq \mathcal{A}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$$

Namely we have a Cartesian diagram of affine schemes:

$$\begin{array}{ccc} \text{Spec } \mathcal{A}(V) & \longleftarrow & \text{Spec } \mathcal{A}(U) \\ \downarrow & & \downarrow \\ V & \hookleftarrow & U \end{array}$$

Now the top map is open since it is base changed from an open map. This proves our claim. Now let's pick an open cover U_i for Y, then we have diagrams like:

$$\begin{array}{ccccc} \text{Spec } \mathcal{A}(U_i) & & & & \\ & \nwarrow & & & \\ & & \text{Spec } \mathcal{A}(U_i \cap U_j) & & \\ & & & \swarrow & \\ & & & & \text{Spec } \mathcal{A}(U_j) \end{array}$$

where each arrow is an open embedding. Thus we're allowed to glue these schemes Spec $\mathcal{A}(U_i)$ to get a scheme $\mathbf{Spec}\ \mathcal{A}$ satisfying the required condition. Now assume $f : X \to Y$ is another Y-scheme satisfying the desired condition. Note we're equiped with a map:

$$\varphi_i : \text{Spec } \mathcal{A}(U_i) \to f^{-1}(U_i)$$

that is an isomorphism. Note these maps φ_i glue to get a map $\varphi : \mathbf{Spec}\ \mathcal{A} \to X$. This is an isomorphism since it is locally given by ring isomorphisms and is a bijection (on the level of points) by construction. This shows the uniqueness of $\mathbf{Spec}\ \mathcal{A}$.

2. Now assume $X = \mathbf{Spec}\ \mathcal{A}$. The fact that $f : X \to Y$ is affine is tautological. Moreover, we see that for each open affine $U \subset Y$:

$$f_*\mathcal{O}_X(U) = \mathcal{O}_X(f^{-1}(U)) = \mathcal{A}(U)$$

therefore we checked $f_*\mathcal{O}_X = \mathcal{A}$. For the converse statement, we only need to check that X satisifies:

(a) For each affine open V, $f^{-1}(V) = \text{Spec } f_*\mathcal{O}_X(V)$. But this is trivial since $f^{-1}(V)$ is affine.

(b) For $U \hookrightarrow V$, the map $f^{-1}(U) \hookrightarrow f^{-1}(V)$ corresponds to restriction:

$$f_*\mathcal{O}_X(V) \to f_*\mathcal{O}_X(U) \iff \mathcal{O}_X(f^{-1}(V)) \to \mathcal{O}_X(f^{-1}(U)) \iff f^{-1}(U) \hookrightarrow f^{-1}(V)$$

again by the affineness of f.

Therefore the uniqueness of $\mathbf{Spec}\ \mathcal{A}$ ensures it must equal X.

□

Exercise 4.12 (Hartshorne II.5.18)**.** *Let Y be a scheme. A geometric vector bundle of rank n over Y is a scheme X and a morphism $f : X \to Y$ together with trivializaitons :*

$$\psi_i : f^{-1}(U_i) \to \mathbb{A}^n_{U_i}$$

for U_i an open covering of Y, such that the transition maps:

$$\psi_j \circ \psi_i^{-1} : \mathbb{A}^n_V \to \mathbb{A}^n_V$$

are **linear***, for $V = \text{Spec } A \subset U_i \cap U_j$. By linear we mean the corresponding ring map $A[x_1, \cdots, x_n] \to A[x_1, \cdots, x_n]$ is A-linear. It is clear what should be the definition of isomorphism of geometric vector bundles.*

1. *Let $\mathcal{E}$ be locally free sheaf of rank n on Y. Denote $S(\mathcal{E})$ the symmetric algebra on $\mathcal{E}$, and let $X = \mathbf{Spec}\ S(\mathcal{E})$. Check the map $f : X \to Y$ is a vector bundle over Y. We denote this vector bundle by $\mathbf{V}(\mathcal{E})$.*

2. *For any morphism $f : X \to Y$, we can talk about the sheaf of sections $\mathcal{S}(X/Y)$. Show that if $f : X \to Y$ is a vector bundle of rank n, then the sheaf of sections $\mathcal{S}(X/Y)$ has a natural structure of $\mathcal{O}_Y$-module, making it a locally free $\mathcal{O}_Y$-module of rank n.*

3. *Again let $\mathcal{E}$ be a locally free sheaf of rank n on Y, let $X = \mathbf{V}(\mathcal{E})$ and let $\mathcal{S} = \mathcal{S}(X/Y)$. Show that $\mathcal{S} \simeq \mathcal{E}^\vee$*

This we establish a one-to-one dictonary between the language of geometric vector bundle and locally free sheaves.

Proof. 1. For any affine open U such that it trivializes $\mathcal{E}$, we have an isomorphism:

$$\mathcal{E}|_U \simeq \mathcal{O}_U^{\oplus n}$$

as $\mathcal{O}_U$-modules. So that this induces a $\mathcal{O}(U)$-algebra isomorphsim:

$$S(\mathcal{E})(U) \simeq \mathrm{Sym}_{\mathcal{O}(U)}(\mathcal{O}(U)^{\oplus n}) = \mathcal{O}(U)[x_1, \cdots, x_n]$$

This translates to a trivialization:

$$\psi_U^{-1} : \mathbb{A}_U^n \simeq \text{Spec } S(\mathcal{E})(U) = f^{-1}(U)$$

Now given an open cover with trivialization (U_i, ψ_i), we must check this is indeed an algebraic bundle. Namely $\psi_j \circ \psi_i^{-1}$ is $\mathcal{O}$-linear, but this is nothing but the statement that the transition function for locally free sheaf is given by an invertible matrix. Finally, we also remark that the vector bundle structure is completely given by the map $f : X \to Y$. In other words, for a space X above Y, whether it is a vector bundle is a property not a structure, and is independent choice of the trivialization.

2. The story is local. Therefore we may restrict to an open affine and assume that:

$$Y = \text{Spec } A, X = \mathbb{A}_Y^n$$

So that $f : X \to Y$ is given by:

$$A \to A[x_1, \cdots, x_n]$$

Now a section of $X \to Y$ is nothing but a A-linear map:

$$A[x_1, \cdots, x_n] \to A$$

which is just an n-tuple of elements in A. So tht we can identify:

$$\mathcal{S}(X/Y)(Y) \simeq A^n$$

as a set, so it carries a natural (free) A-module structure. Moreover, for basic open $V_f = \text{Spec } A[\frac{1}{f}]$ of Y, a similar computation shows:

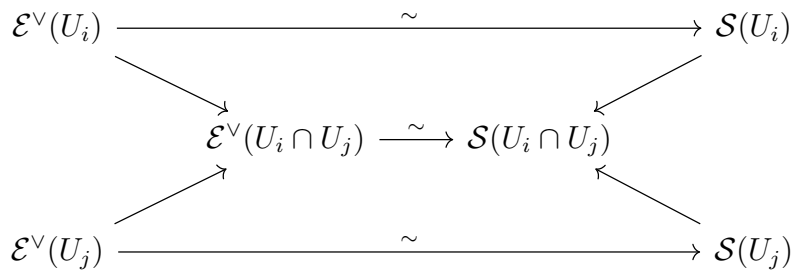
$$\mathcal{S}(X/Y)(V_f) \simeq A[\frac{1}{f}]^n$$

is naturally an $A[\frac{1}{f}]$-module compatible with the restriction maps. This proves $\mathcal{S}(X/Y)$ is actually a **quasi-coherent sheaf**. Now that $\mathcal{S}(X/Y)$ is quasi-coherent on an affine scheme, corresponding to a free module. So that it's actually free.

3. Cover the base Y by trivializing opens affines U_i for $\mathcal{E}$. We see that U_i remains to be trivializing opens for $\mathcal{S}(\mathbf{V}(\mathcal{E})/Y)$ by analyzing the proof of the second part. So that it suffices to present coherent isomorphism:

$$\varphi_i : \mathcal{E}^\vee(U_i) \to \mathcal{S}(U_i)$$

in sense the obvious diagram:



commutes. But this will be evident from our construction. Now let's construct the morphism φ_i. Given $s \in \mathcal{E}^\vee(U_i)$, we can view it as an element:

$$s \in \mathrm{Hom}_{\mathcal{O}\text{-mod}}(\mathcal{E}(U_i), \mathcal{O}(U_i)) \simeq \mathrm{Hom}_{\mathcal{O}\text{-alg}}(S(\mathcal{E})(U_i), \mathcal{O}(U_i))$$

by the universal property of symmetric power. But this is equivalent to a map:

$$\text{Spec } \mathcal{O}(U_i) \to \text{Spec } S(\mathcal{E})(U_i) \iff U_i \to \mathbf{V}(\mathcal{E}|_{U_i})$$

since everything is affine. So we obtain an element $\varphi_i(s) \in \mathcal{S}(X/Y)$. This is clearly a bijection between each step above is an iff. Now This respects the $\mathcal{O}$-module structure since the first identification is $\mathcal{O}$-linear from definition. This exhibits the isomorphism we desired.

□

4.3 Homework 3

Exercise 4.13. *Let $f : X \to Y$ be a morphism of schemes, and $L \in \mathrm{Pic}(Y)$ with a global section $s \in \Gamma(Y, L)$. Show that:*

$$D(f^*(s)) = f^{-1}(D(s))$$

Proof. Cover Y with open $U_i = \mathrm{Spec}\ B_i$ and X with $V_{ij} = \mathrm{Spec}\ A_{ij}$ together with $f : V_{ij} \to U_i$. We may also assume these U_i trivialize the line bundle L. It suffices to show for each U_i:

$$D(f^*(s)) \cap V_{ij} = f^{-1}(D(s)) \cap V_{ij}$$

$$\iff D(f^*(s|_{U_i})) \cap V_{ij} = f^{-1}(D(s) \cap U_i) \cap V_{ij}$$

Therefore we may without loss assume X and Y are both affine, and $L = \mathcal{O}_Y$. Then the problem translate to:

$$f : \mathrm{Spec}\ A \to \mathrm{Spec}\ B \iff \varphi : B \to A$$

Now given $s \in \mathcal{O}_Y(Y) = B$, we need to show,

$$D(f^*(s)) = f^{-1}(D(s))$$

But LHS is simply $\{\mathfrak{q} \in \mathrm{Spec}\ A \mid s \notin \mathfrak{q}\}$, while RHS is:

$$\{\mathfrak{q} \in \mathrm{Spec}\ B \mid s \notin \varphi^{-1}(\mathfrak{q})\}$$

which tautologically equals LHS. □

Exercise 4.14 (Hartshorne II.5.1 (c),(d))**.** *1. Let $\mathcal{E}, \mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. We have:*

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G}))$$

2. (Projection Formula) Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Assume $\mathcal{F}$ is $\mathcal{O}_X$-module and $\mathcal{E}$ is locally free $\mathcal{O}_Y$-module of finite rank, then there is a natural isomorphism:

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \simeq f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{E}$$

Proof. First note that we need to sheafify tensor product sheaf but need not to sheafify Hom sheaf.

1. Given a sheaf morphism:
$$\varphi : \mathcal{F} \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G})$$
For each U open in X, we have:
$$\mathcal{F}(U) \to \mathrm{Hom}(\mathcal{E}(U), \mathcal{G}(U))$$
This gives a map:
$$\mathcal{F}(U) \otimes \mathcal{E}(U) \to \mathcal{G}(U)$$

for each U. Now the compatibility of these morphisms actually induce a morphism of presheaves:

$$\mathcal{E} \otimes_{\text{pre}} \mathcal{F} \to \mathcal{G}$$

This procedure is a bijection by design:

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G})) \simeq \text{Hom}_{\mathcal{O}_X,\text{pre}}(\mathcal{E} \otimes_{\text{pre}} \mathcal{F}, \mathcal{G}) \simeq \text{Hom}_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G})$$

The last isomorphism is the universal property for sheafification.

2. First we assume $\mathcal{E} = \mathcal{O}_Y^n$. Therefore, the projection fomula read:

$$f_*(\mathcal{F} \otimes f^*\mathcal{O}_Y^n) \simeq f_*(\mathcal{F} \otimes \mathcal{O}_X^n) \simeq f_*(\mathcal{F}^n) \simeq f_*(\mathcal{F})^n \simeq f_*(\mathcal{F}) \otimes \mathcal{O}_Y^n$$

Hence is obvious. Now that by choosing a trivializing cover $U_i \subset Y$ for $\mathcal{E}$, we therefore have a bunch of sheaf isomorphisms:

$$f_*(\mathcal{F} \otimes f^*\mathcal{E})|_{U_i} \simeq (f_*\mathcal{F} \otimes \mathcal{E})|_{U_i}$$

Note that $f^*\mathcal{E}$ is trivialized on $f^{-1}(U_i)$. It remains to see these isomorphisms glue. Indeed, since our construction is functorial, for $U_i \cap U_j \neq 0$, the isomorphism on U_i and isomorphism on U_j will descend to the same isomorphism on $U_i \cap U_j$, hence glue. We thus get the global projection formula we desired.

□

Exercise 4.15 (Hartshorne II.5.12)**.** *1. Let X be a scheme over a scheme Y, and let $\mathcal{L}, \mathcal{M}$ be two very ample invertible sheaves on X. Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample.*

2. If $f : X \to Y$ and $g : Y \to Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y, and $\mathcal{M}$ a very ample invertible sheaf on Y relative to Z Show that $\mathcal{L} \otimes f^*\mathcal{M}$ is very ample invertible sheaf on X relative to Z.

Proof. 1. By definition we can find locally closed embeddings:

$$\varphi_{\mathcal{L}} : X \hookrightarrow \mathbb{P}_Y^n, \quad \varphi_{\mathcal{M}} : X \hookrightarrow \mathbb{P}_Y^m$$

such that

$$\mathcal{L} = \varphi_{\mathcal{L}}^*(\mathcal{O}(1)), \quad \mathcal{M} = \varphi_{\mathcal{M}}^*(\mathcal{O}(1))$$

Now we look at Segre embedding for projective space over $\mathbb{Z}$:

$$i : \mathbb{P}_{\mathbb{Z}}^n \times \mathbb{P}_{\mathbb{Z}}^m \hookrightarrow \mathbb{P}_{\mathbb{Z}}^{nm+n+m}$$

Since this embedding has coordinates of homogenous degree $(1, 1)$, we know:

$$i^*(\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^{nm+n+m}}(1)) = \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1) \boxtimes \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^m}(1)$$

Base change to Y, we have:

$$i_Y : \mathbb{P}_Y^n \times \mathbb{P}_Y^m \hookrightarrow \mathbb{P}_Y^{nm+n+m}$$

$$i_Y^*(\mathcal{O}_{\mathbb{P}_Y^{nm+n+m}}(1)) = \mathcal{O}_{\mathbb{P}_Y^n}(1) \boxtimes \mathcal{O}_{\mathbb{P}_Y^m}(1)$$

It follows that:

$$\mathcal{L} \otimes \mathcal{M} = (\varphi_L \boxtimes \varphi_M)^* i_Y^*(\mathcal{O}(1))$$

is very ample. Note:

$$i_Y \circ (\varphi_{\mathcal{L}} \times \varphi_{\mathcal{M}}) : X \to \mathbb{P}_Y^{nm+n+m}$$

is again a locally closed embedding.

2. Very ample $\mathcal{L}, \mathcal{M}$ equips me with locally closed embeddings:

$$\varphi_{\mathcal{L}} : X \to \mathbb{P}_Y^n = \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} Y$$

$$\psi_{\mathcal{M}} : Y \to \mathbb{P}_Z^m = \mathbb{P}_{\mathbb{Z}}^m \times_{\mathbb{Z}} Z$$

Therefore I have locally closed embeding:

$$X \xrightarrow{\varphi_{\mathcal{L}}} \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} Y \xrightarrow{\mathrm{id} \times \psi_{\mathcal{M}}} \mathbb{P}_{\mathbb{Z}}^n \times_{\mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^m \times_{\mathbb{Z}} Z \xrightarrow{i} \mathbb{P}_{\mathbb{Z}}^{nm+n+m} \times_{\mathbb{Z}} Z = \mathbb{P}_Z^{mn+m+n}$$

where the last map is given by Segre embedding. Call the composition λ. Then:

$$\begin{aligned}
\lambda^*(\mathcal{O}(1)) &= \varphi_{\mathcal{L}}^*(\mathrm{id} \times \psi_{\mathcal{M}})^* i^*(\mathcal{O}(1)) \\
&= \varphi_{\mathcal{L}}^*(\mathrm{id} \times \psi_{\mathcal{M}})^*(\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1) \boxtimes \mathcal{O}_{\mathbb{P}_Z^m}(1)) \\
&= \varphi_{\mathcal{L}}^*(\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1) \boxtimes \mathcal{M}) \\
&= \mathcal{L} \otimes f^*\mathcal{M}
\end{aligned}$$

Showing that $\mathcal{L} \otimes f^*\mathcal{M}$ is very ample.

□

Exercise 4.16 (Hartshorne II.6.1)**.** *Assume X is a noetherian integral separated scheme which is regular in codimension* 1*. Show that $X \times \mathbb{P}^n$ has the same property, and:*

$$\mathrm{Cl}(X \times \mathbb{P}^n) \simeq (\mathrm{Cl}X) \times \mathbb{Z}$$

Proof. The facts that $X \times \mathbb{P}^n$ is noetherian, integral separated are all clear. To see it is regular in codimension one, we can just test it locally. Choose any $U = \mathrm{Spec}\ A \subset X$ and $V \simeq \mathbb{A}^n = \mathrm{Spec}\ k[x_1, \cdots, x_n] \subset \mathbb{P}^n$. We need to assess the ring:

$$A[x_1, \cdots, x_n]$$

We need to prove if A is regular in codimension 1, then so is $A[x_1, \cdots, x_n]$. By induction we only need to consider the case for $n = 1$. So we need to classify height one primes in $A[x]$. Let $\mathfrak{q} \in \mathrm{Spec}\ A[x]$ be of height one. There are two possibilities:

1. $\mathfrak{q} \cap A = \mathfrak{p} \neq 0$. Then $\mathfrak{q} = \mathfrak{p} \cdot A[x]$. Look at localization:

$$A[x]_{\mathfrak{q}} \simeq (A_{\mathfrak{p}}[x])_{\mathfrak{p}A_{\mathfrak{p}}[x]}$$

Now if π is the uniformizer for $A_{\mathfrak{p}}$, then it also generate the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}[x]$. Therefore $A[x]_{\mathfrak{q}}$ is a DVR, since its a height one integral local domain with principal maximal ideal, this shows $A[x]$ is regular at $\mathfrak{q}$.

2. $\mathfrak{q} \cap A = 0$. Then:

$$A[x]_{\mathfrak{q}} \simeq K[x]_{\mathfrak{q}K[x]}$$

where $K = \mathrm{Frac}(A)$. Now this is just a PID localized at a prime ideal, which is of course a DVR. So $A[x]$ is regular at $\mathfrak{q}$.

This proves that $X \times \mathbb{P}^n$ is indeed regular in codimension 1. Now compute the class group. Choose $D = X \times \mathbb{P}^{n-1} \subset X \times \mathbb{P}^n$ be a prime divisor. We have exact sequence:

$$\mathbb{Z} \to \mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}(X \times \mathbb{A}^n) \to 0$$

Note that $\mathrm{Cl}(X \times \mathbb{A}^n) \simeq \mathrm{Cl}(X)$, therefore our goal is to prove:

1. The first map is injective.
2. The second map has a splitting.

For the first claim, note that if $nD \simeq 0$, then there exists $f \in K(X \times \mathbb{P}^n)$ such that $n \cdot X \times \mathbb{P}^{n-1} = (f)$. Choose any $x \in X$, and consider:

$$x : \mathrm{Spec}\ \kappa(x) \times \mathbb{P}^n \to X \times \mathbb{P}^n$$

We can pullback f along this map, and it will tell us:

$$(x^* f) = n\mathbb{P}^{n-1}_{\kappa(x)}$$

This contradicts our knowledge for Class group of $\mathbb{P}^n$, hence the first claim follows. Now the second claim is also obvious. Since for each $D \subset X$ a divisor, we can consider:

$$\mathrm{Cl}(X) \to \mathrm{Cl}(X \times \mathbb{P}^n)$$

$$[D] \mapsto [D \times \mathbb{P}^n]$$

Compose with the isomorphism $\mathrm{Cl}(X \times \mathbb{A}^n) \simeq \mathrm{Cl}(X)$ we get the splitting. □

Exercise 4.17. *1. Let R be a noetherian local ring that is a UFD. Show that every minimal nonzero prime ideal is principal.*

2. Same assumption as above, show that R is normal.

3. Same assumption as above, show that:

$$R = \bigcap_{\mathfrak{p} \text{ non-zero, minimal}} R_{\mathfrak{p}}$$

4. Show that if $(A, \mathfrak{m})$ is a DVR, then so is the localization of $A[t]$ at the prime ideal $\mathfrak{m}[t]$.

Proof. 1. Let $\mathfrak{p}$ be a minimal nonzero prime. Choose $x \in \mathfrak{p}$. By looking at the unique factorization of x, we may assume x is irreducible as an element. It follows that $(x) \subset \mathfrak{p}$ is another prime ideal. But the minimality of $\mathfrak{p}$ ensures $(x) = \mathfrak{p}$.

2. Let $t = \frac{x}{y} \in K = \mathrm{Frac}(R)$. Let's assume x, y is coprime and y is a non-unit. is coprime Suppose t is integral over R, then it satisfies an minimal monic polynomial in $R[t]$:

$$t^n + a_{n-1}t^{n-1} + \cdots + a_0 = 0$$

$$\iff x^n + a_{n-1}x^{n-1}y + \cdots + a_0 y^n = 0$$

Modulo y, we see that $y \mid x^n$. Contradition, since we assume x, y are coprime.

3. Let $t \in \bigcap R_{\mathfrak{p}}$. Choose a presentation $t = \frac{x}{y}$ by coprime x, y in R, we see that we must have:

$$y \in \bigcap_{\mathfrak{p}:\mathrm{ht}(\mathfrak{p})=1} R \backslash \mathfrak{p}$$

Now if y is a non-unit, choose any irreducible factor $z \mid y$, then $y \in \mathfrak{p} = (z)$ gives a contradiction. Therefore y is a unit, and $t \in R$ as desired.

4. This was essentially proved in Q4.

□

4.4 Homework 4

Exercise 4.18. *Let X be an integral, separated, regular scheme.*

1. *Given an invertible sheaf L and a rational section s, define the Weil divisor* $\operatorname{div}(s)$ *of zeroes and poles of s, generalizing the case of a rational function.*

2. *Recall for a Weil divisor D, we have:*

$$\mathcal{O}(D) \subset \underline{\mathcal{K}}$$

as a subsheaf. Show that the funciton 1 *defines a global section of $\underline{\mathcal{K}}$ which gives a rational section σ of $\mathcal{O}(D)$.*

3. *Show that* $\operatorname{div}(\sigma) = D$*, in other words, we can recover D from the invertible sheaf and its embedding into $\underline{\mathcal{K}}$.*

4. *Show that if D is effective, then σ is a regular section of $\mathcal{O}(D)$.*

Proof. 1. To define $\operatorname{div}(s)$, we need to assign an integer n_P for each codimension 1 point P. Now let P be one such codimension 1 point. We pick an open affine neighborhood U for P for L so that:

$$\varphi : L|_U \to \mathcal{O}_U$$

This transforms s into $\varphi(s)$ as a rational section on $\mathcal{O}_U$, so that $\varphi(s)$ is just a rational function on U. Therefore it will output an integer n_P at point P. It remains to check it is independent of the choice of our open neighborhood and trivialization. Indeed, suppose I have:

$$\varphi : L|_U \to \mathcal{O}_U$$

and

$$\psi : L|_V \to \mathcal{O}_V$$

be two trivilzations with $P \in U \cap V$. Then we know: $\psi \circ \varphi^{-1} : \mathcal{O}_{U\cap V} \to \mathcal{O}_{U\cap V}$ is given by an element $g \in \mathcal{O}(U \cap V)^\times$. So that:

$$\psi(s|_{U\cap V}) = g\varphi(s|_{U\cap V})$$

This means that $\psi(s), \varphi(s)$ have the same valuation at point P. This shows our expression:

$$\operatorname{div}(s) = \sum n_P P$$

is independent of the choice.

2. The constant function 1 is obviously a section of $\underline{K}$. Now that assume:

$$D = \sum n_P \cdot P$$

for $n_P \neq 0$. We just choose $U = X \setminus \overline{\{P \mid n_P \neq 0\}}$ and 1 will restrict to a regular function on U. The fact that U is nonempty is purely due to dimension reason:

$$\dim \overline{\{P \mid n_P \neq 0\}} \leq \dim X - 1$$

3. Suppose $D = \sum n_P P$ and suppose $\text{div}(\sigma) = \sum m_P P$, we need to show $n_P = m_P$. Indeed, for any P a prime divisor, I can find an open neightborhood U such that:

$$D|_U = (g_P)$$

And therefore:

$$\mathcal{O}_X(D)(U) = \{f \in \underline{\mathcal{K}}(U) \mid (f) + (g_P) \geq 0\}$$

Therefore I have a distinguished trivialization:

$$\varphi : \mathcal{O}_X(D) \to \mathcal{O}_X, f \mapsto f g_P$$

Therefore my image of σ under this trivialization is nothing but $1 \cdot g_P = g_P$. But now it follows from the first part that

$$m_P = \text{val}_P(g_P) = n_P$$

since $D|_U = (g_P)$. We're therefore done.

4. This is clear by contemplating our construction on the second part. Indeed, for any $D = \sum n_P P$, we can always choose:

$$U = X \setminus \overline{\{P \mid n_P < 0\}}$$

this is because $D|_U = \sum_{n_P > 0} n_P P$, and:

$$(1) + D = D \geq 0$$

Now in our special case $U = X$, and we're therefore done.

□

Exercise 4.19 (part of Hartshorne II.6.2)**.** *Let k be algebraically closed field, and $X \subset \mathbb{P}^n$ projective, which is nonsingular in codimension* 1*.*

1. *Let V be an irreducible hypersurface in $\mathbb{P}^n$ which does not contain X, and let Y_i be the irreducible components of $V \cap X$. They all have codimension* 1*. For each i, let f_i be the local equation for V on some open set U_i of $\mathbb{P}^n$ for which $Y_i \cap U_i \neq \emptyset$. Then we define the divisor:*

$$V.X \coloneqq \sum n_i Y_i$$

where $n_i = v_{Y_i}(\overline{f_i})$, where $\overline{f_i}$ is the restriction of f_i to $U_i \cap X$. Show that this induces a well-define group homomorphism:

$$\text{Div}_X(\mathbb{P}^n) \to \text{Div}(X)$$

where the first group is the subgroup of $\text{Div}(\mathbb{P}^n)$ *consisting of elements such that none of whose components contain X.*

2. *If D is a principal divisor on $\mathbb{P}^n$, show that $D.X$ is a principal divisor on X. Thus we get a homomorphism:*

$$\text{Cl}(\mathbb{P}^n) \to \text{Cl}X$$

3. *Show that the map* $\mathrm{Cl}(\mathbb{P}^n) \to \mathrm{Cl}(X)$ *constructed above agrees with pullback of line bundles:*

$$i^*\mathrm{Pic}(\mathbb{P}^n) \to \mathrm{Pic}(X)$$

Proof. 1. Let P be a prime divisor in $\mathbb{P}^n$ not containing X. Then $V_P = \overline{\{P\}}$ defines a hypersurface in $\mathbb{P}^n$, and our desired map is:

$$\mathrm{Div}_X(\mathbb{P}^n) \to \mathrm{Div}(X)$$

$$[P] \mapsto V_P.X$$

This is well-defined since none of the component of V_P contains X by design. This is a group homomorphism purely because we extend linearly. The only ambiguity is that we chose some open subset U_i for Y_i, so that we need to show that the number n_i is independent of the choice of U_i. Now suppose U_i, U_i' both intersects Y_i, without loss we may assume $U_i' \subset U_i$. Now f_i is the local equation defining V on U_i, then f_i is again tautologically the local equation defining V on U_i'. Then:

$$\overline{f_i} = f_i|_{U_i \cap X}, \quad \overline{f_i'} = f_i|_{U_i' \cap X} = \overline{f_i}|_{U_i' \cap X}$$

Namely $\overline{f_i'}$ is simply the restriction of $\overline{f_i}$, so that they have the same valuation at:

$$Y_i \subset U_i' \cap X \subset U_i \cap X$$

Therefore n_i is independent of the choice.

2. Let $f \in k[x_0, \cdots, x_n]$ be an irreducible homogeneous polynomial of degree d. Then $V_f = \{f = 0\}$ defines a hypersurface in $\mathbb{P}^n$. We can view $f \in \mathcal{O}(d)$ as a section of a line bundle. Let $i : X \hookrightarrow \mathbb{P}^n$ be closed embedding. Note that i^*f can be viewed as a section of the line bundle $i^*\mathcal{O}(d)$, so by question one it corresponds to a divisor: $\mathrm{div}(i^*f)$. We claim that $\mathrm{div}(i^*f) = V_f.X$. Indeed, for each $Y \subset X$ a codimension 1 subvariety, the coefficient before the divisor Y in $\mathrm{div}(i^*f)$ is defined to be $n_Y = v_Y(\overline{f})$ for $\overline{f}$ local equation for f in certain open subset intersects Y nontrivially. In particular this is zero for $f \neq 0$ along Y, and otherwise equals our definition of the coefficient beofore Y in $V_f.X$. This proves our claim.

More generally, if $f = f_1^{m_1} \cdots f_n^{m_n}$ has degree d , we can define:

$$V_f = m_1 V_{f_1} + \cdots m_n V_{f_n}$$

as a divisor, and we will have:

$$\mathrm{div}(i^*f) = V_f \cdot X$$

where we again acknowledge $i^*f \in \Gamma(X, i^*\mathcal{O}(d))$. Now any principal divisor will have form:

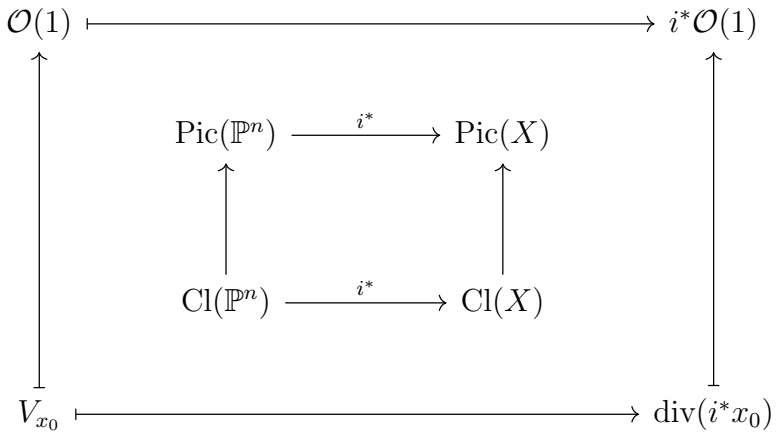
$$D = (f/g)$$

for f, g having same degree d. So that $D = V_f - V_g$ in our previous notation. Then it follows that:

$$D.X = \mathrm{div}(i^* f) - \mathrm{div}(i^* g) = \mathrm{div}(i^*(\frac{f}{g}))$$

where we fix an embedding $\mathcal{O}(d) \hookrightarrow \underline{K}_{\mathbb{P}^n}$ so that we can view $\frac{f}{g}$ as a rational section of the trivial line bundle, and that $D.X$ corresponds to the divisor of the rational section of its pullback, in particular is principal.

3. This follows immediately from the previous part. Recall that $\mathrm{Cl}(\mathbb{P}^n) = \mathbb{Z}$, with a generator V_{x_0} a hyperplane divisor (using the notatoin as in the previous part). We already know V_{x_0} is sent to $\mathrm{div}(i^* x_0) \in \mathrm{Cl}(X)$, and V_{x_0} corresponds line bundle $\mathcal{O}(1)$. Therefore it remains to see $\mathrm{Cl}(X) \to \mathrm{Pic}(X)$ sends $\mathrm{div}(i^* x_0)$ to $i^*\mathcal{O}(1)$. But this is clear since $i^* x_0$ is a section of $i^*\mathcal{O}(1)$.



We're therefore done since V_{x_0} generate $\mathrm{Cl}(\mathbb{P}^n)$.

□

Exercise 4.20 (Hartshonre II.6.6)**.** *Let X be the nonsingular plane cubic curve $y^2 z = x^3 - xz^2$.*

1. *Show that three points of X are collinear iff $P + Q + R = 0$ in the group law of X.*
2. *A point $P \in X$ has order 2 iff the tangent line at P passes through P_0*
3. *A point $P \in X$ has order 3 iff P is an inflection point.*

Proof. 1. Recall we obtain the group law on X via:

$$X(k) \xrightarrow{\sim} \mathrm{Cl}^0(X), \quad P \mapsto [P] - [P_0]$$

and transfering the group law on $\mathrm{Cl}^0(X)$. Let L be the line through P, Q, and L' be the tangent line at P_0. Then $[L] - [L_0] \sim 0$ as divisors in $\mathbb{P}^n$. Now the previous problem tells us we can pullback Weil divisor, so that:

$$[P] + [Q] + [R] - 3[P_0] \sim 0$$

as divisors in X, where R is the third intersection of L with X. This is saying:

$$P + Q + R = 0$$

in the group law of $X(k)$, using the identification above.

2. In fact:
$$P + P = 0 \iff P + P + P_0 = 0$$
since P_0 is the zero element. Now previous question is saying that this precisely happens when the tangent line at P passes through P_0.

3. $3P = 0$ iff there is a line intersecting P with multiplicity 3, which is evidentally equivalent to saying that P is an inflection point.

□

Exercise 4.21 (Hartshorne II.6.10)**.** *Let $K(X) = K(\mathrm{Coh}(X))$ be the Grothendieck group of coherent sheaves on X.*

1. *If $X = \mathbb{A}^1_k$, then $K(X) \simeq \mathbb{Z}$.*

2. *If X is any integral scheme, and $\mathcal{F}$ a coherent sheaf, we define the rank of $\mathcal{F}$ to be:*
$$\dim_K \mathcal{F}_\xi$$
where ξ is the generic point of X, and $K = \mathcal{O}_\xi$ is the function field of X. Show that the rank function defines a surjective homomorphism:
$$\mathrm{rank} : K(X) \to \mathbb{Z}$$

3. *Let $Y \subset X$ be a closed subscheme. Then there is an exact sequence:*
$$K(Y) \to K(X) \to K(X - Y) \to 0$$
where the first map is extension by 0, and the second map is restriction.

Proof. 1. In this case $K(X) = K(k[t]\text{-mod}_{\mathrm{fg}})$. Now the classication of finitely generated modules over PID tells me any finitely generated $k[t]$-module is a direct sum some copies of $k[t]$ and some copies of $k[t]/(f)$. But:
$$0 \to k[t] \xrightarrow{f} k[t] \to k[t]/(f) \to 0$$
So we don't need to care about $k[t]/(f)$. Therefore every element in the K-group takes form $k[t]^{\oplus n}$ for some $n \in \mathbb{Z}$, with the understanding that:
$$k[t]^{\oplus n} = -k[t]^{\oplus -n}$$
for $n < 0$. Now again from classification of modules over PID we see the rank of a module actually factors through its image in the Grothendieck group. So that we see that $k[t]^{\oplus n}$ are pairwise distinct by looking at their ranks.

2. To show the rank functor is well defined, we must show if:

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

then $\text{rank}(\mathcal{F}') + \text{rank}(\mathcal{F}'') = \text{rank}(\mathcal{F})$. Indeed, taking stalk is exact so that:

$$0 \to \mathcal{F}'_\xi \to \mathcal{F}_\xi \to \mathcal{F}''_\xi \to 0$$

and the desired result follows since $\dim_K$ is an additive function. Now to the surjectivity, we note that: $\mathcal{O}_X$ is sent to 1. Therefore 1 is in the image, so must be surjective since rank is a homomorphism.

3. Let $U = X - Y$, with $j : U \hookrightarrow X$. Note that: Exercise II.5.15 tells me:

$$j^* : \text{Coh}(X) \to \text{Coh}(U)$$

is already surjective, hence the exactness at $K(X - Y)$. To show the exactness of the middle, we follow the hint. Let $\mathcal{F}$ be a coherent sheaf with support contained in Y, then we know $\mathcal{F}$ is in fact a $\mathcal{O}_X/\mathcal{I}^n$-module for some n, where $\mathcal{I}$ is the ideal sheaf for $Y \subset X$. Therefore I have filtration:

$$\mathcal{F} \supset \mathcal{I}\mathcal{F} \supset \mathcal{I}^2\mathcal{F}^2 \supset \cdots \mathcal{I}^n\mathcal{F} = 0$$

with each quotient a $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}$-module. Therefore in the Grothendieck group, $\mathcal{F}$ is equivalent to extension by zero of a $\mathcal{O}_Y$-module. Now if $i : Y \to X$, we prove that $i_*K(Y)$ contains all coherent sheaves with support in Y. We have a well-defined map:

$$K(X)/i_*K(Y) \to K(U)$$

given by restriction. We will show this is an isomorphism by constructing an inverse:

$$f : K(U) \to K(X)/i_*K(Y)$$

For $\mathcal{E} \in \text{Coh}(U)$, we know from Exercise II.5.15, there exists $\mathcal{E}' \in K(X)$ such that:

$$\mathcal{E}'|_U = \mathcal{E}$$

Let's try to set:

$$f : \mathcal{E} \mapsto \mathcal{E}'$$

We must show this is well-define. Note first that $\mathcal{E}'|_U = \mathcal{E}$, then the adjunction gives me:

$$\varphi : \mathcal{E}' \to j_*j^*\mathcal{E}' = j_*\mathcal{E}$$

Now that $\ker(\varphi)$ has support in Y, therefore $[\mathcal{E}'] = [\text{im } (\varphi)]$ in $K(X)/i_*K(Y)$. Therfore if $\mathcal{E}'$ and $\mathcal{E}'' \in \text{Coh}(X)$ be such that $\mathcal{E}'|_U \simeq \mathcal{E}''|_U = \mathcal{E}$, then $[\mathcal{E}'] = [\mathcal{E}'']$ in $K(X)/i_*K(Y)$. To see this, we can first replace $\mathcal{E}', \mathcal{E}''$ by certain coherent subsheaf of $j_*\mathcal{E}$, namely their image in id $\Rightarrow j_*j^*$. But then we can consider:

$$\mathcal{E}''' = \mathcal{E}' \cap \mathcal{E}''$$

inside $j_*\mathcal{E}$ and it suffices to show they are all equivalent. Therefore we may assume $\mathcal{E}'' \subset \mathcal{E}'$ is a subsheaf. But then $\mathcal{E}'/\mathcal{E}''$ has support in Y so represents the zero class. So that $[\mathcal{E}'] = [\mathcal{E}'']$ as desired. To summarize, we now obtain a well-defined map:

$$f : \mathrm{Coh}(U) \to K(X)/i_*K(Y)$$

by sending $\mathcal{E}$ to any of its coherent lift. But this map obviously descend to $K(U)$: suppose I have

$$0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$$

in $\mathrm{Coh}(U)$. I pick a lift $\mathcal{F}$ of $\mathcal{E}$ so that $j^*\mathcal{F} = \mathcal{E}$ and $\mathcal{F} \subset j_*\mathcal{E}$. Then I have natural choice for $\mathcal{F}' = \mathcal{F} \cap j_*\mathcal{E}'$, where the intersection is happening inside $j_*\mathcal{E}$ (note that $j_*\mathcal{E}' \subset j_*\mathcal{E}$ since pushforward is left exact), and $\mathcal{F}'' = \mathrm{im}\,(\mathcal{F} \to j_*\mathcal{E} \to j_*\mathcal{E}'')$. I have a well-defined map again by left exactness of j_*:

$$\mathcal{F}/\mathcal{F}' \to \mathcal{F}'' \to 0$$

with

$$\ker \subset \ker(j_*\mathcal{E} \to j_*\mathcal{E}'') \cap \mathcal{F} = j_*(\ker(\mathcal{E} \to \mathcal{E}'')) \cap \mathcal{F} = j_*\mathcal{E}' \cap \mathcal{F} = 0 \in \mathcal{F}/\mathcal{F}'$$

So indeed:

$$\mathcal{F}/\mathcal{F}' \simeq \mathcal{F}''$$

And therefore $f : K(U) \to K(X)/i_*K(Y)$. It is direct to check that this is our desired inverse map.

□

Exercise 4.22. *Let $f : X \to Y$ be finite morphism between smooth projective curves over algebraically closed field k. Show the pullback:*

$$f^* : \mathrm{Cl}(Y) \to \mathrm{Cl}(X)$$

agrees with pullback of invertible sheaves.

Proof. Recall we actually define the pullback on the group of divisors in this setting. So it suffices to show for D a Weil divisor:

$$\mathcal{O}(f^*D) \simeq f^*\mathcal{O}(D)$$

First of all observe that:

$$\mathcal{O}(f^*(D_1 + D_2)) = \mathcal{O}(f^*D_1 + f^*D_2) = \mathcal{O}(f^*D_1) \otimes \mathcal{O}(f^*D_2)$$

and

$$f^*\mathcal{O}(D_1 + D_2) = f^*(\mathcal{O}(D_1) \otimes \mathcal{O}(D_2)) = f^*\mathcal{O}(D_1) \otimes f^*\mathcal{O}(D_2)$$

So that it suffices to prove the above isomorphism for prime divisor $[y]$ for some closed point $y \in Y$. Note that $1 \in \mathcal{O}([p])$ is a regular section, so that $f^*1 = 1 \in f^*\mathcal{O}([p])$ is again regular section. Therefore to show $\mathcal{O}(f^*[y]) \simeq f^*\mathcal{O}([y])$, we only need to show:

$$\mathrm{div}(1) = f^*[y] = \sum_{x \mapsto y} \mathrm{val}_x(f^*(t))[x]$$

where $(t) = \mathfrak{m}_y$, from the first problem. To compute $\operatorname{div}(1)$, we need to find a triviliation of $f^*\mathcal{O}([y])$. Let $U = X \setminus \{y\}$ and V is some affine neighborhood of y, so that U, V cover X with:

$$\mathcal{O}([y])|_U \simeq \mathcal{O}_U, \quad s \mapsto s$$
$$\mathcal{O}([y])|_V \simeq \mathcal{O}_V, s \mapsto st$$

So that:

$$f^*\mathcal{O}([y])|_{f^{-1}U} \to \mathcal{O}_{f^{-1}U}$$

is given by the identity map, and:

$$f^*(\mathcal{O}([y]))|_{f^{-1}V} \to \mathcal{O}_{f^{-1}V}, \quad u \otimes f^*s \mapsto u \otimes f^*s \cdot f^*t$$

for $u \in \mathcal{O}_{f^{-1}V}$. Therefore for the second map 1 is sent to f^*t, and therefore:

$$\operatorname{div}(1) = \sum_{x \mapsto y} \operatorname{val}_x(f^*(t))[x] = f^*[y]$$

as desired. □

4.5 Homework 5

Exercise 4.23. *Let $A \to B$ be a ring morphism, so that we have Kahler differentials $\Omega_{B/A}$ with $d : B \to \Omega_{B/A}$. Given ideal $I \subset B$, let $C = B/I$.*

1. *Show that the restriction of the derivation: $d|_I$: $I \to \Omega_{B/A}$ induces a well-defined map:*

$$\delta : I/I^2 \to C \otimes_B \Omega_{B/A}$$

2. *Use the universal property of Kahler differential to show that $\Omega_{C/A} = \text{coker } (\delta)$.*

Proof. 1. Let $f, g \in I$, we compute:

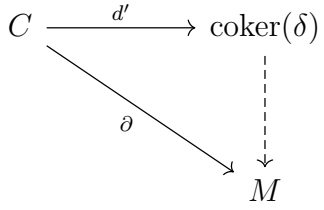
$$\delta(fg) = fdg + gdf = 0 \in C \otimes \Omega_{B/A}$$

because $f, g \in I \implies f, g = 0 \in C = A/I$. Hence δ is well-defined.

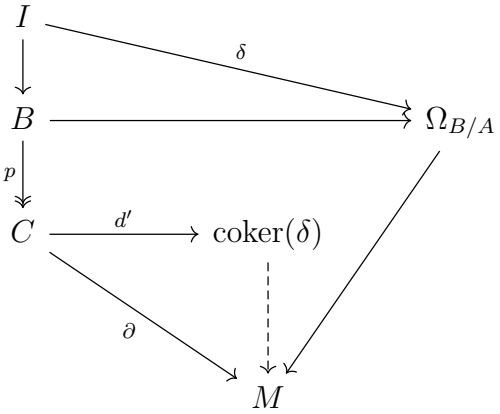
2. Let's start by constructing:

$$d' : C \to \text{coker } (\delta)$$
$$c \mapsto db$$

where b is some lift of c in B. We must show this is well-defined, namely if b is a lift of $0 \iff b \in I$, we must have $db = 0$. But this is clear since in that case $db = \delta(b) \in \text{im } (\delta)$. Moreover, d' is a derivation since d is. It remains to check it is universal. Now assume I have diagram:



with ∂ a deriation. We need to construct the dashed map. However, the diagram fits into:



using the universal property for $\Omega_{B/A}$. Therefore to show the existence of the dashed map it suffices to check $I \xrightarrow{\delta} \Omega_{B/A} \to M$ is zero. But this is clear from the commutativity of the diagram and $I \to B \to C$ is zero. The uniqueness follows by the uniqueness of the morphism $\Omega_{B/A} \to M$. Therefore:

$$\text{coker } (\delta) = \Omega_{C/A}$$

□

Exercise 4.24 (Hartshorne II.7.2)**.** *Let X be a scheme over a field k. Let $\mathcal{L}$ be an invertible sheaf on X, and let $\{s_0, \ldots, s_n\}$ and $\{t_0, \ldots, t_m\}$ be two sets of sections of $\mathcal{L}$, which generate the same subspace $V \subseteq \Gamma(X, \mathcal{L})$, and which generate the sheaf $\mathcal{L}$ at every point. Suppose $n \leqslant m$. Show that the corresponding morphisms $\varphi: X \to \mathbb{P}^n_k$ and $\psi: X \to \mathbb{P}^m_k$ differ by a suitable linear projection $\mathbb{P}^m - L \to \mathbb{P}^n$ and an automorphism of $\mathbb{P}^n$, where L is a linear subspace of $\mathbb{P}^m$ of dimension $m - n - 1$.*

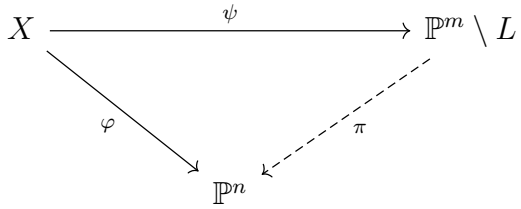
Proof. This is a linear algebra exercise. Let's first arrange s_i, t_j such that:

$$(s_0, \cdots, s_d), (t_0, \cdots, t_d)$$

form an ordered basis basis for V for some $d \leq n \leq m$. We may adjust $\{s_0, \cdots, s_d\}$ by an automorphism $\mathrm{PGL}_d(k) \subset \mathrm{PGL}_n(k)$ so that:

$$(s_0, \cdots, s_d) = (t_0, \cdots, t_d)$$

Now let's set $L = V(s_0, \cdots, s_n)$. We're trying to define:



for some appropriate choices of L and π. Now assume:

$$s_{d+i} = \sum_{j=0}^{d} a_{ij} s_j$$

by expressing s_{d+i} using the chosen basis. To make the diagram commute a natural choice for π is given by:

$$[z_0, \cdots, z_m] \mapsto [z_0, \cdots, z_d, \sum_{j=0}^{d} a_{1j} z_j, \cdots, \sum_{j=0}^{d} a_{n-d,j} s_j]$$

Notably this is well-defined outside $z_0 = z_1 = \cdots z_n = 0$. Therefore our choice of L is simply defined by these equations, it has the right dimension $m - n - 1$. □

Exercise 4.25 (Hartshorne II.7.3)**.** *Let $\varphi\colon \mathbb{P}^n_k \to \mathbb{P}^m_k$ be a morphism. Then:*

1. *Either $\varphi(\mathbb{P}^n) = pt$ or $m \geqslant n$ and $\dim \varphi(\mathbb{P}^n) = n$;*
2. *in the second case, φ can be obtained as the composition of (1) a d-uple embedding $\mathbb{P}^n \to \mathbb{P}^N$ for a uniquely determined $d \geqslant 1$, (2) a linear projection $\mathbb{P}^N - L \to \mathbb{P}^m$, and (3) an automorphism of $\mathbb{P}^m$. Also, φ has finite fibres.*

Proof. 1. Let $\mathcal{L} = \varphi^*\mathcal{O}(1)$. The map φ is equivalent to the data of $m+1$ sections $s_0, \cdots, s_m$ of $\mathcal{L}$ without base point. Note we cannot have $\deg \mathcal{L} < 0$. If $\deg \mathcal{L} = 0$, then $s_0, \cdots, s_m$ are all just constants and therefore $\varphi(\mathbb{P}^n)$ is just a point. Now let's assume $\deg \mathcal{L} = d > 0$. So we can view $s_0, \cdots, s_m \in k[x_0, \cdots, x_n]_d$. Now $s_0, \cdots, s_m$ have no base point means that:

$$\sqrt{(s_0, \cdots, s_m)} = k[x_0, \cdots, x_n] \iff \operatorname{ht}(s_0, \cdots, s_m) = n+1$$

Now we claim that φ has fibres of dimension 0. Indeed, after compositing with an automorphism of $\mathbb{P}^m$, we only need to check the fibre at $[1 : 0 : 0 : \cdots : 0]$ has dimension 0. But then this is just cut out by:

$$s_1 = \cdots s_m = 0$$

However, since polynomial ring is Cohen-Macaulay, we know $\operatorname{ht}(s_1, \cdots, s_m) \geq n$. This implies the fibre has dimension 0. In fact, from this we can see the fibre is in fact finite: they're in bijection with prime ideals in appearing in the primary decomposition of $(s_1, \cdots, s_m)$. But now φ is a morphism of finite dimension, so that $\dim \varphi(\mathbb{P}^n) = \dim(\mathbb{P}^n) = n$, and $m \geq n$.

2. In the second case, we just take $d = \deg \mathcal{L}$ and $\mathbb{P}^n \to \mathbb{P}^N$ is just given by the complete linear system of polynomials of degree d. Use the previous exercise we see that we can arrange (2) and (3) to get a map $\mathbb{P}^N \setminus L \to \mathbb{P}^m$. φ having finite fibres was proved in the previous part.

□

Exercise 4.26. *Let X be a projective variety over k, and let $\mathcal{L}, \mathcal{M} \in \operatorname{Pic}(X)$. Show that if $\mathcal{L}$ is ample and $\mathcal{M}$ is globally generated, then $\mathcal{L} \otimes \mathcal{M}$ is ample. If $\mathcal{L}$ is ample and $\mathcal{M}$ is arbitrary, then $\mathcal{L}^a \otimes \mathcal{M}$ is ample for a sufficiently large.*

Proof. 1. Let's prove the first statement. Choose n such that $\mathcal{L}^n$ is very ample, so that it comes from a closed embedding:

$$\varphi : X \to \mathbb{P}^N$$

for some large N. Now that $\mathcal{M}^n$ is generated by global sections since $\mathcal{M}$ does, so that I have a morphism:

$$\psi : X \to \mathbb{P}^M$$

From these we get:

$$\varphi \times \psi : X \to \mathbb{P}^N \times \mathbb{P}^M$$

this is a closed embedding (since φ is already a closed embedding). Moreover:

$$(\varphi \times \psi)^* \mathcal{O}(1) \boxtimes \mathcal{O}(1) = \mathcal{L}^n \otimes \mathcal{M}^n$$

So that $\mathcal{L}^n \otimes \mathcal{M}^n$ is very ample ($\mathcal{O}(1) \boxtimes O(1)$ is very ample on $\mathbb{P}^N \times \mathbb{P}^M$), hence $\mathcal{L} \otimes \mathcal{M}$ is ample.

2. Let's now prove the second statement. Some tensor power $\mathcal{L}^n$ gives a closed embedding:

$$i : X \hookrightarrow \mathbb{P}^N$$

with $i^*\mathcal{O}(1) = \mathcal{L}^n$. We claim that for any coherent sheaf $\mathcal{F}$ on $\mathbb{P}^N$, we can find m such that

$$\mathcal{F} \otimes \mathcal{O}(m)$$

is generated by global sections. If so, then we take $\mathcal{F} = i_*\mathcal{M}$, then projection formula tells me:

$$i^*(i_*\mathcal{M} \otimes \mathcal{O}(m)) = \mathcal{M} \otimes \mathcal{L}^{nm}$$

is generated by global sections. Now we can tensor some powers of $\mathcal{L}$ by the first part to get an ample sheaf. Therefore it suffices to prove the claim. Indeed this follows from any coherent sheaf on $\mathbb{P}^N$ has the presentation:

$$\bigoplus_{i \in I} \mathcal{O}(a_i) \to \mathcal{M} \to 0$$

for some finite I.

□

Exercise 4.27 (Blow Up)**.** *Let X be a k-variety with line bundle $\mathcal{L}$. Let $s_0, \cdots, s_r \in \Gamma(X, \mathcal{L})$, let $U = \bigcup D(s_i) \subset X$ be the open locus where these sections globally generate. Then we have a map: $f : U \to \mathbb{P}^r$. Let's define $\widetilde{X} \subset X \times \mathbb{P}^r$ to be the closure of the graph of f.*

1. *Check that $\pi : \widetilde{X} \to X$ is an isomorphism over U.*
2. *Calculate the fibre of π over $(0,0)$ when $X = \mathbb{A}^2, \mathcal{L} = \mathcal{O}_X, (s_0, s_1) = (x, y)$*
3. *Do part (b) instead when $X = V(y^2 - x^2(x+1))$ or when $X = V(y^2 = x^3)$.*
4. *In the last case, show that $\widetilde{X}$ is smooth.*

Proof. 1. Let $\widetilde{U} = \{(u, f(u)) \mid u \in U\} \subset U \times \mathbb{P}^r$. Note this is closed inside $U \times \mathbb{P}^r$ since it's given by equation $y = f(u)$ for $(u, y) \in U \times \mathbb{P}^r$. Therefore $\widetilde{X} \cap U = \widetilde{U}$. But then evidentally $\widetilde{U} \to U$ is an isomorphism, with the inverse map given by the graph map $u \mapsto (u, f(u))$.

2. Note that in this case:

$$f : \mathbb{A}^2 \setminus 0 \to \mathbb{P}^1, \quad (u, v) \mapsto [u : v]$$

and thus:

$$\pi^{-1}(U) = \{((u,v),[u:v]) \in U \times \mathbb{P}^1\}$$

Now we write:

$$\mathbb{A}^2 \times \mathbb{P}^1 = \mathbb{A}^2 \times (\mathbb{P}^1 \setminus \infty) \cup \mathbb{A}^2 \times (\mathbb{P}^1 \setminus 0) \simeq \mathbb{A}^3 \cup \mathbb{A}^3$$

In the first copy of $\mathbb{A}^3$, $\pi^{-1}(U)$ is cut by equations (the three coordinates are u, v, w):

$$v \neq 0, vw = u$$

Take closure we see:

$$\widetilde{X} \cap \mathbb{A}^3 = \{(u,v,w) \mid vw = u\} \implies \pi^{-1}(0,0) \cap \mathbb{A}^3 = \mathbb{A}^1$$

for the third copy of $\mathbb{A}^1 = \mathbb{P}^1 \setminus \infty$. Similarly, $\pi^{-1}(0,0) \cap \widetilde{X} = \mathbb{P}^1 \setminus 0$. Hence:

$$\pi^{-1}(0,0) = \mathbb{P}^1$$

3. As hinted we do computation inside $\mathbb{A}^2 \times \mathbb{P}^1$ and using the two standard charts. Let $U = X \setminus 0$ in each case.

   (a) $X = V(y^2 - x^2(x+1))$. In the first copy of $\mathbb{A}^3$, $\pi^{-1}(U)$ is cut out by:

   $$v \neq 0, v^2 = u^2(u+1), vw = u \iff v \neq 0, 1 = w^2(vw+1), vw = u$$

   So $\widetilde{X} \cap \mathbb{A}^3$ is cut out by:

   $$1 = w^2(vw+1), vw = u$$

   Restricting to the fibre, I set $u = v = 0$, I get: $w^2 = 1 \iff w = \pm 1$. So:

   $$\pi^{-1}(0) \cap \mathbb{A}^3 = \{[1:1], [-1,1]\}$$

   In the second copy of $\mathbb{A}^3$, I set:

   $$[1:w] = [u:v] \iff v = uw$$

   So $\pi^{-1}(U)$ is cut out by:

   $$u \neq 0, v^2 = u^2(u+1), v = uw \iff u \neq 0, w^2 = u+1, v = uw$$

   So $\widetilde{X} \cap \mathbb{A}^3$ is cut out by:

   $$w^2 = u+1, v = uw$$

   Restrict to $u = v = 0$, I get: $w = \pm 1$. So I again get:

   $$[1:1], [1:-1]$$

   In summary:

   $$\pi^{-1}(0,0) = \{[1:1], [1:-1]\}$$

(b) $X = V(y^2 - x^3)$. In the first copy, $\pi^{-1}(U)$ is:

$$v \neq 0, v^2 = u^3, vw = u \iff v \neq 0, vw^3 = 1, vw = u$$

Then:

$$\pi^{-1}(0,0) \cap \mathbb{A}^3_1 = \emptyset$$

In the first copy $\pi^{-1}(U)$ is given by:

$$u \neq 0, v^2 = u^3, v = uw \iff u \neq 0, w^2 = u, v = uw$$

Then:

$$\pi^{-1}(0,0) \cap \mathbb{A}^3_2 = \{w^2 = 0\} = \{[1:0]\}$$

So $\pi^{-1}(0,0) = \{[1:0]\}$.

4. Note that in the last case:

$$\widetilde{X} \cap \mathbb{A}^3_1 = \{vw^3 = 1, vw = u\} \simeq \{(v,w) \in \mathbb{A}^2 \mid vw^3 = 1\}$$

is smooth. And:

$$\widetilde{X} \cap \mathbb{A}^3_2 = \{w^2 = u, v = uw\} = \{(u, u^3, u^2) \in \mathbb{A}^3\} \simeq \mathbb{A}^1$$

is smooth. Since smoothness is local property, so that $\widetilde{X}$ is smooth.

□

4.6 Homework 6

Exercise 4.28. *Let A be a k-algebra with maximal ideal $\mathfrak{m}$ such that $A/\mathfrak{m} = k$. Using the universal property, show that $\Omega_{A/k} \otimes_A k = \mathfrak{m}/\mathfrak{m}^2$.*

Proof. Let V be any k-vector space (can be viewed as a $A/\mathfrak{m}$-module, and therefore an A-module). Let's compute:

$$\mathrm{Hom}_k(\Omega_{A/k} \otimes_A k, V) \simeq \mathrm{Hom}_A(\Omega_{A/k}, V) \simeq \mathrm{Der}_A(A, V) \simeq \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, V)$$

the first isomorphisms is just adjunction for extension / restriciton of scalars, the second isomrophism is the universal property of $\Omega_{A/k}$. Let's explain the third isomorphism. Indeed, let $D : A \to V$ be a derivation. For every $a \in A$, we can find $\lambda \in k$ such that $a - \lambda \in \mathfrak{m}$ since $A/\mathfrak{m} \simeq k$. We know:

$$D(a) = D(a - \lambda)$$

So D is uniquely determined by $D|_{\mathfrak{m}}$. Let's see what requirement of D do I have. Indeed, I need for $a = \lambda_1 + x_1, b = \lambda_2 + x_2$ with $\lambda_i \in k, x_i \in \mathfrak{m}$:

$$D(ab) = aD(b)+bD(a) \iff D((\lambda_1+x_1)(\lambda_2+x_2)) = (\lambda_1+x_1)D(\lambda_2+x_2)+(\lambda_2+x_2)D(\lambda_1+x_1)$$

$$\iff D(\lambda_1 x_2) + D(\lambda_2 x_1) + D(x_1 x_2) = (\lambda_1 + x_1)D(x_2) + (\lambda_2 + x_2)D(x_1)$$

Now D is k-linear, and note that $x_1 D(x_2) = x_2 D(x_1) = 0$ since V is a $A/\mathfrak{m} = k$-module, the above condition reduces to:

$$D(x_1 x_2) = 0 \iff D|_{\mathfrak{m}^2} = 0$$

Hence the third isomorphism. We're therefore done by Yoneda. □

Exercise 4.29. *Let $X \hookrightarrow Y$ be a closed embedding, where X and Y are smooth varieties over $k = \overline{k} =$ of dimension d and e. Show that the ideal I_X is locally defined by $r = e - d$ generators. More precisely, let $p \in X$ be a closed point, and pick $f_1, \cdots, f_r \in I_{X,p}$ be elements whose image in $T^\vee_{Y,p}$ is a basis of the kernel of the restriction map $T^\vee_{Y,p} \to T^\vee_{X,p}$. Show that $f_1, \cdots, f_r$ generates $I_{X,p}$.*

Proof. We have exact sequence:

$$0 \to I_X/I_X^2 \to i^*\Omega_Y \to \Omega_X \to 0$$

Localize at p:

$$0 \to I_{X,p}/I_{X,p}^2 \to (i^*\Omega_Y)_p \to \Omega_{X,p} \to 0$$

Since all three terms are free $\mathcal{O}_{X,p}$-modules, so we can tensor with $\mathcal{O}_{X,p}/\mathfrak{m}_p$ and remain exact:

$$0 \to I_{X,p}/\mathfrak{m}_p I_{X,p} \to T^\vee_{Y,p} \to T^\vee_{X,p} \to 0$$

Now $f_1, \cdots, f_r$ are selected spanning $I_{X,p}/\mathfrak{m}_p I_{X,p}$ be assumption, then they must generate $I_{X,p}$ by Nakayama's lemma. □

Exercise 4.30. *Let $C \subset \mathbb{P}^4$ be a smooth curve obtained as scheme theoretic intersection of hypersurface X_d, X_e, X_f of degree d, e, f. Calculate the genus of C.*

Proof. Assume $X_d \cap X_e \cap X_f$ is smooth complete intersection. Consider:

$$C = X_d \cap X_e \cap X_f \overset{i}{\hookrightarrow} X_e \cap X_f \overset{j}{\hookrightarrow} X_f \overset{k}{\hookrightarrow} \mathbb{P}^4$$

We will repeat using the adjunction formula for canonical divisor. First:

$$K_{X_f} = k^*(K_{\mathbb{P}_4} \otimes \mathcal{O}(X_f)) = k^*\mathcal{O}(f-5)$$

So:

$$K_{X_e \cap K_f} = j^*(k^*\mathcal{O}(f-5) \otimes \mathcal{O}(X_e \cap X_f))$$

But observe that:

$$\mathcal{O}(X_e \cap X_f) = k^*\mathcal{O}(X_e) = \mathcal{O}(e) \in \mathrm{Pic}(X_f)$$

So:

$$K_{X_e \cap X_f} = j^*(k^*\mathcal{O}(f-5) \otimes \mathcal{O}(e)) = j^*k^*\mathcal{O}(e+f-5)$$

Similar reasoning shows that:

$$K_C = K_{X_d \cap X_e \cap X_f} = i^*(j^*k^*\mathcal{O}(e+f-5) \otimes \mathcal{O}(X_e \cap X_f \cap X_d)) = i^*j^*k^*\mathcal{O}(d+e+f-5)$$

Let us call $\varphi : C \hookrightarrow \mathbb{P}^4$ the composition. Then:

$$K_C = \varphi^*\mathcal{O}(d+e+f-5)$$

Note that $\varphi^*\mathcal{O}(1)$ has degree def by a generalization of Bezout's theorem, hence: $deg(K_C) = def(d+e+f-5)$. It follows that:

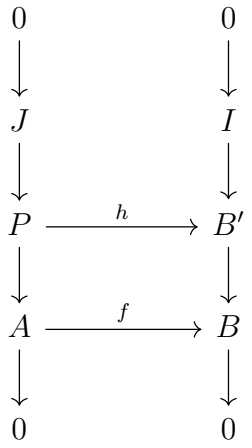
$$g_C = \frac{def(d+e+f-5)}{2} + 1$$

As a sanity check, for $d = e = f = 1$, I indeed have $g = 0$. □

Exercise 4.31. *(Hartshorne II.8.6, Infinitesimal Lifting Property) Let k be algebraically closed field, let A be a finitely generated k algebra such that* Spec A *is nonsingular. Let $0 \to I \to B' \to B \to 0$ be exact, where B' is a k-algebra and $I^2 = 0$. We will prove the* **infinitesimal lifting property** *of A: given a k-algebra map $f : A \to B$, it can be lifted to a map $g : A \to B'$.*

1. *Suppose $g, g' : A \to B'$ are two lifts. Show that $\theta = g - g'$ is a k-derivation of A into I, which we can consider as an element of* $\mathrm{Hom}_A(\Omega_{A/k}, I)$. *Note that $I^2 = 0$ so I has a natural structure of a B-module and hence also of A-module. Conversely, for any $\theta \in \mathrm{Hom}_A(\Omega_{A/k}, I)$, $g' = g + \theta$ is another lift of f.*

2. *Now let $P = k[x_1, \cdots, x_n]$ and that $A = P/J$. Show that there exists a homomorphism $h : P \to B'$ making commutative diagram:*



and show that h induces an A-linear map $\overline{h} : J/J^2 \to I$.

3. *Now use the hypothesis* Spec A *is nonsingular to obtain an exact sequence:*

$$0 \to J/J^2 \to \Omega_{P/k} \otimes A \to \Omega_{A/k} \to 0$$

Show furthermore that apply the functor $\mathrm{Hom}_A(-, I)$ *gives an exact sequence:*

$$0 \to \mathrm{Hom}_A(\Omega_{A/k}, I) \to \mathrm{Hom}_P(\Omega_{P/k}, I) \to \mathrm{Hom}_A(J/J^2, I) \to 0$$

Let $\theta \in \mathrm{Hom}_P(\Omega_{P/k}, I)$ be an element whose image gives $\overline{h} \in \mathrm{Hom}_A(J/J^2, I)$, consider θ as a derivation $P \to B'$ Then let $h' = h - \theta$, and show that $h' : P \to B'$ is a homomorphism such that $h'(J) = 0$. Hence h' induces the desired map $g : A \to B'$.

Proof. 1. First of all it's clear that θ takes value in I since $, g, g'$ are two lifts. It's also clear tht θ is k-linear. To check the derivation property:

$$\begin{aligned}\theta(ab) &= g(ab) - g'(ab) \\ &= g(a)(g(b) - g'(b)) + g'(b)(g(a) - g'(a)) \\ &= f(a) \cdot \theta(b) + f(b) \cdot \theta(a) = a\theta(b) + b\theta(a)\end{aligned}$$

So θ is indeed a derivation. Conversely, for a derivation θ and homomorphism g, we check $g' = g + \theta$ is another homomorphism:

$$\begin{aligned}g'(ab) &= \theta(ab) + g(ab) \\ &= a\theta(b) + b\theta(a) + g(a)g(b) \\ &= g(a)\theta(b) + g'(b)\theta(a) + g(a)g(b) + \theta(a)\theta(b) \quad (\text{note that } \theta(a)\theta(b) = 0) \\ &= (\theta(a) + g(a))(\theta(b) + g(b)) = g'(a)g'(b)\end{aligned}$$

2. Let $\overline{x_i} \in A$ be the image of x_i in A. Choose $y_i \in B'$ be any lift of $f(\overline{x_i}) \in B'$. We can just insist $x_i \mapsto y_i$ and this defines a map h. It is easy to check that the diagram

commutes. Now h induces a map $J \to I$ from the commutativity of the diagram (and columns are exact), but since h is ring homomorphism so that:

$$h(J^2) \subset I^2 = 0$$

Hence h descends to: $\overline{h} : J/J^2 \to I$.

3. The first exact sequence comes is just the conormal sequence for Spec $A \subset$ Spec P. Applying $\mathrm{Hom}_A(-, I)$ preserves the exactness because all terms are locally free (projective). Now consider the composition:

$$P \to A \xrightarrow{f} B$$

Then we know h is a lift of one such map. Now θ is a derivation so from the first part we know $h' = h - \theta$ is a ring homomorphism. Moreover, if $(f_1, \cdots, f_r) = J$, we compute:

$$\theta(f_i) = \theta(df_i) = \theta(df_i \otimes 1) = \overline{h}(f_i) = h(f_i)$$

from the commutative diagram:

$$f_i \longmapsto df_i \otimes 1$$

$$\begin{array}{ccc} J/J^2 & \longrightarrow & \Omega_{P/k} \otimes A \\ & \searrow^{\bar{h}} & \downarrow \theta \\ & & I \end{array}$$

So we check $h'|_J = 0$.

□

Exercise 4.32. *Let $C \subset \mathbb{P}^2_{\mathbb{C}}$ be a smooth plane curve of degree d. Fix a closed point $p \in C$. Show that the number of tangent lines to C that pass through p is $(d+1)(d-2)$ (not counting the tangent line at p itself). Assume that any tangency line through p is tangent to C at exactly one point, with multiplicity exactly 2.*

Proof. Choose a generic line $L \subset \mathbb{P}^2$, and consider the projection map from p:

$$\pi : \mathbb{P}^2 \setminus \{p\} \to L \simeq \mathbb{P}^1$$

This restricts to:

$$\pi : C \setminus \{p\} \to \mathbb{P}^1$$

Since any rational morphism between curves can always extend to a regular morphism, so we have an extended map:

$$\pi : C \to \mathbb{P}^1$$

By Bezout's theorem, π has degree $d-1$. Indeed, for any point $x \neq \pi(p)$, we have $\pi^{-1}(x)$ consists of points on the line px except for p (counting multiplicity), and there are $d-1$ of such by Bezout. Now by Riemann-Hurwtiz:

$$2g_C - 2 = (d-1)(-2) + \deg R$$

Now $2g_C - 2 = (d-1)(d-2) - 2 = d^2 - 3d \implies \deg R = d^2 - d + 2 = (d+1)(d-2)$. Now $\deg R$ precisely dictates the number of tangent lines, we're therefore done. $\square$

4.7 Homework 7

Exercise 4.33. *(Hartshonre III.4.3) Let $X = \mathbb{A}^2_k = \text{Spec } k[x,y]$ and let $U = X \setminus \{(0,0)\}$. Use Cech cohomology to show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k-vector space spanned by $\{x^i y^j \mid i,j < 0\}$. In particular, it is infinite dimensional.*

Proof. Let $V_1 = D(x), V_2 = D(y)$ be an open cover for U. Moreover $V_1, V_2, V_1 \cap V_2$ are all affine:

$$V_1 = \text{Spec } k[x,y]_x, V_2 = k[x,y]_y, V_3 = k[x,y]_{xy}$$

So the Čech complex is:

$$0 \to k[x,y]_x \oplus k[x,y]_y \xrightarrow{d} k[x,y]_{xy} \to 0$$

with $d(f,g) = f - g$. It is therefore clear that:

$$H^1(U, \mathcal{O}_U) = \text{coker } (d) = k \cdot \{x^i y^j \mid i,j \in \mathbb{Z}_{<0}\}$$

as required. □

Exercise 4.34. *(Hartshorne III.4.7) Let X be a degree d curve inside $\mathbb{P}^2$ (do not assume f is irreducible). Assume $(1,0,0) \notin X$. Then show that X can be covered by two open affines:*

$$U = X \cap \{x_1 \neq 0\}, \quad V = X \cap \{x_2 \neq 0\}$$

Use the Čech cover $\{U, V\}$ to compute $H^*(X, \mathcal{O}_X)$,

Proof. Let $S = k[x_0, x_1, x_2]$, $A = S/(f)$ be the homogeneous coordinate ring of X. Also note that f contains a monomial x_0^d (WLOG, since $f(1,0,0) \neq 0$). Then:

$$\Gamma(U, \mathcal{O}_X) = (A_{x_1})_{\deg=0}, \quad \Gamma(V, \mathcal{O}_X) = (A_{x_2})_{\deg=0}, \quad \Gamma(U \cap V, \mathcal{O}_X) = (A_{x_1 x_2})_{\deg=0}$$

Let's consider the diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & S(-d)_{x_1} \oplus S(-d)_{x_2} & \xrightarrow{f} & S_{x_1} \oplus S_{x_2} & \longrightarrow & A_{x_1} \oplus A_{x_2} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle d^1} & & \downarrow{\scriptstyle d^2} & & \downarrow{\scriptstyle d^3} & & \\
0 & \longrightarrow & S(-d)_{x_1 x_2} & \xrightarrow{f} & S_{x_1 x_2} & \longrightarrow & A_{x_1 x_2} & \longrightarrow & 0
\end{array}$$

We need to compute $\ker(d^3), \text{coker } (d^3)$, and their degree zero parts will give the cohomlogy group. By snake lemma, we have a LES:

$$0 \to \ker d^1 \to \ker d^2 \to \ker d^3 \to \text{coker } (d^1) \to \text{coker } (d^2) \to \text{coker } (d^3) \to 0$$

Note that:

$$\ker d^1 = S(-d)_{x_1} \cap S(-d)_{x_2} = S(-d)$$

and similarly $\ker d^2 = S$. We claim that coker $d^1 \to$ coker d^2 is injective in degree 0. If so, then in degree 0 we have:

$$H^0(X, \mathcal{O}_X) = (\ker d^3)_{\deg=0} = (S/(f))_0 = k$$

Now the image of d^2 is generated by monomials of the form $x_0^a x_1^b x_2^c$ with $a \geq 0$ and one of $b, c \geq 0$. So coker (d^2) has a basis of monomials of the form $x_0^a x_1^b x_2^c$ with $a \geq 0, b, c < 0$. Similar statement holds for coker (d^1) with the shift in degree. In particular:

$$(\text{coker } d^1)_{\deg=0} = k\{x_0^{b+c-d} x_1^{-b} x_2^{-c} \mid b, c \geq 1, b+c-d \geq 0\}$$

$$(\text{coker } d^2)_{\deg=0} = k\{x_0^{b+c} x_1^{-b} x_2^{-c} \mid b, c \geq 1\}$$

Now assume $\alpha \in (\text{coker } d^1)_{\deg=0}$ with $f\alpha = 0$. Recall f has a term x_0^d, and let e be the largest power of x_0 occuring in ay monomial in α, and we can write:

$$\alpha = x_0^e \beta + \gamma$$

for $x_0 \nmid \beta$ and γ contains no term dividing x_0^e. Therefore $f\alpha = x_0^{d+e}\beta + \gamma'$ since f contains a term x_0^d. Now $f\alpha = 0 \implies \beta = 0 \implies \alpha = 0$. This checks the injectivity. Hence we computed H^0. To compute H^1 one needs to understand the degree 0 part of of the cokernel of:

$$f : \text{coker } (d^1) \to \text{coker } (d^2)$$

Note that element in coker $(f)_{\deg=0}$ can be written in a way such that it only involves $x_0^{a+b} x_1^{-a} x_2^{-b}$ with $a + b < d$. Indeed, if $a + b \geq d$, we can consider:

$$x_0^{a+b} x_1^{-a} x_2^{-b} - f \cdot x_0^{a+b-d} x_1^{-a} x_2^{-b}$$

and that the degree of x_0 decreases. Moreover, every element in the image of f must have the property that $a + b \geq d$. Hence coker $(f) =$ coker (d^3) has a basis:

$$\{x_0^{a+b} x_1^{-a} x_2^{-b} \mid a + b < d, a \geq 1, b \geq 1\}$$

which has dimension $\frac{(d-2)(d-1)}{2}$ as desired. □

Exercise 4.35. *(Vakil 18.2.I). Let X be a quasi-compact separated k-scheme, and $\mathcal{L} \in \text{Pic}(X)$. Assume that K/k is a field extension, show that $\mathcal{L}$ is base-point-free iff $\mathcal{L}_K \in \text{Pic}(X_K)$ is base-point-free.*

Proof. In this setting we have an isomorphism:

$$\Gamma(X_K, \mathcal{L}_K) \xrightarrow{\sim} \Gamma(X, \mathcal{L}) \otimes_k K$$

Now let $(s_i)_{i \in I}$ be a basis for $\Gamma(X, \mathcal{L})$. Let $X = \bigcup_j \text{Spec } A_j$ be an affine open covering of X trivializing $\mathcal{L}$. Then $\mathcal{L}$ being globally generated reads: for every $A = A_j$, we have

$$\bigoplus_{i \in I} A \xrightarrow{\oplus s_i} A \to 0$$

is exact. Tensoring with K, we know:

$$\bigoplus_{i \in I} A \otimes_k K \xrightarrow{\oplus s_i} A \otimes_k K \to 0$$

is exact, implying that $\mathcal{L}_K$ is globally generated. Conversely, assume $\mathcal{L}_K$ is globally generated, so that the sequence in the previous line is exact. Let's write:

$$\bigoplus_{i\in I} A \xrightarrow{\oplus s_i} A \to \text{coker } \to 0$$

tensoring with K gives exact sequence:

$$\bigoplus_{i\in I} A\otimes_k K \xrightarrow{\oplus s_i} A\otimes_k K \to \text{coker } \otimes_k K \to 0$$

By assumption we know coker $\otimes_k K = 0 \implies$ coker $= 0$ since K/k is flat. It follows that $\mathcal{L}$ is globally generated. □

Exercise 4.36. *Let $S_* = k[x_0, \cdots, x_n]$ be the graded polynomial ring. Let E be a coherent sheaf on $\mathbb{P}^n_k$, corresponding to graded S_*-module $\Gamma_*(E) = \bigoplus_n \Gamma(E\otimes\mathcal{O}(n))$.*

1. *For every $i \geq 0$, genearlize this to define $\Gamma^i_*(E) = \bigoplus H^i(E\otimes\mathcal{O}(n))$. Show this has the structure of of a graded S_*-module.*
2. *Using descending induction show $\Gamma_*(E)$ is a finitely generated S_*-module.*
3. *Show if $i > 0$, then $\Gamma^i_*(E)$ only nonzero for finitely many degrees.*
4. *Harthsorne II.5.9. (a,b)*

Proof. Let $M = \Gamma_*(E)$.

1. Let's consider the Čech complex $C^\bullet_n$ for each $E\otimes\mathcal{O}(n)$:

$$C^i_n = \prod_{I\subset\{0,\cdots,n\},|I|=i+1} \Gamma(U_I, E\otimes\mathcal{O}(n)) = \prod_I (M(n)_{\prod_I x_i})_{\deg=0} = \prod_I M_{\prod x_i,\deg=n}$$

   Therefore for each $f \in S_d$, we have a multiplication by f map:

$$f : C^i_n \to C^i_{n+d}$$

   which obviously commutes with the Čech differentials. Therefore we have a chain map:

$$f : C^\bullet_n \to C^\bullet_{n+d}$$

   which induces a map on cohomology $f : H^*(E\otimes\mathcal{O}(n)) \to H^*(E\otimes\mathcal{O}(n+d))$. This gives $\Gamma^i_*(E)$ a graded S_*-module structure.

2. We will show that eaah $\Gamma^i_*(E)$ is finitely generated by descending induction. The statement is clearly true for $i \geq n+1$ since in this case $\Gamma^i_*(E) = 0$. Now assume the statement is true for $i+1$, let's look at $\Gamma^i_*(E)$. Since E is coherent, one can find a, N such that:

$$0 \to \ker \to \mathcal{O}(a)^{\oplus N} \to E \to 0$$

Let $n \geq n_0$ be sufficiently large so that $H^i(\mathcal{O}(a+n)) = 0$ for all $i > 0$. So we have:

$$0 \to \ker \otimes \mathcal{O}(n) \to \mathcal{O}(a+n)^{\oplus N} \to E \otimes \mathcal{O}(n) \to 0$$

Take cohomology gives me:

$$0 \to H^i(E \otimes \mathcal{O}(n)) \to H^{i+1}(\ker \otimes \mathcal{O}(n)) \to 0$$

This tells me:

$$\bigoplus_{n \geq n_0} H^i(E \otimes \mathcal{O}(n)) \simeq \bigoplus_{n \geq n_0} H^{i+1}(\ker \otimes \mathcal{O}(n))$$

as graded S_*-modules. In particular LHS is finitely generated. This implies $\Gamma_*^i(E)$ is finitely geneated, by adding a basis for:

$$\bigoplus_{n < n_0} H^i(E \otimes \mathcal{O}(n))$$

Note each term is finite dimensional since our space is projective. The induction process will carry whenever $i > 0$. For $i = 0$ the cohomology sequence is slightly different:

$$0 \to H^0(\ker \otimes \mathcal{O}(n)) \to H^0(\mathcal{O}(a+n)^{\oplus N}) \to H^0(E \otimes \mathcal{O}(n)) \to H^1(\ker \otimes \mathcal{O}(n)) \to 0$$

Gather all such n we get an exact sequence of graded S_*-modules:

$$0 \to \Gamma_*(\ker \otimes \mathcal{O}(n)) \to \Gamma_*(\mathcal{O}(a+n)^{\oplus N}) \to \Gamma_*(E \otimes \mathcal{O}(n)) \to \Gamma_*^1(\ker \otimes \mathcal{O}(n)) \to 0$$

Note the last term is finitely generated by induction hypothesis, the second term is finitely generated since we explicitly understand these cohomology spaces. The first term is finitely generated since it's a submodule of a finitely generated module over a noetherian ring. It follows that the third term is also finitely generated. We can thus proceed as before.

3. This essentially follows from our argument in the previous part. We still do descending induction. The statement is obviously true for $i \geq n+1$. Now suppose the statement is true for $i+1$, we now have:

$$\bigoplus_{n \geq n_0} H^i(E \otimes \mathcal{O}(n)) \simeq \bigoplus_{n \geq n_0} H^{i+1}(\ker \otimes \mathcal{O}(n))$$

Therefore $\Gamma_*^i(E)$ has only finitely many nonzero terms after degree n_0 by induction hypothesis. The result follows.

4. This was done in Homework 2. Maybe the contents here is that:

$$\Gamma_* : \mathrm{Coh}(\mathbb{P}^n) \to S_*\text{-mod}/\sim$$

has image lying in the subcategory consisting of quasi-finitely generated modules. Together with the inverse map $M \mapsto \widetilde{M}$, we establish the equivalence of categories as desired.

□

Exercise 4.37. *Vakil 18.1.C*

Proof. This is almost obvious. We have a short exact sequence:

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

with $\mathcal{F}$ coherent. Let n_0 be such that for all $n \geq n_0$, $H^1(\mathcal{F}(n)) = 0$. Now for all $n \geq n_0$, we have short exact sequence:

$$0 \to \mathcal{F}(n) \to \mathcal{G}(n) \to \mathcal{H}(n) \to 0$$

The result follows by taking long exact sequence on cohomology. □

4.8 Homework 8

Exercise 4.38. *1. (Vakil 12.3.D(c)). Let $X \hookrightarrow \mathbb{P}^n_k$ be a closed subset of dimension r. Show that there is an intersection of $r+1$ nonempty hypersurfaces missing X. Moreover, when k is infinite we can choose these hypersurfaces to be hyperplanes.*

2. Also prove that any projective variety of dimension d can be coverd by $d+1$ open affines.

Proof. 1. Let's prove the stronger statement by replacing the condition X having dimension r by X having dimension $\leq r$. We will prove (both statments) by induction on r. For $r = 0$ it is clear. Suppose the result is true for $r-1$. Now let X be a dimension r subvariety. Write $X = \cup X_i$ into irreducible components, with $\eta_i \in X_i$ the generic point. By (Vakil Exercise 12.3.C), we know we can find hypersurface H_1 such that $\eta_i \notin H_1$. Therefore: $H_1 \cap X_i \subsetneq X_i$ and thus $\dim(H_1 \cap X) < \dim(X) = r$. Thus by induction hypothesis, we can find $H_2 \cdots H_{r+1}$ such that $(H_2 \cap \cdots \cap H_{r+1}) \cap (H_1 \cap X) = \emptyset$, we're done in this case. Now if we further impose the condition that k is infinite, it suffices to see that we can choose H_1 to be a hyperplane. Indeed, consider the space of hyperplanes $\mathbb{P}(V^\vee) = \{H_{a_0,\cdots,a_n} = (a_0x_0 + \cdots a_nx_n = 0)\}$. Now the condition $\eta_i \in H$ is a closed condition for $H \in \mathbb{P}(V^\vee)$, so it corresponds to some proper closed subvariety $Z_i \subset \mathbb{P}(V^\vee)$. We can now choose H_1 to be any element in:

$$\mathbb{P}(V^\vee) \setminus (\cup Z_i)$$

This space is nonempty because if $Z_i \subset V(f_i)$, then $\cup Z_i \subset V(f)$ for $f = \prod f_i$. But then $f \neq 0$, over an infinite field we can find a nonzero place of this polynomial.

2. Again let's prove by induction on d. For $d = 0$ this is clear. Now assume we know the result for $\leq d-1$. Let $X \hookrightarrow \mathbb{P}^n$ be a projective variety of dimension d. Choose a hyperplane H such that $X \cap H$ is a proper subvariety of X, so that $\dim(X \cap H) \leq d-1$. So we can find open affine $U_2, \cdots, U_{d+1}$ such that $(\cup_{2 \leq i \leq d+1} U_i) \supset X \cap H$. Now take $U_1 = \mathbb{P}^n \setminus H \simeq \mathbb{A}^n$. Then it is clear that $\cup_{i \geq 1} U_i \supset X$. But then $U_i \cap X$ are affine, since they're closed subvariety of an affine variety. So that they form an affine open cover of X. We're done.

□

Exercise 4.39. *(Vakil 18.4 N) If C is a projective curve, and $\mathcal{L}$ is ample line bundle on C, show* $\deg \mathcal{L} > 0$

Proof. Let m be such that $\mathcal{L}^{\otimes m}$ is very ample. So that we can form:

$$i : C \hookrightarrow \mathbb{P}^n$$

and $i^*\mathcal{O}(1) = \mathcal{L}^{\otimes m}$. Now $\deg(i^*\mathcal{O}(1)) \geq |C \cap H| > 0$ for any choice of hyperplane H. It follows that:

$$\deg(\mathcal{L}) = \frac{1}{m}\deg(\mathcal{L}^{\otimes m}) = \frac{1}{m}\deg(i^*\mathcal{O}(1)) > 0$$

□

Exercise 4.40. *(Vakil 18.5 B) Suppose X is a connected smooth projective variety over $k = \overline{k}$ of dimension at least 2. Suppose D is an effective ample divisor. Show that D is connected.*

Proof. Let $\mathcal{L} = \mathcal{O}(D)$ be ample, and s a regular section of $\mathcal{L}$. So that we can write $D = V(s)$. Now let n be such that $\mathcal{L}^{\otimes n}$ is effective very ample. Then $V(s^n) = V(s)$ as a set. It suffices to show $V(s^n)$ is connected, since it equals D as topological spaces. Now we have a short exact sequence:

$$0 \to \mathcal{O}_X(-nD) \to \mathcal{O}_X \to \mathcal{O}_{V(s^n)} \to 0$$

which gives me long exact sequence on cohomology:

$$0 \to H^0(\mathcal{O}_X(-nD)) \to H^0(\mathcal{O}_X) \to H^0(\mathcal{O}_{V(s^n)}) \to H^1(\mathcal{O}_X(-nD))$$

We know $H^0(\mathcal{O}_X(-nD)) = 0$, and by Serre duality:

$$H^1(\mathcal{O}_X(-nD)) = H^{\dim X - 1}(\mathcal{O}_X(nD) \otimes \omega_X)^\vee$$

which vanishes for n sufficiently large, by Serre's vanishing theorem. Therefore we replace n by a larger n and get:

$$k \simeq H^0(\mathcal{O}_X) \simeq H^0(\mathcal{O}_{V(s^n)}) \simeq k^c$$

where c is the number of connected components of $V(s^n)$. This shows $D = V(s^n)$ is connected. □

Exercise 4.41. *(Vakil 18.6 D) Recall the d-th Veronese embedding of $\mathbb{P}^n$ is the embedding*

$$i : \mathbb{P}^n \hookrightarrow \mathbb{P}^N$$

with $N = \binom{n+d}{d} - 1$, and $i^*\mathcal{O}(1) = \mathcal{O}(d)$. Compute the Hilbert polynomial for this.

Proof. Let $p_{n,d}$ be the Hilbert polynomial for this. Then we know by definition:

$$p_{n,d}(m) = \chi(\mathbb{P}^n, \mathcal{O}(md)) = \binom{md+n}{n}$$

□

Exercise 4.42. *(Vakil 18.6 E) Suppose we have:*

$$X \overset{i}{\hookrightarrow} Y \hookrightarrow \mathbb{P}^n_k$$

a sequence of closed embeddings.

1. *Show that $p_X(m) \leq p_Y(m)$ for $m \gg 0$.*
2. *If $p_X(m) = p_Y(m)$ for $m \gg 0$, show that $X = Y$.*

Proof. 1. Recall that $p_X(m) = \chi(\mathcal{O}_X(m)), p_Y(m) = \chi(\mathcal{O}_Y(m))$. We want to compare these two. Now if $\mathcal{I}_{X/Y}$ is the ideal sheaf of X in Y, we have exact sequence:

$$0 \to \mathcal{I}_{X/Y}(m) \to \mathcal{O}_Y(m) \to i_*\mathcal{O}_X(m) \to 0$$

Therefore:

$$\chi(\mathcal{O}_Y(m)) = \chi(i_*\mathcal{O}_X(m)) + \chi(\mathcal{I}_{X/Y}(m)) = \chi(\mathcal{O}_X(m)) + \chi(\mathcal{I}_{X/Y}(m))$$

It suffices to show that $\chi(\mathcal{I}_{X/Y}(m)) > 0$ for m sufficiently large. But again by Serre vanishing, for m sufficiently large:

$$\chi(\mathcal{I}_{X/Y}(m)) = h^0(\mathcal{I}_{X/Y}(m))0$$

2. By our discussion from last time, we know in this case:

$$\chi(\mathcal{I}_{X/Y}(m)) = h^0(\mathcal{I}_{X/Y}(m)) = 0$$

for all m sufficiently large. But this forces $\mathcal{I}_{X/Y} = 0$, since otherwise for m large $\mathcal{I}_{X/Y}(m)$ will be generated by global sections and h^0 will not vanish. This shows $X = Y$.

□

Exercise 4.43. *(Vakil 18.6.J, Generalized Bezout) Suppose $X \subset \mathbb{P}^n_k$ be a projective scheme of dimension at least 1, and H is a hypersurface not containing any associated points of X. Show that* $\deg(H \cap X) = (\deg H)(\deg X)$.

Proof. Let $\deg H = d$. We have the exact sequence:

$$0 \to \mathcal{I}_{(X\cap H)/X} \hookrightarrow \mathcal{O}_X \to \mathcal{O}_{X\cap H} \to 0$$

If H is define by degree d homogenous polynomial f, then: $\mathcal{I}_{(X\cap H)/X}$ is generated by f, and hence $\mathcal{I}_{(X\cap H)/X} = \mathcal{O}_X(-d)$. We thus have exact sequence:

$$0 \to \mathcal{O}_X(-d) \to \mathcal{O}_X \to \mathcal{O}_{X\cap H} \to 0$$

Tensoring with m:

$$0 \to \mathcal{O}_X(m-d) \to \mathcal{O}_X(m) \to \mathcal{O}_{X\cap H}(m) \to 0$$

Therefore:

$$p_{X\cap H}(m) = \chi(\mathcal{O}_{X\cap H}(m)) = \chi(\mathcal{O}_X(m)) - \chi(\mathcal{O}_X(m-d))$$

which reads:

$$\frac{\deg X \cdot m^r}{r!} - \frac{\deg X \cdot (m-d)^r}{r!} + (\text{lower terms}) = \frac{\deg X \cdot d \cdot m^{r-1}}{(r-1)!} + (\text{lower terms})$$

Hence $\deg(X \cap H) = \deg X \cdot d = (\deg X)(\deg H)$. □

Exercise 4.44. *(Vakil 18.6 K) Let C be a degree one curve in $\mathbb{P}^3_k$. Show that C is a line.*

Proof. Firxst note that $C_{\overline{k}} \subset \mathbb{P}^3_{\overline{k}}$ is again a degree one curve. Suppose we can prove that $\overline{C} = C_{\overline{k}}$ is a line, we must show that C is also a line. Now $C_{\overline{k}}$ is the intersection of two hyperplanes, so that

$$\dim_{\overline{k}} H^0(\overline{C}, \mathcal{I}_{\overline{C}}(1)) = 2$$

Flat base change shows that $\dim_k H^0(C, \mathcal{I}_C(1)) = 2$. So that we can choose H_1, H_2 hyperplanes that their defining equations form a basis for $H^0(C, \mathcal{I}_C(1))$. It follows that $L = H_1 \cap H_2 \cap C$. But L, C are both curves, so that $L = C$ is a line. Hence we reduce the problem to the case when $k = \overline{k}$.

Assume $k = \overline{k}$. Choose $p, q \in C$ be two distinct points, and let H be any hyperplane containing p, q. We claim $C \subset H$. Indeed if not, then by Bezout's theorem

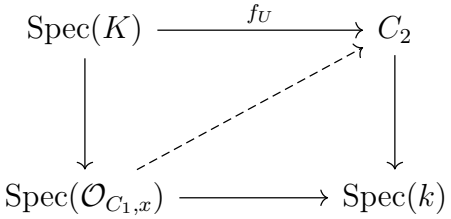
$$\deg(C \cap H) = (\deg C) \cdot \deg(H) = 1$$

This implies $C \cap H$ is a single point, a contradiction. Therefore $C \subset H$. Now intersecting all possible H, we see $C \subset L$ for L the line connecting p, q. Hence $C = L$. □

4.9 Homework 9

Exercise 4.45. *Let C_1, C_2 be smooth irreducible projective curves over k, and let $U \subset C_1$ be a nonempty open set. Show a rational morphism $f_U : U \to C_2$ extends uniquely to a morphism $f : C_1 \to C_2$.*

Proof. Uniqueness follows from the fact that everything is reduced and separated. To prove existence we use valuation criterion for properness. Let $x \in C_1 \setminus U$. Let's consider the diagram:



So there exists a map Spec $(\mathcal{O}_{C_1,x}) \to C_2$ making the diagram commute. In other words, their exists $g_x : V_x \to C_2$ defined at an neighborhood of $x \in C_1$, which agrees with f_U. Therefore, f is extendable at point x. But x is arbitrary, we're done. □

Exercise 4.46 (Vakil 19.2 C)**.** *Let C be a curve defined over an algebraically closed field k. Show that for any closed point p, $C \setminus p$ is affine.*

Proof. Note for $j \gg 0$, $\mathcal{O}(jp)$ is very ample and $h^0(jp) - h^0((j-1)p) = 1$ by Riemann-Roch. Choose j large enough such that the previously two properties hold and in addition $h^0(jp) \geq 2$. Then I can choose:

$$s_0 \in H^0(\mathcal{O}(jp)) \setminus H^0(\mathcal{O}((j-1)p))$$

and $s_1 = 1 \in H^0(\mathcal{O}(jp))$ linearly independent to s_0. In particular, s_0 has a pole of order j at p, and no other poles. Note that s_0, s_1 generates globally generated linear system, so we can use that to define a map:

$$f : C \to \mathbb{P}^1$$
$$x \mapsto [s_0(x) : 1]$$

This is a nonconstant morphism between curves, so that f is finite. It is clear from this presentation that $f^{-1}(\infty) = p$ as a set. Therefore f restricts to a map:

$$f : C \setminus p \to \mathbb{A}^1$$

and this restriciton is finite, in particular affine. Therefore $C \setminus p$ is affine. □

Exercise 4.47. *Let C be a smooth projective curve over an algebraically closed field k, and let $p_1, \cdots, p_r$ denote closed points of C. Show there exists a rational function on C having poles (of some positive order) at each p_i and regular elsewhere.*

Proof. By Riemann-Roch and Serre vanishing, we know we can choose $n_1, \cdots, n_r > 0$,with $n_1 + \cdots + n_r$ large enough such that for every $q \in C$, we have:

$$h^0(n_1p_1 + \cdots + n_rp_r) - h^0(n_1p_1 + \cdots + n_rp_r - q) = 1$$

Now let:

$$V = H^0(n_1p_1 + \cdots + n_rp_r)$$

$$V_q = H^0(n_1p_1 + \cdots n_rp_r - q)$$

So that $V_q \subset V$ is a codimension 1 subspace for each q. Now choose:

$$s \in V \setminus \cup_{i=1}^r V_{p_i}$$

It follows that s has pole of order n_i at each p_i and regular else where. □

Exercise 4.48 (Vakil 19.3 B)**.** *Suppose C is a genus 0 curve. Show that C has a point of degree at most 2.*

Proof. It is shown that every genus 0 curve is a conic in $\mathbb{P}^2_k$. Let's write:

$$C : ax^2 + by^2 + cz^2 + dxy + exz + fyz = 0$$

Let's first assume $(a, b, d) \neq (0, 0, 0)$. We set $z = 0$ and find:

$$ax^2 + by^2 + dxy = 0$$

If $a \neq 0$, we can set $y = 1$ and solve x in a quadratic extension of k. This gives a degree at most 2 point. Otherwise $a = 0$, then we can set:

$$[x : y : z] = [-b : d : 0]$$

and find a degree one point. The remaining case is $(a, b, d) = (0, 0, 0)$, so that:

$$C : cz^2 + exz + fyz = 0$$

One again has a degree one point:

$$[1 : 1 : 0]$$

□

Exercise 4.49 (Vakil 19.4 C)**.** *Suppose C is a regular plane curve of degree $e > 2$, and D_1, D_2 are two plane curves of the same degree d not containing C. By Bézout's theorem we know D_i meet C at de points, count with multiplicity. Suppose D_1, D_2 meet C at $de - 1$ of the same points, plus one more. Show that the remaining points are the same as well. More precisely, suppose there is a divisor E on C of degree $de - 1$, and degree 1 points p_1, p_2 such that: $D_i \cap C = E + p_i$. Show that $p_1 = p_2$.*

Proof. Note that $\mathcal{O}(D_1) \simeq \mathcal{O}(D_2) \simeq \mathcal{O}(d)$, it follos that:

$$\mathcal{L} = \mathcal{O}(D_1 \cap C) \simeq \mathcal{O}(D_2 \cap C)$$

is a line bundle of degree de. We need to show that given two sections $s_1, s_2 \in \Gamma(C, \mathcal{L})$ with:

$$\operatorname{div}(s_i) = E + [p_i]$$

then $p_1 = p_2$. Indeed, otherwise we can consider:

$$\frac{s_1}{s_2} \in \Gamma(C, \mathcal{O}_C)$$

and therefore:

$$\mathcal{O}_C \simeq \mathcal{O}(\text{div}(\frac{s_1}{s_2})) = \mathcal{O}(p_1 - p_2) \implies \mathcal{O}(p_1) \simeq \mathcal{O}(p_2)$$

This implies that $p_1 = p_2$ since $g_C > 0$. □

Exercise 4.50 (Vakil 19.5 A)**.** *Suppose* $\text{char}(k) \neq 2$*. Show that* x *does not have a square root in* $k(x)[y]/(y^2 - f(x))$*, where* f *is a polynomial with nonzero roots* $p_1, \cdots, p_r$*. Explain how this proves Propoisiton 19.5.2 in the case where* r *is odd.*

Proof. Note any element in $k(x)[y]/(y^2 - f(x))$ can be written as:

$$h_1(x) + yh_2(x)$$

for $h_i(x) \in k(x)$. Suppose I have:

$$(h_1(x) + yh_2(x))^2 = x \iff h_1^2 + fh_2^2 + 2yh_1(x)h_2(x) = x$$

This forces $h_1 h_2 = 0$. If $h_1(x) = 0$, then:

$$fh_2(x)^2 = x$$

But note that $f(0) \neq 0$, so $h_2(0) = 0$ and therefore fh_2^2 has a zero of order at least 2 at 0. This is a contradiction. Therefore, $h_1 = 0$. But then $h_1^2 = x$, a similar argument shows it's still a contradiction. This shows x doesn't have a square root. For the application for Proposition 19.5.2, we saw there we need to choose z as a regular function on $\mathbb{P}^1 \setminus 0$ such that:

$$z^2 = y^2/x^r$$

But z can be viewed as an element in the function field $k(x)[y](y^2 - f(x))$, so that: y^2/x^r is a square in the function field. But this implies x is a square, contradiction. □

Exercise 4.51 (Vakil 19.5 B)**.** *Let* C *be a curve with* $g_C \geq 1$*. Verify that* C *admits a degree 2 cover of* $\mathbb{P}^1$ *if and only if it admits a degree 2 invertible sheaf* $\mathcal{L}$ *with* $h^0(C, \mathcal{L}) = 2$*. Possibly in the course of doing this, verify that if* C *has a degree* 2 *invertible sheaf with at least* 2 *linearly independent sections, then* $\mathcal{L}$ *has precisely two sections, and that this* $\mathcal{L}$ *is base point free and gives a hyperelliptic map.*

Proof. First assume $f : C \to \mathbb{P}^1$ is a degree 2 map. Let's set $\mathcal{L} = f^*\mathcal{O}(1)$ has degree 2. It is clear that:

$$f^* : \Gamma(\mathbb{P}^1, \mathcal{O}(1)) \to \Gamma(C, \mathcal{L})$$

is injection, so the later space has dimension ≥ 2. On the other hand, we know there exists p, q such that $\mathcal{O}(p + q) \simeq \mathcal{L}$. Therefore by Riemann-Roch:

$$h^0(\mathcal{L}) \leq h^0(\mathcal{O}(p)) + 1 = 2$$

since $g_C \geq 1$. Therefore $h^0(\mathcal{L}) = 2$, as desired. Conversely, assume $\mathcal{L}$ is a degree 2 line bundle with $h^0(C, \mathcal{L}) \geq 2$. By similar argument as before we see $h^0(\mathcal{L}) \leq 2$, so that we must have $h^0(C, \mathcal{L}) = 2$. Let $p \in C$ be any closed point, then $\mathcal{L}(-p)$ is a degree 1 line bundle, and we know for every degree one line bundle it has only 1 section. Therefore:

$$h^0(C, \mathcal{L}) - h^0(C, \mathcal{L}(-p)) = 1$$

for all p. Therefore $\mathcal{L}$ is globally generated and define a degree 2 map $C \to \mathbb{P}^1$ as desired. □

Exercise 4.52. *Consider an effective Weil divisor D on a smooth projective curve of genus g. Show that $h^0(\mathcal{O}(D)) \leq \deg(D) + 1$ and equality holds iff $D = 0$ or $g = 0$.*

Proof. This is another immediate consequence of Riemann-Roch. We know:

$$\chi(\mathcal{O}(D)) - \chi(\mathcal{O}) = \deg D$$

$$\iff (h^0(\mathcal{O}(D)) - h^0(\mathcal{O})) + (h^0(K_C) - h^0(K_C \otimes \mathcal{O}(-D))) = \deg D$$

It is evident that $h^0(K_C) - h^0(K_C \otimes \mathcal{O}(-D)) \geq 0$, and $h^0(\mathcal{O}) = 1$. Therefore:

$$h^0(\mathcal{O}(D)) \leq \deg D + 1$$

with equality happens iff:

$$h^0(K_C) = h^0(K_C \otimes \mathcal{O}(-D))$$

Obviously if $g = 0$ then then both sides equal zero. Otherwise, use Riemann-Roch we know:

$$h^0(K_C \otimes \mathcal{O}(-D)) = h^0(\mathcal{O}(D)) - (\deg D + 1 - g)$$

So that:

$$h^0(\mathcal{O}(D)) - \deg D = h^0(\mathcal{O}) = 1$$

Now if $\deg D \geq 1$, then we can find p such that $D - p$ is again effective. But then:

$$h^0(\mathcal{O}(D)) \leq h^0(\mathcal{O}(p)) + \deg(D - p) = 1 + \deg D - 1 = \deg D$$

using C has genus ≥ 1. This is a contradiction. Therefore we must have $D = 0$, when $g \neq 0$. □

4.10 Homework 10

Exercise 4.53. *Let $k = \overline{k}$ be an algebraically closed field, and let $K \supset k(t)$ be a finite extension. Let $R_0 = k[t]$ and S_0 the integral closure of R_0 in K, and similarly $R_\infty = k[\frac{1}{t}]$ and S_∞. Show that S_0, S_∞ glue to get a smooth integral curve C with a finite map $C \to \mathbb{P}^1$*

Proof. It is clear that Spec S_0, Spec S_∞ are both smooth affine curves with a map to Spec R_0 and Spec (R_∞) respectively. They're curve because integral extension preserves dimension; it is smooth because normal and smooth are equivalent for curves over algebraically closed field; the natural map to Spec R_0 and Spec R_∞ is given by the ring extensions. Moreover, by Noether's theorem we also know S_0/R_0 and S_∞/R_∞ are finite, so once we show they glue then we're done. But the gluing is simply due to the fact that integral closure behaves well with localization:

$$S_0[\frac{1}{t}] \simeq \mathrm{cl}(R_0[\frac{1}{t}]) \simeq \mathrm{cl}(R_\infty[t]) \simeq S_\infty[t]$$

and this is an isomorphism of $R[t, \frac{1}{t}]$ algebras. We're done. □

Exercise 4.54 (Vakil 19.8 C)**.** *Suppose C is a nonhyperelliptic genus 5 curve. Show that the canonical curve is degree 8 in $\mathbb{P}^4$. Show that it lies on a three-dimensional vector space of quadric surfaces. Show that a regular complete intersection of three quadric surfaces is a canonical genus 5 curve.*

Proof. Since C is not hyperelliptic curve, so K_C is very ample and gives an embedding:

$$C \hookrightarrow \mathbb{P}^{g-1} = \mathbb{P}^4$$

It has degree equals $\deg K_C = 2g - 2 = 8$. Let's look at the sequence:

$$0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^4} \to \mathcal{O}_C \to 0$$

We wish to find quadric surfaces containing C, namely we wish to find degree 2 homogeneous polynomials f such that $C \subset \{f = 0\} \iff f \in \Gamma(\mathcal{I}_C(2))$. Therefore we need to compute the global sections of $\mathcal{I}_C(2)$. Therefore tensoring the original sequence with $\mathcal{O}(2)$, we get:

$$0 \to \mathcal{I}_C(2) \to \mathcal{O}(2) \to K_C^{\otimes 2} \to 0$$

$$\implies 0 \to \Gamma(\mathcal{I}_C(2)) \to \Gamma(\mathcal{O}(2)) \to \Gamma(K_C^{\otimes 2})$$

The second term has dimension $\binom{6}{2} = 15$, and the third term is given by Riemann Roch:

$$\ell(K_C^{\otimes 2}) = \deg(K_C^{\otimes 4}) + 1 - g = 8 \cdot 2 + 1 - 5 = 12$$

It follows that $\Gamma(\mathcal{I}_C(2))$ has dimension ≥ 3, which is the first part. For the second part, we proved in previous exercises that a triple complete interesection in $\mathbb{P}^4$ has genus:

$$g_C = \frac{def(d + e + f - 5)}{2} + 1$$

Plug in $d = e = f = 2$, we get $g_C = 5$ as desired. □

Exercise 4.55 (Vakil 19.8 E)**.** *Show that if $C \subset \mathbb{P}^{g-1}$ is a canonical curve of genus $g \geq 6$, then C is not a complete intersection.*

Proof. Assume C is a complete intersection of hypersurfaces of degree $d_1, \cdots, d_{g-2}$. Then a similar calculation as in the previous part shows that:

$$g = \frac{\prod_{i \leq g-2} d_i (\sum_{i \leq g-2} d_i - g)}{2} + 1$$

We claim that $d_i \geq 2$ for each i. This is because the construction of canonical embedding ensures that it cannot lie in any hyperplane. But then:

$$\text{RHS} \geq \frac{2^{g-2}(2(g-2)-g)}{2} + 1 = 2^{g-3}(g-4) + 1 > g$$

for $g \geq 6$. This is a contradiction. □

Exercise 4.56. *Let C be a smooth projective curve of genus g over an algebraically closed field k of characteristic not 2. Suppose C is not hyperelliptic, so that we have canonical embedding: $i : C \hookrightarrow \mathbb{P}^{g-1}$. We say C has three colinear points if there is a line $\ell \subset \mathbb{P}^{g-1}$ such that $\ell \cap C$ contains a subscheme $Z \subset C$ of length 3.*

Show that C has three colinear points iff C has a degree 3 map to $\mathbb{P}^1$ (i.e. the curve is trigonal*). Deduce that genus 3 non-hyperelliptic curves are all trigonal. Reprove the deduction by using a projection $\mathbb{P}^2 \setminus p \to \mathbb{P}^1$ for $p \in C$.*

Proof. Note C admitting a degree 3 map to $\mathbb{P}^1$ is equivalent to saying that there is a globally generated line bundle $\mathcal{L}$ on C such that:

$$\deg \mathcal{L} = 3, h^0(\mathcal{L}) = 2$$

Suppose I have three colinear points, so that I have a subscheme Z. Now Z is contained in a line ℓ, which is a complete interesection of $g-2$ hyperplanes. Therefore:

$$h^0(\mathcal{I}_Z(1)) \geq g - 2$$

Note it must equal to $g-2$ since $g-1$ hyperplanes will cut out only one point not three. Therefore we know $h^0(\mathcal{I}_Z(1)) = g-2$. Now we view Z as a degree 3 divisor on C and set $\mathcal{L} = \mathcal{O}(Z)$. It is clearly globally generated, and we only need to show $h^0(\mathcal{L}) = 2$. But by Riemann-Roch:

$$h^0(\mathcal{O}(Z)) - h^0(\mathcal{O}_C(-Z) \otimes K_C) = \deg Z + 1 - g = 4 - g$$

and $\mathcal{O}_C(-Z) \simeq \mathcal{I}_Z \otimes \mathcal{O}_C$, so that we only need to show:

$$h^0(\mathcal{I}_Z \otimes \mathcal{O}_C \otimes K_C) = h^0(\mathcal{I}_Z \otimes K_C) = g - 2$$

However projection formula reads:

$$i_*(\mathcal{I}_Z \otimes K_C) = i_*(\mathcal{I}_Z \otimes i^*\mathcal{O}(1)) = i_*\mathcal{I}_Z \otimes \mathcal{O}(1) = \mathcal{I}(Z)(1)$$

Hence $h^0(\mathcal{I}_Z \otimes K_C) = g - 2$ by assumption.

Conversely, suppose I have a globally generated line bundle $\mathcal{L}$ with $\deg \mathcal{L} = 3$ and $h^0(\mathcal{L}) = 2$. Let's choose $Z \subset C$ be any effective divisor such that $\mathcal{O}_C(Z) \simeq \mathcal{L}$. Then similar computation shows that $h^0(\mathcal{I}_Z(1)) = g - 2$, namely $\mathcal{I}_Z$ is contained in $g - 2$ linearly independent hyperplanes. Therefore Z is contained in a line. This proves the equivalence.

Now given any genus 3 non-hyperelliptic curve C, the canonical embedding $i : C \hookrightarrow \mathbb{P}^2$ ensures that we can have colinear points, by intersecting C with any line in $\mathbb{P}^2$. Therefore C must be trigonal. Alternatively, let $p \in \mathbb{P}^2 \setminus C$ and consider:

$$C \hookrightarrow \mathbb{P}^2 \setminus \{p\} \xrightarrow{\pi} \mathbb{P}^1$$

This map is again degree 3. Indeed, take any point $x \in \mathbb{P}^1$, then $\pi^{-1}(x) \in \mathbb{P}^2$ is a line, which intersects C at three points. □

Exercise 4.57. *Given an irreducible degree* 2 *homogeneous polynomial* $f(x_0, x_1, x_2, x_3)$*, show that the hypersurface* $V(f) \subset \mathbb{P}^3$ *contains a line* $\ell \subset \mathbb{P}^3$*.*

Using this and the previous exercises, show that every non-hyperelliptic genus 4 *curve is trigonal.*

Proof. By change of coordinates we can diagonalize f to assume $f = x_0^2 - x_1^2 + x_2^2 - x_3^2$. Then $V(f)$ will contain the line:

$$\ell = \{x_0 = x_1\} \cap \{x_2 = x_3\}$$

Now let C be a non-hyperelliptic genus 4 curve with canonical embedding $i : C \hookrightarrow \mathbb{P}^3$. We know $C = X_2 \cap X_3$ for a degree 2 hypersurface X_2 and some degree 3 hypersurface X_3. Now choose a line $\ell \subset X_2$ as explained in the previous part. Then:

$$\ell \cap C = \ell \cap X_3$$

consists of three points counting multiplicity. □